

A Speed Limit for Cheap Monitoring

Sequential detection of adversarial behaviour in stochastic streams

Working draft

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Abstract

A trusted monitor scores every action of an untrusted system, and a scarce human audit can adjudicate only a small fraction of them; the monitor must decide online which to escalate, so as to catch misbehaviour while almost never spending an audit on benign activity. How cheaply can this be done — what is the smallest detection delay at a given false-alarm budget?

We treat the question as sequential change detection on a temporally dependent score stream. A single scalar governs the answer: the relative-entropy *rate* R at which the attack-active stream separates from the benign one — not the single-step divergence d that score histograms display. The two agree when the scores are independent; temporal dependence in general severs them, and the gap is a safety hazard rather than a rounding error.

We prove a speed limit: every detector whose mean time between false alarms is at least γ has worst-case expected delay at least $(1 - o(1))(\log \gamma)/R$; a CUSUM detector on the score log-likelihood attains it, so the limit is exact; and in finite-window form, an attack of length L is caught with probability tending to one if $LR > (1 + o(1))\log \gamma$ and missed, by every budget-compliant detector, if $LR < (1 - o(1))\log \gamma$ — the statement a safety case needs, with R where an independence analysis would wrongly put d . Closed forms make the stakes concrete: a Gaussian AR(1) mean shift has $R = d(1 - \phi)/(1 + \phi)$, so positive correlation inflates the optimal delay over the independence prediction by exactly $(1 + \phi)/(1 - \phi)$, while a correlation-only attack has $d = 0 < R$ — detectable in principle, invisible to any analysis that treats the scores as independent. The development is self-contained for finite-memory models and marks the boundaries of the clean theory: a closed-loop attacker who times its malice in response to the monitor, and unbounded state spaces with state-dependent increments.

1 Introduction

1.1 Watching an untrusted agent

Suppose we deploy a capable AI agent to do real and consequential work — writing and merging code, administering systems, taking thousands of actions a day with little direct human oversight — while granting the possibility that somewhere in that long stream it may try to do something harmful. We cannot read every action: there are too many of them, and we are too slow. So we do what any large organisation does with a small security team. We attach a cheap, tireless *monitor* that scores each action for suspicion, and we hold in reserve a scarce and expensive resource — careful human audit, or adjudication by a far more capable model — for the few actions the monitor flags. This is the setting of *AI control* (Greenblatt et al. 2023): to argue for safety even on the pessimistic assumption that the model is adversarial, by surrounding an untrusted model with a trusted monitor and a strict budget for escalation.

The binding constraint in this picture is not computation but human attention. The monitor is cheap and can score everything; the audit is what costs, and it does not scale. Every false

alarm spends a unit of the one resource in fixed supply, so a monitor that escalated even a small fraction of all benign actions would exhaust its budget long before catching anything. The right figure of merit is therefore a *catch rate at a fixed audit budget*: of the attacks an adversary might mount, what fraction does the monitor escalate in time, if the deployment can afford to audit only, say, one action in γ ? Recent work has turned this question into an experimental science — measuring catch rates of trusted monitors on backdoored code (Greenblatt et al. 2023), extending the protocols to multi-step agents (Bhatt et al. 2025), modelling the interaction as a game between a blue team and a red team (Griffin et al. 2024), and stress-testing monitors against adaptive attacks (Terekhov et al. 2025).

1.2 The natural estimate, and the assumption inside it

To estimate a catch rate one must measure how a malicious action looks to the monitor. The recipe is natural: run the monitor on a corpus of benign actions and a corpus of malicious ones, and compare the two suspicion-score histograms. They overlap — a single malicious action is often indistinguishable from an unlucky benign one — but they are not identical, and their separation measures how much one flagged action tells you. Write d for that single-step separation (made precise as the Kullback–Leibler divergence between the two score distributions). The rest looks like arithmetic. If one action is worth d in evidence and an attack runs for L actions, the attack should expose about Ld worth, and the monitor should catch it once Ld exceeds the bar that the budget γ sets.

Concealed in the phrase “about Ld ” is an assumption: that the L actions contribute evidence the way L *independent* samples would, additively and without redundancy. It is the kind of assumption one adopts without noticing, and it is built into the standard tools: the original control setting reasons about one backdoor decision at a time (Greenblatt et al. 2023), game-theoretic analyses model the attacker as striking independently and identically across opportunities (Griffin et al. 2024), and the trajectory-level monitors most used in practice aggregate the per-step suspicion scores with order-independent rules — a maximum or a mean over steps (Bhatt et al. 2025; Terekhov et al. 2025).

But the actions of a single agent executing a single plan are no more independent than the successive frames of a film. A scheming agent does not draw fresh intentions each step; it carries a hidden purpose forward, and the suspicion scores it generates inherit that continuity — they drift, they persist, they arrive in clusters. To treat them as independent draws is to estimate how long a fuse will burn by examining a single spark. The empirical shadow of this is already on record: when Bhatt et al. (2025) monitored multi-step agent attacks and aggregated the suspicion of several actions instead of judging each in isolation, the fraction of attacks caught roughly doubled. Their explanation is the point of departure for this paper. An attack that must be carried out over many steps has to look suspicious *repeatedly*, and a consistent signal across correlated steps is far harder to fake than any single flag. The difficulty of agentic monitoring is exactly that the individual actions can each look benign in isolation (Bhatt et al. 2025; Terekhov et al. 2025); whatever betrays the attacker lives not in any one score but in how the scores accumulate. A recent line of work has begun to act on this empirically, building *stateful* monitors that carry evidence across steps rather than scoring each alone (TRACE 2026; TRACES 2026; Kale et al. 2025) — corroborating the premise of this paper, while leaving open the analytic question it answers: which scalar governs the accumulation, and when the cheap order-independent analysis is safe. We return to this in §7.

1.3 The rate, not the histogram

Once evidence accumulates over dependent steps, the quantity that governs detection is not the single-step separation d but the *rate* at which evidence builds: the asymptotic growth rate of the log-likelihood ratio between the attack-active and benign streams, a relative-entropy rate we denote R . The two coincide when the scores are independent; dependence in general severs them, and in either direction. The dangerous direction is $d > R$: when benign and attack scores share temporal persistence — positive autocorrelation, say — the rate falls below the single-step divergence, so the histograms, which see only marginal distributions, credit the monitor with evidence that accrues more slowly than they suggest. The opposite slack, $d < R$, is the reverse hazard: an attacker that spaces its malicious actions among benign ones, or routes them through a latent state, can drive d to a small fraction of R — even to zero — so that the histograms see almost nothing while a detectable signal accumulates in the temporal structure they discard. Either way the histogram misreads the rate.

This gap is not a quantitative nicety but a safety hazard, and naming it is the purpose of the paper. The quantity a safety case must respect is the total evidence an attack emits over its lifetime, which is LR and not Ld ; the monitor catches the attack once LR clears $\log \gamma$ and misses it otherwise (§4.3). An analyst who measures d from histograms and reasons as though $R = d$ therefore mis-estimates the catch threshold by the ratio d/R . In the benign direction this merely wastes effort. In the dangerous direction — whenever $d > R$ — it produces a safety case that is not just loose but *unsound*: it certifies as safe, and beyond the reach of the audit budget, attacks that no budget-compliant detector can catch, by a multiplicative margin in attack length that we compute in closed form and that grows without bound as the correlation strengthens (§6.1).

1.4 What this paper does

The mathematical engine is classical. Sequential change detection — the theory of catching, as quickly as possible and subject to a false-alarm constraint, the moment a stochastic stream changes its law — has taken the cumulative log-likelihood ratio as its central object since Page (1954), and its central result for independent observations is that the detection delay scales like $\log \gamma$ divided by the per-step divergence (Lorden 1971; Moustakides 1986; Lai 1998). We claim no credit for that machinery, nor for the now-recognised fact that monitoring must reckon with dependence across steps. The contribution is the analytic layer that this empirical recognition still lacks — the scalar that governs the accumulation and the safety reading it forces:

- a *speed limit* (§3–§4): every monitor whose mean time between false alarms is at least γ suffers worst-case detection delay at least $(1 - o(1))(\log \gamma)/R$, and a cumulative-sum monitor attains it, so the optimal delay is exactly $(\log \gamma)/R$ — governed by the rate R , not the single-step divergence d ;
- a *sharp finite-window threshold* (§4.3): an attack of length L is caught with probability tending to one if LR exceeds $\log \gamma$, and missed by every budget-compliant monitor if it falls short — the statement a safety case actually needs;
- *closed forms* (§5) for the three levers by which an attacker holds R below d — temporal correlation, pacing, and hiding behind a latent state — each with the exact factor it costs the defender and the limit beyond which it cannot help; for a Gaussian first-order model,

positive autocorrelation ϕ inflates the optimal delay over the independence prediction by exactly $(1 + \phi)/(1 - \phi)$;

- a *soundness characterisation* (§6): the histogram-based safety case is sound if and only if $d \leq R$, together with a conservative way to certify safety from quantities that can be estimated in the safe direction.

We are equally explicit about what the theory does not cover, collecting the boundaries in §8: an attacker who adapts its timing in closed loop to the monitor’s running output, for whom the rate need not even exist; the verification of the technical mixing hypotheses for hidden-state models; and detection against an attack whose statistics are unknown in advance. The thesis that survives these caveats is a single sentence. The difficulty of cheap monitoring is set not by how different a malicious action looks in isolation, but by the rate at which that difference accumulates against the budget for human attention — and that rate, the leading-order scalar under the stated assumptions, is what a safety case must measure first, before the constants, mixing times, and finite-budget corrections that the body also tracks.

What is proved, and where the edges are.

- The speed limit (§3–§4) and the finite-window threshold (§4.3) are exact for a *known* attack law under two deviation conditions, (U) for the lower bound and (L) for the upper. Both are discharged with *explicit* constants for positive finite-state chains and the Gaussian AR(1) mean shift; the general stationary-ergodic case invokes the Shannon–McMillan–Breiman theorem for relative entropy.
- The closed forms (§5) — AR(1), lazy resampler, correlation-only, Gaussian, and the hidden-state deficit identity $R = \bar{\kappa} - \iota$ — are exact. For hidden-state pacings the *rate* identities are proved, but discharging (U)/(L) there (hence the conversion of a rate to a delay guarantee) is left open.
- Soundness of the independence safety case (§6.1) holds exactly in the safe direction $d \leq R$. Where the text says an attack is “catchable” on the strength of a positive rate or an oracle (known-attack) detector alone, that is a statement about evidence, not yet a deployed guarantee; the gap is closed for finite and smoothly parametrised attack classes (§6.3) and left open nonparametrically.
- A fully adaptive, closed-loop attacker who times its malice against the monitor’s running output is non-stationary and outside scope (§8).

2 Model and foundations

2.1 What we are modelling, and the choices that matter

Picture a deployment that runs for many steps. At each step an untrusted system acts; a cheap, trusted *monitor* inspects the action and emits a *suspicion score*. Most of the time the system behaves and the scores fluctuate around a benign baseline. At some unknown moment the system may begin to misbehave, and during that stretch the law of the scores shifts. A human reviewer can adjudicate any single step definitively, but human attention is scarce, so we may escalate only a small fraction of steps. The monitor must escalate *during* misbehaviour, before

it completes, while almost never escalating during benign operation. The quantity of interest is the exchange rate between two costs: how often we escalate spuriously (the budget) and how long misbehaviour can run before we catch it (the delay).

This is the classical problem of *quickest change detection*, and two choices decide whether the resulting theory says anything true about real monitors. The first is the choice not to assume independence. In the textbook version the scores are independent and identically distributed before and after the change, and the difficulty of detection is governed by one number, the Kullback–Leibler divergence between the post- and pre-change laws of a single observation. Real monitor scores are strongly dependent across steps: a suspicious moment is often followed by another, benign lulls cluster, and — crucially — an adversary can make its malign steps resemble the system’s ordinary correlated fluctuations. Once observations are dependent, the difficulty of detection is no longer a divergence between one-step laws but a divergence *rate*, and the two can differ by a large factor. Proving that it is the rate, not the one-step divergence, that sets the speed limit is the point of the paper.

The second choice concerns how much structure to impose on the stream, and it is worth dwelling on. It is tempting to model the score stream as a finite-state Markov chain: finite, concrete, and easy to compute with. But an attacker will not oblige us by acting at a steady cadence; it will insert long, irregular, possibly random gaps between malicious steps, precisely to dilute the signal. Does a finite Markov chain forbid this? The answer is: it forbids less than it first appears, but it does forbid something, and we must locate exactly what.

- *Long mean gaps are fine.* In a finite irreducible chain the gap between malicious steps can have arbitrarily large *mean*: give the benign-to-malicious transition a small probability p and the mean gap is $1/p$, unbounded as $p \rightarrow 0$. So “long delays between malicious steps” is not excluded.
- *Heavy-tailed gaps are not fine.* What a finite chain cannot produce, with a bounded number of states, is a gap distribution with a heavy (sub-exponential, e.g. power-law) tail. The time between visits to a fixed state of a finite irreducible chain is *phase-type*, hence exponentially tailed: $\mathbb{P}(\text{gap} > k) \leq C\rho^k$ for some $\rho < 1$. Phase-type laws are weakly dense in all laws on $[0, \infty)$, so finite-state is not restrictive in *expressiveness*; it is restrictive in *uniformity*. As one demands heavier tails (within the exponential family, $\rho \rightarrow 1$), the chain’s mixing time diverges, and it is the mixing time — not Markovianity — through which the constants of the theory degrade.

The resolution is to make the modelling commitment exactly as strong as the proof needs and no stronger. The detection lower bound needs only that the normalised log-likelihood ratio between the attack-active and benign streams obeys a law of large numbers with limit R ; this holds for any stationary ergodic stream with adequate mixing, of which finite-state Markov chains are the special case where the law of large numbers is a one-line application of Birkhoff’s theorem. We therefore prove at the stationary-ergodic level, where R is the only survivor, and use finite-state chains solely to exhibit closed forms. The realistic “bursty” attacker — arbitrary gap distribution, including unbounded mean — is then captured by a renewal model (§2.7), for which the transparent quantity “information per malicious event divided by mean gap” is the exact *oracle* rate and an upper bound on R . The one regime genuinely beyond this theory, the fully adaptive attacker who times malice as a function of the monitor’s own behaviour, is non-stationary; we delimit it in §8.

2.2 The stochastic stream and the two regimes

Scores take values in a standard Borel space $(\mathcal{X}, \mathcal{B})$, which covers a finite alphabet, $\mathcal{X} = \mathbb{R}$ (continuous scores, e.g. a calibrated log-odds), or $\mathbb{R}^k$. The sample space is the sequence space $\Omega := \mathcal{X}^{\mathbb{N}}$ with coordinate maps $S_t(\omega) = \omega_t$, natural filtration $\mathcal{F}_t := \sigma(S_1, \dots, S_t)$ (with $\mathcal{F}_0 := \{\emptyset, \Omega\}$), and $\mathcal{F} := \sigma(\bigcup_t \mathcal{F}_t)$. We write $x_{1:t} := (x_1, \dots, x_t)$ for a history, so the conditioning history at step t is $x_{1:t-1}$. Lowercase letters x, s denote fixed values in $\mathcal{X}$ and the uppercase S_t the coordinate random variables; a function evaluated at a fixed history is thus a number, while the same function evaluated at the coordinate maps is a random variable — so the predictive densities $f_i(\cdot \mid x_{1:t-1})$ introduced next are deterministic, whereas $f_i(S_t \mid S_{1:t-1})$ is random. For a probability law $\mathbb{P}$ on $(\Omega, \mathcal{F})$ we write $\mathbb{P}^{(n)}$ for its restriction to $\mathcal{F}_n$, equivalently the law of the prefix $S_{1:n} = (S_1, \dots, S_n)$.

The benign and attack regimes are specified, in the manner standard for dependent-data change detection (Lai 1998), by their one-step *predictive densities*.

Assumption 2.1 (Predictive densities and absolute continuity). There is a σ -finite measure μ on $(\mathcal{X}, \mathcal{B})$ and, for each $t \geq 1$ and each history $x_{1:t-1} \in \mathcal{X}^{t-1}$, probability densities $f_0(\cdot \mid x_{1:t-1})$ and $f_1(\cdot \mid x_{1:t-1})$ with respect to μ , the benign and attack one-step predictive laws. We assume $f_1(\cdot \mid x_{1:t-1}) \ll f_0(\cdot \mid x_{1:t-1})$ for every history, so that the log-likelihood increment defined below is almost surely finite.

Definition 2.2 (The benign, attack, and change-point laws). Let $\mathbb{P}_\infty$ be the law on $(\Omega, \mathcal{F})$ whose every conditional $S_t \mid \mathcal{F}_{t-1}$ has density $f_0(\cdot \mid S_{1:t-1})$; let $\mathbb{P}_{\text{att}}$ be the law whose every conditional has density $f_1(\cdot \mid S_{1:t-1})$. For a change time $\nu \in \mathbb{N}$ let $\mathbb{P}_\nu$ be the law that uses f_0 for the transitions producing $S_1, \dots, S_{\nu-1}$ and f_1 for those producing $S_\nu, S_{\nu+1}, \dots$; that is, $S_t \mid \mathcal{F}_{t-1}$ has density $f_0(\cdot \mid S_{1:t-1})$ for $t < \nu$ and $f_1(\cdot \mid S_{1:t-1})$ for $t \geq \nu$. Set $\mathbb{P}_\nu := \mathbb{P}_\infty$ for $\nu = \infty$. Each law exists and is unique on $(\Omega, \mathcal{F})$ by the Ionescu–Tulcea theorem.

We impose the regularity that makes $\mathbb{P}_\infty$ and $\mathbb{P}_{\text{att}}$ carry a well-defined rate; everything in §3 is proved from Assumptions 2.1–2.4 alone.

Assumption 2.3 (Stationarity and ergodicity). Both $\mathbb{P}_\infty$ and $\mathbb{P}_{\text{att}}$ are stationary and ergodic. (Equivalently, each predictive family is shift-invariant and the induced shift on Ω is ergodic.)

Assumption 2.4 (Information rate). Writing $Z_t := \log \frac{f_1(S_t \mid S_{1:t-1})}{f_0(S_t \mid S_{1:t-1})}$ for the per-step log-likelihood increment and $\Lambda_a^b := \sum_{t=a}^b Z_t$ for its cumulative sum over steps a through b (so $\Lambda_1^n = \sum_{t=1}^n Z_t$), the limit

$$R := \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=1}^n Z_t$$

exists $\mathbb{P}_{\text{att}}$ -almost surely and in $L^1(\mathbb{P}_{\text{att}})$, with R a finite constant (the stationary mean of the increments; the precise identification is given in §3).

Remark 2.5 (What Assumption 2.4 costs, and where it is discharged). Assumption 2.4 is the analytic engine of the paper, and it is mild. Under Assumption 2.3 and integrability of the increments, the existence of the $\mathbb{P}_{\text{att}}$ -a.s. and L^1 limit, equal to the stationary mean of the increments, is Birkhoff’s ergodic theorem applied to the additive functional $\sum_t Z_t$ of the stationary ergodic attack stream; equivalently it is the Shannon–McMillan–Breiman theorem for relative entropy (Barron 1985; Algoet–Cover 1988); a self-contained derivation — the definitions, the

classical theorems with proof sketches, and the precise hypotheses — is given in Appendix A. Note also that the L^1 statement already identifies R with the divergence rate: by Lemma 2.7, $D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_{\infty}^{(n)}) = \mathbb{E}_{\mathbb{P}_{\text{att}}} \Lambda_1^n$, so $\frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_{\infty}^{(n)}) \rightarrow R$. For finite-memory models (finite-state chains, AR(1)) the assumption is verified by a short Birkhoff computation, carried out in §3; for hidden-state models such as the renewal example it is the cited Shannon–McMillan–Breiman theorem, which we do not reprove — nothing later depends on it. For the lower bound we will additionally use a uniform upper-deviation condition on the post-change log-likelihood ratio, uniform in the change time and in the pre-change history (Condition (U), §3.2); the rate at which that uniformity holds is governed by the mixing time, which is the precise sense in which the constants, but not the leading term R , see how heavy the gap tails are (cf. §2.1).

2.3 The change-point family: agreement and factorisation

Two facts about the family $\{\mathbb{P}_{\nu}\}$ underlie every later estimate. The first says an attack is invisible before it starts; the second computes its visibility afterward.

Lemma 2.6 (Agreement before the change). *For every $\nu \in \mathbb{N}$, the restrictions of $\mathbb{P}_{\nu}$ and $\mathbb{P}_{\infty}$ to $\mathcal{F}_{\nu-1}$ coincide.*

Proof. $\mathcal{F}_{\nu-1} = \sigma(S_{1:\nu-1})$, whose law is fixed by the predictive densities of the transitions producing $S_1, \dots, S_{\nu-1}$. By Definition 2.2 those are f_0 under both $\mathbb{P}_{\nu}$ and $\mathbb{P}_{\infty}$. Hence the two measures agree on every set in $\mathcal{F}_{\nu-1}$. For $\nu = 1$ the statement is vacuous, as $\mathcal{F}_0$ is trivial. $\square$

Lemma 2.7 (Likelihood-ratio factorisation). *Fix $\nu \geq 1$ and $n \geq \nu$. On $\mathcal{F}_n$, $\mathbb{P}_{\nu} \ll \mathbb{P}_{\infty}$ with*

$$\left. \frac{d\mathbb{P}_{\nu}}{d\mathbb{P}_{\infty}} \right|_{\mathcal{F}_n} = \prod_{t=\nu}^n \frac{f_1(S_t \mid S_{1:t-1})}{f_0(S_t \mid S_{1:t-1})} = \exp \left(\sum_{t=\nu}^n Z_t \right).$$

Proof. Under $\mathbb{P}_{\nu}$ the joint density of $S_{1:n}$ with respect to the n -fold product $\mu^{\otimes n}$ telescopes through the predictive densities:

$$p_{\nu}(x_{1:n}) = \prod_{t=1}^{\nu-1} f_0(x_t \mid x_{1:t-1}) \prod_{t=\nu}^n f_1(x_t \mid x_{1:t-1}),$$

and under $\mathbb{P}_{\infty}$, $p_{\infty}(x_{1:n}) = \prod_{t=1}^n f_0(x_t \mid x_{1:t-1})$. By Assumption 2.1 the factors of p_{∞} are positive $\mu^{\otimes n}$ -a.e. on the support, so the ratio $p_{\nu}/p_{\infty} = \prod_{t=\nu}^n f_1(x_t \mid x_{1:t-1})/f_0(x_t \mid x_{1:t-1})$ is well defined there and vanishes off it. For any $A \in \mathcal{F}_n$, integrating this ratio against $\mathbb{P}_{\infty}$ recovers $\mathbb{P}_{\nu}(A)$ by the definition of the densities; hence the ratio, evaluated at $(S_{1:n})$, is a version of $d\mathbb{P}_{\nu}/d\mathbb{P}_{\infty}$ on $\mathcal{F}_n$. The exponential form is the definition of Z_t . $\square$

Remark 2.8 (Eventual certainty; the question is the rate). Lemma 2.7 gives absolute continuity on each finite horizon but not on $\mathcal{F}$. Indeed, by Assumption 2.4 the normalised log-likelihood ratio $\frac{1}{n} \sum_{t \leq n} Z_t \rightarrow R > 0$ under $\mathbb{P}_{\text{att}}$ (when $R > 0$) and to $-R^* \leq 0$ under $\mathbb{P}_{\infty}$ for a companion benign-side rate $R^* \in [0, \infty]$ (Propositions 3.5 and 3.6), so $\mathbb{P}_{\text{att}}$ and $\mathbb{P}_{\infty}$ are mutually singular on $\mathcal{F}$. An attack is eventually certain to reveal itself; the whole question is *how fast*, and at what budget.

2.4 The two divergences

Definition 2.9 (Single-step divergence and rate). Let μ_b denote the stationary one-dimensional marginal law of S_1 under $\mathbb{P}_b$ ($b \in \{\infty, \text{att}\}$, i.e. benign and attack). The *single-step divergence* is $d := D(\mu_{\text{att}} \parallel \mu_\infty)$, the divergence an analyst reads from the monitor's score histograms. The *divergence rate* is the constant R of Assumption 2.4, equivalently $\lim_n \frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_\infty^{(n)})$ (Remark 2.5).

Proposition 2.10 (Memorylessness forces $R = d$). *If the predictive densities depend only on the current argument — $f_b(\cdot \mid x_{1:t-1}) = g_b(\cdot)$ for fixed marginals g_b , so the scores are i.i.d. — then $R = d$. In general $R \neq d$, and $R - d$ may take either sign.*

Proof. In the i.i.d. case the stationary marginal μ_b has density g_b with respect to the reference measure μ of Assumption 2.1 — that is, $d\mu_b/d\mu = g_b$ — and the increment is $Z_1 = \log \frac{g_1(S_1)}{g_0(S_1)}$, whence $R = \mathbb{E}_{\mathbb{P}_{\text{att}}}[Z_1] = \int g_1 \log \frac{g_1}{g_0} d\mu = D(\mu_{\text{att}} \parallel \mu_\infty) = d$. That $R \neq d$ in general, with either sign, is exhibited by the two-state computations of §5.1: persistence gives $R < d$ (Proposition 5.2), while an attack altering only the correlation structure gives $d = 0 < R$ (Proposition 5.4). $\square$

Remark 2.11 (Memory, not observability). What forces $R = d$ in Proposition 2.10 is the absence of *memory* in the score stream, not the observability of any underlying state. An i.i.d. score sequence has $R = d$ whether or not it is produced by some hidden mechanism, while a temporally dependent stream has $R \neq d$ even when it is fully observed. The attacker's levers for separating the two — correlation, pacing, and hiding — all act through this memory, and the panorama of §5.5 sorts them accordingly.

Definition 2.12 (The independence surrogate). The *independence safety case* is the analysis performed by an observer who, having measured the marginals $\mu_\infty, \mu_{\text{att}}$, models the stream as i.i.d. It replaces $\mathbb{P}_\infty, \mathbb{P}_{\text{att}}$ by the product laws with those marginals; under these surrogates the rate equals d by Proposition 2.10, so the surrogate predicts every detection quantity with d in the role that R plays under the true laws. The discrepancy between using d and using R is what §4 quantifies and §6.1 interprets as a possible failure of the safety case.

2.5 Detectors, the false-alarm budget, and delay

Definition 2.13 (Detector and false-alarm budget). A *detector* is a stopping time τ for $(\mathcal{F}_t)_{t \geq 0}$: a map $\tau : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ with $\{\tau = t\} \in \mathcal{F}_t$ for all t (equivalently $\{\tau \leq t\} \in \mathcal{F}_t$; in discrete time the two coincide). Its *average run length to false alarm* is $\text{ARL}(\tau) := \mathbb{E}_\infty[\tau]$, and for a budget $\gamma \geq 1$ the admissible class is $\mathcal{C}_\gamma := \{\tau : \mathbb{E}_\infty[\tau] \geq \gamma\}$. We call $\alpha := 1/\gamma$ the *audit rate*; cheap monitoring is $\alpha \rightarrow 0$, i.e. $\gamma \rightarrow \infty$.

Definition 2.14 (Worst-case detection delay). For a detector τ ,

$$J(\tau) := \sup_{\nu \geq 1} \text{ess sup } \mathbb{E}_\nu[(\tau - \nu + 1)^+ \mid \mathcal{F}_{\nu-1}], \quad x^+ := \max(x, 0),$$

the essential supremum taken under $\mathbb{P}_\nu$ (equivalently $\mathbb{P}_\infty$ on $\mathcal{F}_{\nu-1}$, by Lemma 2.6). The optimum at budget γ is $J^*(\gamma) := \inf_{\tau \in \mathcal{C}_\gamma} J(\tau)$.

Remark 2.15 (The definition is worst-case over adversarially timed, history-dependent launches). The pairing of $\sup_\nu$ with ess sup over $\mathcal{F}_{\nu-1}$ is exactly a supremum over attack-launch rules that

may depend on the benign history. Let the attacker pick a launch time ξ with $\{\xi = \nu\} \in \mathcal{F}_{\nu-1}$ for each ν — the decision to start at ν uses only what is visible before the first attack-affected score S_ν . Then $J(\tau) = \sup_{\xi} \text{ess sup } \mathbb{E}[(\tau - \xi + 1)^+ \mid \mathcal{F}_{\xi-1}]$: for any such ξ the conditional delay on each history is among the values $J(\tau)$ maximises over, hence $\leq J(\tau)$; and the maximising pair $(\nu^*, \text{history } h^*)$ is realised by the degenerate rule “launch at ν^* on exactly h^* ,” giving $\geq J(\tau)$. Randomising the launch cannot beat a worst case, so deterministic history-adapted launches suffice. Thus J is the guarantee that survives an attacker free to strike on the most confusing prefixes, not a restriction on when the attack may occur (Lorden 1971; Moustakides 1986).

Definition 2.16 (Finite-window attack and safety). For an attack length $L \geq 1$, under $\mathbb{P}_\nu$ the attack occupies steps $\nu, \dots, \nu + L - 1$; the detector *catches* it on $\{\tau \leq \nu + L - 1\}$ and *misses* it on $\{\tau \geq \nu + L\}$. The worst-case miss probability is

$$\beta_L(\tau) := \sup_{\nu \geq 1} \text{ess sup } \mathbb{P}_\nu(\tau \geq \nu + L \mid \mathcal{F}_{\nu-1}),$$

and τ is (L, δ) -safe at budget γ if $\tau \in \mathcal{C}_\gamma$ and $\beta_L(\tau) \leq \delta$.

Remark 2.17 (Delay versus window). A bound on the expected delay controls the miss probability only through Markov’s inequality, $\beta_L(\tau) \leq J(\tau)/L$, which is too lossy to be the whole story; conversely small miss probability for every L encodes the full delay tail. §4 bounds β_L directly (§4.3), because it is β_L , not the expected delay, that the safety case of §6 must certify. We record the crude inequality only to fix the direction.

2.6 Finite-state Markov chains: the source of closed forms

The finite-state case is where Assumptions 2.1–2.4 hold transparently and R, d are explicit. Let $\mathcal{X} = \mathcal{S}$ be finite, $|\mathcal{S}| = m$, and let the predictive densities be Markov: $f_b(j \mid x_{1:t-1}) = Q_b(x_{t-1}, j)$ for row-stochastic kernels Q_0, Q_1 with strictly positive entries (hence primitive, with unique full-support stationary laws π_0, π_1). Then Assumption 2.3 holds (a primitive chain is stationary and ergodic in steady state), Assumption 2.1 holds with μ counting measure, and Assumption 2.4 is Birkhoff applied to the function $(i, j) \mapsto \ell(i, j) := \log \frac{Q_1(i, j)}{Q_0(i, j)}$ of the stationary pair (S_{t-1}, S_t) :

$$R = \sum_{i \in \mathcal{S}} \pi_1(i) \sum_{j \in \mathcal{S}} Q_1(i, j) \log \frac{Q_1(i, j)}{Q_0(i, j)}, \quad d = D(\pi_1 \parallel \pi_0) = \sum_i \pi_1(i) \log \frac{\pi_1(i)}{\pi_0(i)}.$$

Remark 2.18 (Phase-type gaps: long mean, light tail). The time between successive visits of a finite irreducible chain to a fixed state (or set) A is phase-type: if T_A is the return time, $\mathbb{P}(T_A > k) = v^\top M^k \mathbf{1}$ for the substochastic restriction M of the kernel to $\mathcal{S} \setminus A$, whence $\mathbb{P}(T_A > k) \leq C\rho^k$ with $\rho = \|M\| < 1$. The mean return time $\mathbb{E}[T_A] = 1/\pi(A)$ is unbounded as $\pi(A) \rightarrow 0$, so arbitrarily long *mean* gaps between malicious steps are available; but the tail is always exponentially light. To approach a heavy tail one drives $\rho \rightarrow 1$, and the mixing time $\sim 1/(1 - \rho)$ diverges, degrading the uniformity constant of Remark 2.5 while leaving the leading rate R intact. This is the precise content of the claim in §2.1 that finite-state restricts the tail, not the mean, and pays in mixing time, not in R .

2.7 The running example: a bursty attacker as a renewal process

To represent an attacker that spaces malicious steps by arbitrary, possibly random gaps — including gaps of unbounded mean, beyond any finite chain — we use a stationary marked

renewal process. Malicious steps occur at renewal epochs whose inter-arrival gaps are i.i.d. with law G on $\mathbb{N}$ and mean $\bar{g} := \mathbb{E}_G[\text{gap}]$; at a malicious epoch the score is drawn from an attack emission q_1 , and at every other step from the benign emission q_0 , with $q_1 \ll q_0$, per-event information $\kappa := D(q_1 \| q_0) < \infty$, and $\mathbb{E}_{q_1} |\log \frac{q_1}{q_0}| < \infty$. The benign law uses q_0 throughout. This is stationary and ergodic, started from the stationary delay (equilibrium-age) law, provided $\bar{g} < \infty$ and G is *aperiodic* — the greatest common divisor of its support $\{k \geq 1 : G(k) > 0\}$ is 1, so the gaps are not confined to a proper sublattice $d\mathbb{N}$ with $d \geq 2$. (Were G supported on such a lattice, malicious epochs would recur only on a fixed residue cycle and the renewal mass would oscillate with period d rather than settle to $1/\bar{g}$, costing ergodicity; the finite-mean requirement is what makes the equilibrium-age law normalisable.)

One must be careful about which likelihood ratio is which. The *oracle* increment $\mathbf{1}\{t \text{ malicious}\} \log \frac{q_1(S_t)}{q_0(S_t)}$ presumes the malicious epochs are revealed to the observer. The *observable* increment Z_t of Assumption 2.4 conditions on past scores only; since $q_1 \ll q_0$, the epochs are hidden, the predictive density $f_1(\cdot | x_{1:t-1})$ is a mixture over the posterior phase of the renewal process, and the observable log-likelihood ratio is *not* the sum of oracle increments. Conflating the two overstates the observable rate. What survives is an exact accounting:

Proposition 2.19 (Oracle bound and information deficit). *For the renewal model, with $X_t \in \{0, 1\}$ the (latent) indicator of a malicious epoch:*

(i) *the oracle rate — the almost-sure limit of $\frac{1}{n} \sum_{t \leq n} \mathbf{1}\{X_t = 1\} \log \frac{q_1}{q_0}(S_t)$ under the attack law — equals $\kappa/\bar{g}$;*

(ii) *the observable rate of Assumption 2.4 satisfies*

$$R = \frac{\kappa}{\bar{g}} - \iota, \quad \iota := \lim_{n \rightarrow \infty} \frac{1}{n} I(X_{1:n}; S_{1:n}) \in \left[0, \frac{\kappa}{\bar{g}}\right],$$

where the mutual information is computed under the attack law and the limit exists; in particular $R \leq \kappa/\bar{g}$, with equality iff the scores reveal asymptotically nothing about the location of the malicious epochs;

(iii) *if $\bar{g} = \infty$, no stationary version of the process exists (the stationary delay law requires $\bar{g} < \infty$); for the process started fresh, with R read directly as $\lim_n \frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \| \mathbb{P}_{\infty}^{(n)})$, one has $R = 0$.*

In general $\iota > 0$ and the inequality is strict; the score stream is then a hidden-state process whose exact rate admits no closed form (compare Blackwell 1957 on entropy rates of functions of Markov chains). The deficit cannot exceed the epochs' own entropy rate, $\iota \leq H(G)/\bar{g}$ (Proposition 5.13); revealing the epochs *completely* costs R exactly that rate, leaving $R = (\kappa - H(G))/\bar{g}$, which vanishes only in the boundary case $\kappa = H(G)$ — not whenever the epochs are observable, and distinct from the $\bar{g} = \infty$ mechanism of part (iii).

Proof. Make the latent structure explicit. Let $X = (X_t)_{t \geq 1}$ be the $\{0, 1\}$ -valued stationary renewal-indicator process, with law $\mathbb{Q}$ (so $X_t = 1$ marks a malicious epoch). Adjoining it to the scores, define two laws for the pair (X, S) , each written with an overbar to signal that it is a *joint* law on the pair whose marginal on the scores S is the corresponding un-barred law. Under $\bar{\mathbb{P}}_{\text{att}}$ we have $X \sim \mathbb{Q}$ and, given X , the scores independent across t with $S_t \sim q_1$ when $X_t = 1$ and $S_t \sim q_0$ otherwise. Under $\bar{\mathbb{P}}_{\infty}$ we have $X \sim \mathbb{Q}$ and, independently, $S_t \sim q_0$ i.i.d. The marginal law of S is then $\mathbb{P}_{\text{att}}$ under $\bar{\mathbb{P}}_{\text{att}}$ and $\mathbb{P}_{\infty}$ under $\bar{\mathbb{P}}_{\infty}$, and $\bar{\mathbb{P}}_{\infty}$ is the product of its marginals.

(i) Let $N(n) := \#\{t \leq n : X_t = 1\}$. By the renewal strong law, $N(n)/n \rightarrow 1/\bar{g}$ a.s.; the marked increments $\log \frac{q_1}{q_0}(S_t)$ at malicious epochs are i.i.d. with mean κ and finite first absolute moment, so their running average converges to κ a.s.; the product of the two converges to $\kappa/\bar{g}$.

(ii) On the first n coordinates, factor the joint density:

$$\frac{d\bar{\mathbb{P}}_{\text{att}}^{(n)}(x, s)}{d\bar{\mathbb{P}}_{\infty}^{(n)}(x, s)} = \frac{d\mathbb{P}_{\text{att}}^{(n)}(s)}{d\mathbb{P}_{\infty}^{(n)}(s)} \cdot \frac{d\bar{\mathbb{P}}_{\text{att}}^{(n)}(x | s)}{d\mathbb{Q}^{(n)}(x)},$$

using that under $\bar{\mathbb{P}}_{\infty}$ the conditional law of X given S is the prior $\mathbb{Q}$. Taking $\mathbb{E}_{\bar{\mathbb{P}}_{\text{att}}} \log(\cdot)$ of both sides,

$$D(\bar{\mathbb{P}}_{\text{att}}^{(n)} \| \bar{\mathbb{P}}_{\infty}^{(n)}) = D(\mathbb{P}_{\text{att}}^{(n)} \| \mathbb{P}_{\infty}^{(n)}) + I(X_{1:n}; S_{1:n}),$$

the last term being the mutual information under $\bar{\mathbb{P}}_{\text{att}}$, by its definition as the mean divergence of posterior from prior. The left side is computed by the chain rule in the other order: the X -marginals agree (contributing 0), and given X the score coordinates are independent, contributing $D(q_1 \| q_0)$ exactly at malicious epochs; hence it equals $\kappa \mathbb{E}[N(n)]$. Therefore

$$\frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \| \mathbb{P}_{\infty}^{(n)}) = \kappa \frac{\mathbb{E}[N(n)]}{n} - \frac{1}{n} I(X_{1:n}; S_{1:n}).$$

As $n \rightarrow \infty$, the left side converges to R by the L^1 statement of Assumption 2.4 together with Lemma 2.7 (Remark 2.5; the verification of the assumption for this hidden-state model is the cited Shannon–McMillan–Breiman theorem), and $\mathbb{E}[N(n)]/n \rightarrow 1/\bar{g}$ by the elementary renewal theorem. Hence $\frac{1}{n} I(X_{1:n}; S_{1:n})$ converges as well, to $\iota := \kappa/\bar{g} - R \geq 0$; nonnegativity of mutual information gives $\iota \geq 0$ directly and $R \geq 0$ gives $\iota \leq \kappa/\bar{g}$.

(iii) The latent-coupling identity above is valid for the fresh-start process verbatim (no stationarity was used in deriving it). By nonnegativity of the mutual information, $0 \leq D(\mathbb{P}_{\text{att}}^{(n)} \| \mathbb{P}_{\infty}^{(n)}) \leq \kappa \mathbb{E}[N(n)]$, and the elementary renewal theorem — valid also for $\bar{g} = \infty$ — gives $\mathbb{E}[N(n)]/n \rightarrow 1/\bar{g} = 0$. Hence $\frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \| \mathbb{P}_{\infty}^{(n)}) \rightarrow 0$, which is the claim. $\square$

Remark 2.20 (The genuine cliff, and why it needs leaving finite state). Proposition 2.19 bounds the speed limit by “information per malicious event divided by mean spacing,” with the gap law G a free parameter: the attacker buys stealth by increasing $\bar{g}$, and the lower bound $J^* \geq (1 - o(1))(\log \gamma)/R \geq (1 - o(1))(\log \gamma) \bar{g}/\kappa$ of §3 says detection slows *at least* in proportion. Hiding the epochs lowers R further still, by exactly the information rate ι the scores leak about their location — in the attacker’s favour, strengthening the thesis. The boundary case is real: if G has infinite mean (a heavy-tailed gap), then the oracle rate $\bar{\kappa} = \kappa/\bar{g}$ vanishes and $R = 0$ in the fresh-start reading of Proposition 2.19(iii), so the logarithmic speed limit $(\log \gamma)/R$ becomes *vacuous*: the rate-based bound carries no information about delay at the boundary, and a separate, non-asymptotic analysis is needed there. (This does not assert that detection is impossible — individual malicious events still each carry evidence κ — only that the rate, the quantity this paper tracks, has degenerated; consistently, the per-event certificate of Corollary 5.15 also declines, since $\log(e\bar{g}) \rightarrow \infty$.) The degeneration is invisible to finite-state models, whose gaps always have finite mean (Remark 2.18); representing it is exactly why the theory is built at the stationary-ergodic level.

2.8 Scope: the adaptive-timing frontier, and an estimation warning

Remark 2.21 (Where the clean theorem ends). The development assumes the attack stream is stationary (Assumption 2.3); the attacker may choose its stationary law and, by Remark 2.15, the

history-dependent launch time, but not react in closed loop to the monitor’s running output. A fully adaptive attacker who throttles its malice up or down in response to the monitor’s behaviour produces a non-stationary stream, for which R need not exist and the rate law need not hold. Two fallbacks remain provable. First, take the guarantee to be the supremum over a *class* $\mathcal{A}$ of stationary attack strategies, $\inf_{\tau \in \mathcal{C}_\gamma} \sup_{P_1 \in \mathcal{A}}(\cdot)$; this is the adversarial objective $\min_{P_1 \in \mathcal{A}} LR$ developed in §6.2 and covers “the attacker picks the most evasive stationary pacing.” Second, retreat to a minimax-regret or competitive guarantee against arbitrary timing, a weaker and separate result. We state the stationary-class theorem as the main result and flag fully adaptive, closed-loop timing as open.

Remark 2.22 (Estimating R : a point estimate can be anti-conservative). The modelling concern of §2.1 reappears as an estimator-design concern. A fixed low-order fit of the score process estimates, at order zero, exactly the marginal divergence d , and converges to R only as the order grows; its bias therefore carries the sign of $d - R$. When benign and attack share temporal redundancy and the attack shifts the marginal — positive autocorrelation common to both laws, as in the AR(1) mean shift or the lazy resampler — a short-memory fit misses the redundancy and *overestimates* R , hence underestimates the delay, biased in the same dangerous direction as the independence surrogate it was meant to improve on. When instead the attack introduces structure absent under the benign law — clustered renewal events, or a correlation-only shift — the marginal understates the rate and a short fit *underestimates* R , the safe direction. The hazardous case is the former, and it coincides with the unsound regime $d > R$ of the safety case; a sound certificate therefore rests not on a point estimate but on a conservative lower bound on R , for which the deficit bound of §5.4 supplies a model-light, guaranteed instance. We develop this in §6.4.

2.9 Notation and roadmap

Scores in a standard Borel space $\mathcal{X}$; predictive densities f_0, f_1 (Assumption 2.1); laws $\mathbb{P}_\infty, \mathbb{P}_{\text{att}}, \mathbb{P}_\nu$ on $\Omega = \mathcal{X}^\mathbb{N}$ (Definition 2.2), stationary and ergodic (Assumption 2.3); increment $Z_t = \log \frac{f_1(S_t|S_{1:t-1})}{f_0(S_t|S_{1:t-1})}$; rate R (Assumption 2.4, Definition 2.9) and single-step divergence $d = D(\mu_{\text{att}}\|\mu_\infty)$; detectors τ , budget class $\mathcal{C}_\gamma$, audit rate $\alpha = 1/\gamma$ (Definition 2.13); worst-case delay $J(\tau)$, optimum $J^*(\gamma)$ (Definition 2.14); attack length L , miss probability $\beta_L(\tau)$, (L, δ) -safety (Definition 2.16). Finite-state closed forms in §2.6; the renewal example in §2.7.

From these and Assumptions 2.1–2.4 the paper proves:

- **The rate law and the speed-limit lower bound (§3, self-contained).** The strong law of Assumption 2.4 proved for finite-memory models via Birkhoff’s theorem, with the general stationary-ergodic case cited (Barron 1985; Algoet–Cover 1988) and nothing later depending on it; the benign-side rate and mutual singularity; Condition (U), discharged with explicit constants for positive finite-state chains and the Gaussian AR(1) mean shift; and the lower bound $J^*(\gamma) \geq (1 - o(1))(\log \gamma)/R$ from Lemma 2.7 by a fixed-horizon change of measure and a geometric chaining argument, every hypothesis discharged.
- **The matching upper bound and finite-window safety (§4).** A CUSUM detector on $\sum Z_t$ attains the rate, and a non-asymptotic bound on β_L shows catching an attack of length L at miss probability δ is possible iff $LR \gtrsim \log \gamma$, with explicit constants in the mixing time.

- **Closed forms (§5).** Two-state and Gaussian-AR(1) evaluations giving $R/d = (1-\phi)/(1+\phi)$ and the delay-inflation factor $(1+\phi)/(1-\phi)$; the renewal oracle bound $\kappa/\bar{g}$ and its information deficit ι ; the hidden-chain generalisation.
- **The adversarial objective and the safety case (§6).** The attacker’s problem $\min_{P_1 \in \mathcal{A}} LR$, the characterisation $d > R$ of when the independence surrogate is unsafe, and the estimation of the sign of $d - R$ on deployed monitors (§6.4).

3 The rate law and the speed-limit lower bound

This section proves the “monitoring is hard” half of the speed limit:

$$J^*(\gamma) \geq (1 - o(1)) \frac{\log \gamma}{R} \quad (\gamma \rightarrow \infty).$$

The development is in three steps, each elementary in the best sense. First (§3.1) we make the rate R itself precise: Assumption 2.4 postulates a law of large numbers, and here we prove it, with an explicit identification of the constant, for the class of *finite-memory* models — which covers every closed-form example in the paper — together with the benign-side companion rate and the mutual singularity anticipated in Remark 2.8. Second (§3.2) we isolate the single hypothesis the lower bound needs beyond §2: the post-change log-likelihood ratio must not grow *faster* than its rate, uniformly over change times and pre-change histories; we discharge it with explicit constants for positive finite-state chains and for the Gaussian AR(1) mean shift. Third (§3.3) we prove the theorem by a fixed-horizon change of measure and a geometric chaining argument — no optional stopping, no renewal theory, no appeal to the change-detection literature. Assumptions 2.1–2.4 are in force throughout.

3.1 The rate law, made precise

Definition 3.1 (Finite memory). The model has *memory* $k \geq 0$ if for $b \in \{0, 1\}$ and all $t > k$,

$$f_b(\cdot \mid x_{1:t-1}) = f_b(\cdot \mid x_{t-k:t-1}),$$

the predictive density depending only on the last k scores. (Memory 0 is the i.i.d. case; the finite-state model of §2.6 and the AR(1) model have memory 1.)

For a memory- k model write

$$\ell_k(x_{1:k+1}) := \log \frac{f_1(x_{k+1} \mid x_{1:k})}{f_0(x_{k+1} \mid x_{1:k})}, \quad \text{so that} \quad Z_t = \ell_k(S_{t-k:t}) \text{ for all } t > k.$$

For $t \leq k$ the increments Z_t are governed by the initial predictive densities, whatever makes the law stationary; they will be visible below only as a vanishing boundary term.

Theorem 3.2 (Rate law for finite-memory models). *Assume 2.1, 2.3, memory k , and the integrability conditions $\mathbb{E}_{\mathbb{P}_{\text{att}}} |Z_t| < \infty$ for $t \leq k$ and $\mathbb{E}_{\mathbb{P}_{\text{att}}} |\ell_k(S_{1:k+1})| < \infty$. Then Assumption 2.4 holds, with the constant identified as*

$$R = \mathbb{E}_{\mathbb{P}_{\text{att}}} [\ell_k(S_{1:k+1})] :$$

$\mathbb{P}_{\text{att}}$ -almost surely and in $L^1(\mathbb{P}_{\text{att}})$, $\frac{1}{n} \sum_{t \leq n} Z_t \rightarrow R$; and moreover $\frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_{\infty}^{(n)}) \rightarrow R$.

Proof. Split the sum at $t = k$. The head $\sum_{t \leq k} Z_t$ is a fixed, $\mathbb{P}_{\text{att}}$ -integrable random variable; divided by n it converges to 0 both almost surely and in L^1 . For the tail, define $\psi : \Omega \rightarrow \mathbb{R}$ by $\psi(\omega) := \ell_k(\omega_{1:k+1})$, so that $Z_t = \psi(\theta^{t-k-1}\omega)$ for $t \geq k+1$, where θ is the left shift; the hypothesis says $\mathbb{E}_{\mathbb{P}_{\text{att}}}|\psi| < \infty$. Since $\mathbb{P}_{\text{att}}$ is stationary and ergodic (Assumption 2.3), Birkhoff's pointwise and L^1 ergodic theorem gives

$$\frac{1}{n} \sum_{t=k+1}^n Z_t = \frac{n-k}{n} \cdot \frac{1}{n-k} \sum_{j=0}^{n-k-1} \psi(\theta^j \omega) \rightarrow \mathbb{E}_{\mathbb{P}_{\text{att}}}[\psi] = R,$$

almost surely and in L^1 . Adding the two pieces proves the first claim. For the divergence identification, Lemma 2.7 with $\nu = 1$ gives $D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_{\infty}^{(n)}) = \mathbb{E}_{\mathbb{P}_{\text{att}}}[\Lambda_1^n]$, which is finite under the integrability hypotheses; the L^1 convergence just proved yields $\frac{1}{n} \mathbb{E}_{\mathbb{P}_{\text{att}}} \Lambda_1^n \rightarrow R$. $\square$

Remark 3.3 (Reconciling with the phrasing of Assumption 2.4). The constant is the *stationary* mean of the increments — the expectation of ℓ_k over a stationary $(k+1)$ -block. By stationarity, $\mathbb{E}_{\mathbb{P}_{\text{att}}}[Z_t] = R$ for *every* $t > k$; the time-1 increment Z_1 , whose conditioning history is shorter than k , need not have mean R (the two coincide exactly in the memory-0 case). For the finite-state model of §2.6 the hypotheses of Theorem 3.2 hold automatically — $\ell_1 = \ell$ is bounded by positivity of the kernels, and the initial increment $\log \frac{\pi_1(S_1)}{\pi_0(S_1)}$ is bounded too — and the identified constant is exactly the display of §2.6. The AR(1) model also has memory 1, with Gaussian (hence integrable) increments; its closed form is computed in §5.2.

Proposition 3.4 (Nonnegativity, and the degenerate case). *Under the hypotheses of Theorem 3.2, $R \geq 0$; and $R = 0$ if and only if $f_1(\cdot \mid S_{1:k}) = f_0(\cdot \mid S_{1:k})$ μ -almost everywhere, for $\mathbb{P}_{\text{att}}$ -almost every history $S_{1:k}$.*

Proof. Condition on the history: by the tower property and the memory- k structure,

$$R = \mathbb{E}_{\mathbb{P}_{\text{att}}} \left[\mathbb{E}[\ell_k(S_{1:k+1}) \mid S_{1:k}] \right] = \mathbb{E}_{\mathbb{P}_{\text{att}}} \left[D(f_1(\cdot \mid S_{1:k}) \parallel f_0(\cdot \mid S_{1:k})) \right],$$

since, given $S_{1:k}$, the score S_{k+1} has density $f_1(\cdot \mid S_{1:k})$ under $\mathbb{P}_{\text{att}}$, and the inner expectation of $\log(f_1/f_0)$ under f_1 is by definition the conditional Kullback–Leibler divergence. Each conditional divergence is nonnegative, and zero exactly when the two densities agree μ -a.e. (Gibbs' inequality); averaging preserves both statements. $\square$

Proposition 3.5 (The benign-side rate). *Assume 2.1, 2.3, and memory k . Define*

$$R^* := \mathbb{E}_{\mathbb{P}_{\infty}} \left[D(f_0(\cdot \mid S_{1:k}) \parallel f_1(\cdot \mid S_{1:k})) \right] \in [0, \infty]$$

(no integrability hypothesis is needed: the integrand is nonnegative). Then, $\mathbb{P}_{\infty}$ -almost surely, the limit $\lim_n \frac{1}{n} \Lambda_1^n$ exists in $[-\infty, 0]$ and equals $-R^$ on the event that the initial increments $Z_1, \dots, Z_k$ are all finite; on the complementary event the limit is $-\infty$. In particular, if additionally $f_0(\cdot \mid x_{1:t-1}) \ll f_1(\cdot \mid x_{1:t-1})$ for all histories, the limit is $-R^*$ almost surely.*

Proof. Let $\psi(\omega) = \ell_k(\omega_{1:k+1})$ as before, now viewed under $\mathbb{P}_{\infty}$, where it takes values in $[-\infty, \infty)$ (the value $-\infty$ occurs where f_1 vanishes and f_0 does not, which Assumption 2.1 permits). Its positive part is integrable with no further hypotheses: $x^+ \leq e^x$ for all $x \in [-\infty, \infty)$, and

$$\mathbb{E}_{\mathbb{P}_{\infty}}[e^{\psi}] = \mathbb{E}_{\mathbb{P}_{\infty}} \left[\mathbb{E}_{\mathbb{P}_{\infty}}[e^{Z_{k+1}} \mid \mathcal{F}_k] \right] = \mathbb{E}_{\mathbb{P}_{\infty}} \left[\int f_1(\cdot \mid S_{1:k}) d\mu \right] = 1,$$

so $\mathbb{E}_{\mathbb{P}_\infty}[\psi^+] \leq 1$. Birkhoff's theorem extends to such one-sidedly integrable functions by truncation: for $M > 0$ put $\psi_M := \max(\psi, -M)$, which is integrable since $|\psi_M| \leq M + \psi^+$; Birkhoff gives $\frac{1}{n} \sum_{j < n} \psi_M(\theta^j \omega) \rightarrow \mathbb{E}_{\mathbb{P}_\infty}[\psi_M]$ a.s., and since $\psi \leq \psi_M$,

$$\limsup_n \frac{1}{n} \sum_{j < n} \psi(\theta^j \omega) \leq \mathbb{E}_{\mathbb{P}_\infty}[\psi_M] = \mathbb{E}_{\mathbb{P}_\infty}[\psi^+] - \mathbb{E}_{\mathbb{P}_\infty}[\psi^- \wedge M] \xrightarrow{M \rightarrow \infty} \mathbb{E}_{\mathbb{P}_\infty}[\psi] \in [-\infty, 1].$$

If $\mathbb{E}_{\mathbb{P}_\infty}[\psi] = -\infty$ this shows the averages tend to $-\infty$ a.s.; if $\mathbb{E}_{\mathbb{P}_\infty}[\psi]$ is finite, ψ is integrable and plain Birkhoff applies. In either case $\frac{1}{n} \sum_{t=k+1}^n Z_t \rightarrow \mathbb{E}_{\mathbb{P}_\infty}[\psi]$ a.s. Conditioning as in Proposition 3.4, but now with $S_{k+1} \sim f_0(\cdot \mid S_{1:k})$ under $\mathbb{P}_\infty$, identifies $\mathbb{E}_{\mathbb{P}_\infty}[\psi] = -R^*$. Finally, the head $\sum_{t \leq k} Z_t$ is a fixed random variable with values in $[-\infty, \infty)$: on the event that it is finite, dividing by n contributes 0 to the limit; where some $Z_t = -\infty$ ($t \leq k$), the whole sum Λ_1^n equals $-\infty$ from time t on, and the limit is $-\infty$. $\square$

Proposition 3.6 (Mutual singularity). *Assume 2.1, 2.3, and Assumption 2.4 with $R > 0$. Then $\mathbb{P}_{\text{att}}$ and $\mathbb{P}_\infty$ are mutually singular on $\mathcal{F}$.*

Proof. Let $E := \{\lim_n \frac{1}{n} \Lambda_1^n = R\} \in \mathcal{F}$. Assumption 2.4 gives $\mathbb{P}_{\text{att}}(E) = 1$. For the benign side, no law of large numbers is needed — a Chernoff bound and Borel–Cantelli suffice. By Lemma 2.7 with $\nu = 1$, $e^{\Lambda_1^n}$ is the $\mathcal{F}_n$ -density of $\mathbb{P}_{\text{att}}$ with respect to $\mathbb{P}_\infty$, so $\mathbb{E}_{\mathbb{P}_\infty}[e^{\Lambda_1^n}] = 1$, whence by Markov's inequality

$$\mathbb{P}_\infty\left(\Lambda_1^n \geq \frac{nR}{2}\right) \leq e^{-nR/2} \mathbb{E}_{\mathbb{P}_\infty}[e^{\Lambda_1^n}] = e^{-nR/2}.$$

The right-hand side is summable, so by Borel–Cantelli, $\mathbb{P}_\infty$ -almost surely $\Lambda_1^n < nR/2$ for all large n ; on that event $\limsup_n \frac{1}{n} \Lambda_1^n \leq R/2 < R$, so the event E fails. Hence $\mathbb{P}_\infty(E) = 0$, and E witnesses the mutual singularity. $\square$

Remark 3.7 (How little singularity needs). Proposition 3.6 consumed only the factorisation (Lemma 2.7) and the almost-sure half of Assumption 2.4; no finite memory, no integrability under $\mathbb{P}_\infty$, no benign-side rate. Proposition 3.5 is therefore not a prerequisite for singularity but a refinement of the picture: it locates the benign-side limit at $-R^*$, and shows that the two normalised log-likelihood ratios separate to *constants* of opposite sign — the geometric picture behind every detection argument to come. The exponential bound in the proof of Proposition 3.6 is itself a preview: it is the benign-side false-alarm estimate that the CUSUM analysis reuses in maximal form (Lemma 4.3).

Remark 3.8 (Scope: the general stationary-ergodic case). For models with infinite memory — the renewal example of §2.7, and hidden-state models generally — the existence of the limit in Assumption 2.4 is the Shannon–McMillan–Breiman theorem for relative entropy, due to Barron (1985) and Algoet–Cover (1988): for stationary ergodic $\mathbb{P}_{\text{att}}, \mathbb{P}_\infty$ with the appropriate domination, the normalised log-likelihood ratio converges almost surely and in L^1 to a constant. We cite this rather than reprove it; the dependency is then as follows: every theorem in §§3–4 is proved from Assumptions 2.1–2.4 plus Condition (U), so for finite-memory models the entire development is self-contained via Theorem 3.2, while for hidden-state models Assumption 2.4 rests on the cited result and Condition (U) is currently discharged only in special cases (Remark 3.13).

3.2 Uniform deviations: Condition (U)

The lower bound concerns the worst case over change times *and pre-change histories* (Definition 2.14, Remark 2.15). The rate law of §3.1 is a statement about the *stationary* attack stream; under $\mathbb{P}_\nu$, however, the post-change process is the attack dynamics initialised from a history generated under the benign law — and the essential supremum hands that history to the adversary. The stationary law of large numbers is therefore not, by itself, enough: we need the log-likelihood ratio not to accumulate evidence *faster* than its rate, uniformly over when and after what history the attack begins. The following condition is the precise requirement, the formal descendant of the uniformity anticipated in Remark 2.5. It asks only for *upper* deviations, and only at a fixed endpoint — weaker than the running-maximum (“*r*-quick”) conditions standard in the change-detection literature (Lai 1998) — because the proof of the lower bound will need no more.

Condition 3.9 ((U)). For $\delta > 0$ and $n \geq 1$ set

$$\varepsilon_n(\delta) := \sup_{\nu \geq 1} \text{ess sup } \mathbb{P}_\nu \left(\Lambda_\nu^{\nu+n-1} \geq (1+\delta)nR \mid \mathcal{F}_{\nu-1} \right),$$

the essential supremum taken under $\mathbb{P}_\nu$ (equivalently $\mathbb{P}_\infty$: the conditional probability is $\mathcal{F}_{\nu-1}$ -measurable and the two laws agree there, Lemma 2.6). *Condition (U) holds* if $\varepsilon_n(\delta) \rightarrow 0$ as $n \rightarrow \infty$, for every $\delta > 0$.

Note that $\Lambda_\nu^{\nu+n-1}$ has exactly n increments, the first being the transition *into* the attack. Lower deviations — evidence accumulating too slowly — are the upper bound’s concern and belong to §4; we record them below where they come free. We now discharge (U), with explicit constants, for the two model classes carrying this paper’s closed forms. This is where mixing enters, exactly as forecast in §2.1.

Lemma 3.10 (Dobrushin contraction). *Let Q be a transition kernel on the finite set $\mathcal{S}$, $|\mathcal{S}| = m$, with $Q(i, j) \geq q_{\min} > 0$ for all i, j , and put $\rho := 1 - m q_{\min} \in [0, 1)$. Then for every $\psi : \mathcal{S} \rightarrow \mathbb{R}$ and $k \geq 0$,*

$$\text{osc}(Q^k \psi) \leq \rho^k \text{osc}(\psi), \quad \text{where} \quad \text{osc}(\psi) := \max \psi - \min \psi.$$

If moreover π is the stationary law of Q and $\pi\psi = 0$, then $\|Q^k \psi\|_\infty \leq \rho^k \text{osc}(\psi)$.

Proof. For two rows i, i' , the total-variation distance is $\text{TV}(Q(i, \cdot), Q(i', \cdot)) = 1 - \sum_j \min(Q(i, j), Q(i', j)) \leq 1 - m q_{\min} = \rho$. For any ψ , subtracting the constant $c := \frac{1}{2}(\max \psi + \min \psi)$ (which integrates to zero against the difference of two probability vectors) gives

$$(Q\psi)(i) - (Q\psi)(i') = \int (\psi - c) d(Q(i, \cdot) - Q(i', \cdot)) \leq 2\|\psi - c\|_\infty \cdot \text{TV}(Q(i, \cdot), Q(i', \cdot)) \leq \rho \text{osc}(\psi),$$

using $\|\psi - c\|_\infty = \frac{1}{2} \text{osc}(\psi)$ and the elementary bound $\int f d(\mu - \nu) \leq 2\|f\|_\infty \text{TV}(\mu, \nu)$. Maximising over i, i' gives $\text{osc}(Q\psi) \leq \rho \text{osc}(\psi)$; iterate. For the last claim: $\pi(Q^k \psi) = \pi\psi = 0$, so $Q^k \psi$ either vanishes identically or takes values of both signs, whence $\|Q^k \psi\|_\infty \leq \text{osc}(Q^k \psi)$. $\square$

Proposition 3.11 ((U) for positive finite-state chains, with exponential rate). *In the finite-state model of §2.6 with $R > 0$, set*

$$B := \max_{i,j} |\ell(i, j)|, \quad B' := \max_i \left| \log \frac{\pi_1(i)}{\pi_0(i)} \right|, \quad q_{\min} := \min_{i,j} Q_1(i, j), \quad \rho := 1 - m q_{\min},$$

and $\tilde{B} := 2B + \frac{4B}{1-\rho}$. Then for every $\delta > 0$ and all $n \geq n_0(\delta) := \lceil \frac{2}{\delta R} (\frac{4B}{1-\rho} + B') \rceil$,

$$\varepsilon_n(\delta) \leq \exp\left(-\frac{\delta^2 R^2 n}{8\tilde{B}^2}\right),$$

so Condition (U) holds with exponential rate, the constants degrading only through the contraction parameter $1/(1-\rho)$ — a mixing time. The same bound holds for the lower deviations $\sup_\nu \text{ess sup } \mathbb{P}_\nu(\Lambda_\nu^{\nu+n-1} \leq (1-\delta)nR \mid \mathcal{F}_{\nu-1})$, recorded here for reuse in §4.

Proof. Reduction. Fix $\nu \geq 2$. By the Markov structure of the change-point law, conditionally on $\mathcal{F}_{\nu-1}$ the post-change segment $(S_{\nu-1+u})_{u \geq 0}$ is the Q_1 -chain started from $S_{\nu-1}$, and $\Lambda_\nu^{\nu+n-1} = \sum_{u=1}^n \ell(S_{\nu-2+u}, S_{\nu-1+u})$ is its n -step additive functional. The conditional probability in Condition (U) is therefore a function of the state $S_{\nu-1}$ alone, and the essential supremum is a maximum over states. It suffices to prove, uniformly in the initial state i ,

$$\mathbb{P}_i\left(\sum_{t=1}^n \ell(S_{t-1}, S_t) \geq (1+\delta)nR\right) \leq \exp\left(-\frac{\delta^2 R^2 n}{8\tilde{B}^2}\right) \quad (n \geq n_0), \quad (*)$$

where $\mathbb{P}_i$ is the law of the Q_1 -chain from $S_0 = i$.

Poisson equation. Let $h(s) := \sum_j Q_1(s, j) \ell(s, j)$, the one-step conditional mean, so that $\pi_1 h = R$ (this is the display of §2.6) and $\text{osc}(h) \leq 2B$. Define $g := \sum_{k \geq 0} Q_1^k(h - R\mathbf{1})$. Each summand is mean-zero against π_1 , so by Lemma 3.10, $\|Q_1^k(h - R\mathbf{1})\|_\infty \leq \rho^k \text{osc}(h) \leq 2B\rho^k$; the series converges absolutely with $\|g\|_\infty \leq \frac{2B}{1-\rho}$, and telescoping gives the Poisson equation $g - Q_1 g = h - R\mathbf{1}$.

Martingale decomposition. Under $\mathbb{P}_i$,

$$\sum_{t=1}^n [\ell(S_{t-1}, S_t) - R] = \sum_{t=1}^n \underbrace{[\ell(S_{t-1}, S_t) - h(S_{t-1})]}_{:= D_t} + \sum_{t=1}^n [h(S_{t-1}) - R],$$

and, by the Poisson equation, the second sum rearranges as $[g(S_0) - g(S_n)] + \sum_{t=1}^n D'_t$ with $D'_t := g(S_t) - (Q_1 g)(S_{t-1})$. Each D_t and D'_t has zero conditional mean given $\mathcal{F}_{t-1}$, so $M_n := \sum_{t \leq n} (D_t + D'_t)$ is a martingale with $M_0 = 0$ and increments bounded by $|D_t| + |D'_t| \leq 2B + \text{osc}(g) \leq 2B + 2\|g\|_\infty \leq \tilde{B}$. The boundary term obeys $|g(S_0) - g(S_n)| \leq 2\|g\|_\infty \leq \frac{4B}{1-\rho}$. Hence $\sum_{t \leq n} \ell(S_{t-1}, S_t) - nR \leq M_n + \frac{4B}{1-\rho}$.

Concentration. For $n \geq n_0(\delta)$ we have $\frac{4B}{1-\rho} + B' \leq \frac{\delta n R}{2}$ (the B' slack is spent only at $\nu = 1$ below), so the event in $(*)$ forces $M_n \geq \delta n R - \frac{4B}{1-\rho} \geq \frac{\delta n R}{2}$, and the Azuma–Hoeffding inequality — for a martingale with $M_0 = 0$ and increments bounded by c , $\mathbb{P}(M_n \geq s) \leq \exp(-s^2/(2nc^2))$ (Hoeffding 1963; Azuma 1967) — gives

$$\mathbb{P}_i(M_n \geq \frac{\delta n R}{2}) \leq \exp\left(-\frac{(\delta n R/2)^2}{2n\tilde{B}^2}\right) = \exp\left(-\frac{\delta^2 R^2 n}{8\tilde{B}^2}\right),$$

uniformly in i . This proves $(*)$, hence the claim for all $\nu \geq 2$.

The case $\nu = 1$. Here $\Lambda_1^n = \log \frac{\pi_1(S_1)}{\pi_0(S_1)} + \sum_{t=2}^n \ell(S_{t-1}, S_t)$: the first term is bounded by B' , and the sum is the $(n-1)$ -step functional of the Q_1 -chain from S_1 . Repeating the decomposition for that chain,

$$\Lambda_1^n - nR \leq B' - R + M_{n-1} + \frac{4B}{1-\rho} \leq M_{n-1} + \left(B' + \frac{4B}{1-\rho}\right),$$

using $R \geq 0$, where M_{n-1} is the corresponding martingale with the same increment bound $\tilde{B}$. For $n \geq n_0(\delta)$ the additive term is at most $\frac{\delta n R}{2}$, and Azuma–Hoeffding gives $\mathbb{P}(M_{n-1} \geq \frac{\delta n R}{2}) \leq \exp\left(-\frac{(\delta n R/2)^2}{2(n-1)\tilde{B}^2}\right) \leq \exp\left(-\frac{\delta^2 R^2 n}{8\tilde{B}^2}\right)$.

Lower deviations. Symmetrically, $\sum_{t \leq n} \ell - nR \geq M_n - \frac{4B}{1-\rho}$, and the same Azuma bound applied to the martingale $-M_n$ yields the identical estimate for $\mathbb{P}_i(\sum_{t \leq n} \ell \leq (1-\delta)nR)$ when $n \geq n_0(\delta)$, uniformly in i , and likewise at $\nu = 1$. $\square$

Proposition 3.12 ((U) for the Gaussian AR(1) mean shift). *Let the benign and attack laws be the stationary AR(1) processes*

$$S_t = \phi S_{t-1} + (1-\phi)\mu_b + \varepsilon_t, \quad \varepsilon_t \text{ i.i.d. } N(0, \sigma_\varepsilon^2), \quad |\phi| < 1,$$

with $\mu_0 = 0$ (benign), $\mu_1 = \mu \neq 0$ (attack), common stationary variance $\sigma^2 = \sigma_\varepsilon^2/(1-\phi^2)$, and write $a := (1-\phi)\mu$. Then:

- (i) for $t \geq 2$ the increment is $Z_t = \frac{a}{\sigma_\varepsilon^2}(S_t - \phi S_{t-1} - \frac{a}{2})$; under any $\mathbb{P}_\nu$, conditionally on $\mathcal{F}_{\nu-1}$, the post-change increments $(Z_t)_{t \geq \nu \vee 2}$ are i.i.d. $N(R, 2R)$, independent of the history, with

$$R = \frac{a^2}{2\sigma_\varepsilon^2} = \frac{\mu^2}{2\sigma^2} \cdot \frac{1-\phi}{1+\phi} = d \cdot \frac{1-\phi}{1+\phi}, \quad d = \frac{\mu^2}{2\sigma^2};$$

- (ii) Condition (U) holds with exponential rate: $\varepsilon_n(\delta) \leq \exp\left(-\frac{\delta^2 n R}{32}\right)$ for all $n \geq n_1(\delta) := \left\lceil 1 + \frac{2|R-d|}{\delta R} + \frac{d}{R} \right\rceil$, and the same bound holds for the lower deviations.

Proof. (i) The one-step predictive laws are $N(\phi x + (1-\phi)\mu_b, \sigma_\varepsilon^2)$, so with $u := y - \phi x$, $\log \frac{f_1(y|x)}{f_0(y|x)} = \frac{u^2 - (u-a)^2}{2\sigma_\varepsilon^2} = \frac{a(u-a/2)}{\sigma_\varepsilon^2}$. Under any $\mathbb{P}_\nu$, for $t \geq \nu \vee 2$ the transition producing S_t is the attack kernel, so $u_t := S_t - \phi S_{t-1} = a + \varepsilon_t$ whatever the value of S_{t-1} , and

$$Z_t = \frac{a(\frac{a}{2} + \varepsilon_t)}{\sigma_\varepsilon^2} \sim N\left(\frac{a^2}{2\sigma_\varepsilon^2}, \frac{a^2}{\sigma_\varepsilon^2}\right) = N(R, 2R),$$

a function of the innovation ε_t alone; since the innovations are i.i.d. and independent of $\mathcal{F}_{\nu-1}$, so are these increments, uniformly in the history. The constant: $R = \frac{a^2}{2\sigma_\varepsilon^2} = \frac{(1-\phi)^2 \mu^2}{2\sigma^2(1-\phi^2)} = \frac{\mu^2}{2\sigma^2} \cdot \frac{1-\phi}{1+\phi}$, while the stationary marginals are $N(\mu_b, \sigma^2)$, whose divergence is $d = \mu^2/(2\sigma^2)$ (equal-variance Gaussians). This verifies the hypotheses of Theorem 3.2 (memory 1, Gaussian hence integrable increments) and identifies the constant; the systematic treatment of closed forms belongs to §5.

(ii) For $\nu \geq 2$: conditionally on $\mathcal{F}_{\nu-1}$, $\Lambda_\nu^{\nu+n-1} \sim N(nR, 2nR)$ exactly, so by the Gaussian tail bound $\bar{\Phi}(x) \leq e^{-x^2/2}$,

$$\mathbb{P}_\nu(\Lambda_\nu^{\nu+n-1} \geq (1+\delta)nR \mid \mathcal{F}_{\nu-1}) = \bar{\Phi}\left(\frac{\delta n R}{\sqrt{2nR}}\right) = \bar{\Phi}\left(\delta \sqrt{\frac{nR}{2}}\right) \leq e^{-\delta^2 n R/4},$$

uniformly in the history. For $\nu = 1$: $\Lambda_1^n = W + T$ with $W := \log \frac{dN(\mu, \sigma^2)}{dN(0, \sigma^2)}(S_1) = \frac{\mu}{\sigma^2}(S_1 - \frac{\mu}{2}) \sim N(d, 2d)$ under the attack initial law $S_1 \sim N(\mu, \sigma^2)$, and $T := \sum_{t=2}^n Z_t \sim N((n-1)R, 2(n-1)R)$, independent of W . Hence $\Lambda_1^n \sim N(m_n, v_n)$ with mean $m_n = d + (n-1)R$ and variance $v_n =$

$2d + 2(n-1)R$, where $\bar{\Phi}(x) := \mathbb{P}(N(0,1) > x)$. The mean differs from the reference value nR by $m_n - nR = d - R$, so both deviation events sit at distance at least $\delta nR - |R - d|$ from the mean:

$$\mathbb{P}(\Lambda_1^n \geq (1 + \delta)nR) = \bar{\Phi}\left(\frac{\delta nR + (R - d)}{\sqrt{v_n}}\right), \quad \mathbb{P}(\Lambda_1^n \leq (1 - \delta)nR) = \bar{\Phi}\left(\frac{\delta nR - (R - d)}{\sqrt{v_n}}\right).$$

The two numerators are $\delta nR \pm (R - d)$; here the lower tail is the binding one exactly when $R > d$ (that is, $\phi < 0$), so the symmetry across the mean does *not* make the two tails interchangeable, and the initial term $R - d$ must be carried. For $n \geq n_1(\delta)$ we have $|R - d| \leq \frac{\delta nR}{2}$, so each numerator is at least $\frac{\delta nR}{2}$, and $d \leq nR$, so $v_n \leq 4nR$; each tail is thus at most $\bar{\Phi}\left(\frac{\delta\sqrt{nR}}{4}\right) \leq e^{-\delta^2 nR/32}$. With the i.i.d. $N(R, 2R)$ bound $e^{-\delta^2 nR/4}$ for the cases $\nu \geq 2$, this proves (ii) for both the upper deviations (Condition (U)) and the lower deviations (its twin), the latter being what §4 consumes. $\square$

Remark 3.13 (Hidden state: an open hypothesis). For hidden-state models — the renewal attacker of §2.7, and HMMs generally — the observable increment Z_t involves the predictive density, a mixture over the posterior law of the latent phase given past scores, and verifying Condition (U) uniformly over pre-change histories amounts to a stability property of that posterior filter under the change-point law. Such filter-stability arguments exist in the literature (in the spirit of exponential forgetting for hidden-Markov filters), but they are not elementary, and we do not develop them here: for hidden-state models, Condition (U) is an explicitly open hypothesis of Theorem 3.16. Two observations bound the consequence. First, the theorem of §3.3 is modular — it applies verbatim to any model in which (U) is verified. Second, its proof yields the weaker conclusion $J^*(\gamma) \geq (1 - o(1))(\log \gamma)/\bar{R}$ whenever the deviation condition is verified at *any* threshold rate $\bar{R} \geq R$ (Remark 3.18), so partial control of the observable log-likelihood ratio’s upper deviations — short of the exact rate — already buys a valid, if weaker, speed limit. For the renewal model neither form is currently verified.

3.3 The speed-limit lower bound

The argument has a simple shape, worth seeing before the formalities. Suppose a detector has small worst-case delay J . Then after any change point, whatever the history, it stops within a window of roughly J steps with conditional probability bounded away from zero — that is Markov’s inequality. The factorisation lemma converts this statement about the attack law into one about the benign law: the conversion costs a factor e^Λ , and Condition (U) caps Λ at about $(1 + \delta)nR$ over a window of length n , so detection-within-the-window retains probability at least roughly $e^{-(1+\delta)nR}$ under the *benign* law. But a detector that, after every multiple of n steps, stops within the next n steps with conditional probability at least p under the benign law cannot have mean run length to false alarm better than n/p : false alarms arrive geometrically. Stringing the three facts together forces $n/p \geq \gamma$, and unwinding the definitions of n and p yields $J \gtrsim (\log \gamma)/R$. Everything below is bookkeeping around this skeleton, arranged so that every constant is explicit and every exceptional null set is accounted for.

Lemma 3.14 (Conditional change of measure). *Let $\nu \geq 1$, $n \geq 1$, $c > 0$, and $A \in \mathcal{F}_{\nu+n-1}$. Then, almost surely (the exceptional set lies in $\mathcal{F}_{\nu-1}$, on which $\mathbb{P}_\nu$ and $\mathbb{P}_\infty$ agree, so the qualifier is unambiguous),*

$$\mathbb{P}_\nu(A \mid \mathcal{F}_{\nu-1}) \leq e^c \mathbb{P}_\infty(A \mid \mathcal{F}_{\nu-1}) + \mathbb{P}_\nu(\Lambda_\nu^{\nu+n-1} > c \mid \mathcal{F}_{\nu-1}).$$

Proof. Write $\Lambda := \Lambda_\nu^{\nu+n-1}$ and $D := A \cap \{\Lambda \leq c\} \in \mathcal{F}_{\nu+n-1}$. Let $B \in \mathcal{F}_{\nu-1}$ be arbitrary. By the factorisation (Lemma 2.7), applied on $\mathcal{F}_{\nu+n-1}$ to the event $D \cap B$,

$$\mathbb{P}_\nu(D \cap B) = \mathbb{E}_\infty[\mathbf{1}_{D \cap B} e^\Lambda] \leq e^c \mathbb{P}_\infty(D \cap B) \leq e^c \mathbb{P}_\infty(A \cap B),$$

while $(A \cap B) \setminus D \subseteq B \cap \{\Lambda > c\}$ gives $\mathbb{P}_\nu(A \cap B) \leq e^c \mathbb{P}_\infty(A \cap B) + \mathbb{P}_\nu(B \cap \{\Lambda > c\})$. Rewrite each term through conditional expectations given $\mathcal{F}_{\nu-1}$: the left side is $\mathbb{E}_\nu[\mathbf{1}_B \mathbb{P}_\nu(A \mid \mathcal{F}_{\nu-1})]$; the last term is $\mathbb{E}_\nu[\mathbf{1}_B \mathbb{P}_\nu(\Lambda > c \mid \mathcal{F}_{\nu-1})]$; and the middle term is $e^c \mathbb{E}_\infty[\mathbf{1}_B \mathbb{P}_\infty(A \mid \mathcal{F}_{\nu-1})] = e^c \mathbb{E}_\nu[\mathbf{1}_B \mathbb{P}_\infty(A \mid \mathcal{F}_{\nu-1})]$, the equality because the integrand is bounded and $\mathcal{F}_{\nu-1}$ -measurable and the two laws agree on $\mathcal{F}_{\nu-1}$ (Lemma 2.6). Thus, with $X := \mathbb{P}_\nu(A \mid \mathcal{F}_{\nu-1})$ and $Y := e^c \mathbb{P}_\infty(A \mid \mathcal{F}_{\nu-1}) + \mathbb{P}_\nu(\Lambda > c \mid \mathcal{F}_{\nu-1})$, both $\mathcal{F}_{\nu-1}$ -measurable and bounded, $\mathbb{E}_\nu[\mathbf{1}_B X] \leq \mathbb{E}_\nu[\mathbf{1}_B Y]$ for every $B \in \mathcal{F}_{\nu-1}$. Taking $B = \{X > Y\} \in \mathcal{F}_{\nu-1}$ shows $\mathbb{P}_\nu(X > Y) = 0$; and this null set is $\mathbb{P}_\infty$ -null too, by agreement on $\mathcal{F}_{\nu-1}$. $\square$

Lemma 3.15 (Geometric chaining). *Let τ be a detector and suppose there exist $n \in \mathbb{N}$ and $p \in (0, 1]$ such that for every $k \geq 0$, almost surely on $\{\tau \geq kn + 1\}$,*

$$\mathbb{P}_\infty(\tau \geq (k+1)n + 1 \mid \mathcal{F}_{kn}) \leq 1 - p.$$

Then $\mathbb{E}_\infty[\tau] \leq n/p$.

Proof. We claim $\mathbb{P}_\infty(\tau \geq kn + 1) \leq (1-p)^k$ for all $k \geq 0$, by induction. The base $k = 0$ is trivial. For the step, note $\{\tau \geq kn + 1\} = \{\tau \leq kn\}^c \in \mathcal{F}_{kn}$ and $\{\tau \geq (k+1)n + 1\} \subseteq \{\tau \geq kn + 1\}$, so

$$\mathbb{P}_\infty(\tau \geq (k+1)n + 1) = \mathbb{E}_\infty[\mathbf{1}_{\{\tau \geq kn + 1\}} \mathbb{P}_\infty(\tau \geq (k+1)n + 1 \mid \mathcal{F}_{kn})] \leq (1-p) \mathbb{P}_\infty(\tau \geq kn + 1).$$

Then, summing the tail by blocks of length n ,

$$\mathbb{E}_\infty[\tau] = \sum_{t \geq 1} \mathbb{P}_\infty(\tau \geq t) = \sum_{k \geq 0} \sum_{t=kn+1}^{(k+1)n} \mathbb{P}_\infty(\tau \geq t) \leq \sum_{k \geq 0} n \mathbb{P}_\infty(\tau \geq kn + 1) \leq n \sum_{k \geq 0} (1-p)^k = \frac{n}{p}.$$

$\square$

Theorem 3.16 (Speed limit, lower half). *Assume 2.1–2.4 with $R \in (0, \infty)$, and Condition (U). Then*

$$\liminf_{\gamma \rightarrow \infty} \inf_{\tau \in \mathcal{C}_\gamma} \frac{R J(\tau)}{\log \gamma} \geq 1, \quad \text{equivalently} \quad J^*(\gamma) \geq (1 - o(1)) \frac{\log \gamma}{R} \quad (\gamma \rightarrow \infty).$$

Proof. If every $\tau \in \mathcal{C}_\gamma$ has $J(\tau) = \infty$, the infimum is $+\infty$ and there is nothing to prove for that γ ; so fix $\tau \in \mathcal{C}_\gamma$ with $J := J(\tau) < \infty$. (Note $J \geq 1$: taking $\nu = 1$ in Definition 2.14 gives $J \geq \mathbb{E}_1[\tau \mid \mathcal{F}_0] \geq 1$, since $\tau \geq 1$.) Fix $\delta, \eta \in (0, 1)$; every constant below depends only on (δ, η) . Set $\bar{\kappa} := \frac{\eta}{2(1+\eta)}$, and let

$$n_{\delta, \eta} := \min\{n_0 \geq 1 : \varepsilon_m(\delta) \leq \bar{\kappa} \text{ for all } m \geq n_0\},$$

which is finite by Condition (U). Put

$$n := \max\{\lceil (1+\eta)J \rceil + 1, n_{\delta, \eta}\}, \quad c := (1+\delta)nR, \quad p := \bar{\kappa} e^{-c} \in (0, 1).$$

Step 1: after the change, detection within n steps is conditionally likely. By Definition 2.14, $\mathbb{E}_\nu[(\tau - \nu + 1)^+ | \mathcal{F}_{\nu-1}] \leq J$ almost surely, for every ν . Since $\{\tau \geq \nu + n\} = \{(\tau - \nu + 1)^+ \geq n + 1\}$, the conditional Markov inequality gives, almost surely,

$$\mathbb{P}_\nu(\tau \geq \nu + n | \mathcal{F}_{\nu-1}) \leq \frac{J}{n+1} < \frac{1}{1+\eta} = 1 - 2\bar{\kappa},$$

the strict inequality because $n+1 \geq (1+\eta)J+2 > (1+\eta)J$ and $J \geq 1$. Since $\{\tau \geq \nu\} \in \mathcal{F}_{\nu-1}$, on that event $\mathbb{P}_\nu(\tau \geq \nu | \mathcal{F}_{\nu-1}) = 1$, whence almost surely on $\{\tau \geq \nu\}$,

$$\mathbb{P}_\nu(\nu \leq \tau \leq \nu + n - 1 | \mathcal{F}_{\nu-1}) = 1 - \mathbb{P}_\nu(\tau \geq \nu + n | \mathcal{F}_{\nu-1}) \geq 2\bar{\kappa}.$$

Step 2: transfer to the benign law. The event $A_\nu := \{\nu \leq \tau \leq \nu + n - 1\}$ lies in $\mathcal{F}_{\nu+n-1}$ (indeed $\{\tau \leq \nu + n - 1\} \in \mathcal{F}_{\nu+n-1}$ and $\{\tau \geq \nu\} \in \mathcal{F}_{\nu-1}$). Lemma 3.14 with this c , together with $\mathbb{P}_\nu(\Lambda_\nu^{\nu+n-1} > c | \mathcal{F}_{\nu-1}) \leq \varepsilon_n(\delta) \leq \bar{\kappa}$ (the first inequality since $\{\Lambda > c\} \subseteq \{\Lambda \geq c\}$ and by the definition of ε_n ; the second since $n \geq n_{\delta,\eta}$), gives almost surely $\mathbb{P}_\nu(A_\nu | \mathcal{F}_{\nu-1}) \leq e^c \mathbb{P}_\infty(A_\nu | \mathcal{F}_{\nu-1}) + \bar{\kappa}$. Combining with Step 1: almost surely on $\{\tau \geq \nu\}$,

$$\mathbb{P}_\infty(A_\nu | \mathcal{F}_{\nu-1}) \geq (2\bar{\kappa} - \bar{\kappa})e^{-c} = p, \quad \text{hence} \quad \mathbb{P}_\infty(\tau \geq \nu + n | \mathcal{F}_{\nu-1}) \leq 1 - p,$$

where we used again that on $\{\tau \geq \nu\}$ one has $\mathbb{P}_\infty(A_\nu | \mathcal{F}_{\nu-1}) = 1 - \mathbb{P}_\infty(\tau \geq \nu + n | \mathcal{F}_{\nu-1})$.

Step 3: chaining. Apply Step 2 at $\nu = kn + 1$ for $k = 0, 1, 2, \dots$, noting $\mathcal{F}_{\nu-1} = \mathcal{F}_{kn}$; the countably many exceptional null sets union into one. Lemma 3.15 yields $\mathbb{E}_\infty[\tau] \leq n/p$.

Step 4: bookkeeping. The budget constraint gives

$$\gamma \leq \mathbb{E}_\infty[\tau] \leq \frac{n}{\bar{\kappa}} e^{(1+\delta)nR}, \quad \text{i.e.} \quad (1+\delta)nR \geq \log \gamma - \log n - \log \frac{1}{\bar{\kappa}}. \quad (\star)$$

Let $\gamma_0 := (n_{\delta,\eta}/\bar{\kappa}) e^{(1+\delta)n_{\delta,\eta}R}$ and suppose $\gamma > \gamma_0$. If n equalled $n_{\delta,\eta}$, then $(\star)$ would read $\gamma \leq \gamma_0$; so necessarily $n = \lceil (1+\eta)J \rceil + 1 \leq (1+\eta)J + 2$. Distinguish two cases. (a) $n > (\log \gamma)^2$. Then $J \geq \frac{n-2}{1+\eta} > \frac{(\log \gamma)^2 - 2}{1+\eta}$, so $\frac{RJ}{\log \gamma} \rightarrow \infty$ in this case; in particular it exceeds $\frac{1}{(1+\delta)(1+\eta)}$ once γ is large. (b) $n \leq (\log \gamma)^2$. Then $\log n \leq 2 \log \log \gamma$, and $(\star)$ gives

$$J \geq \frac{n-2}{1+\eta} \geq \frac{1}{1+\eta} \left[\frac{\log \gamma - 2 \log \log \gamma - \log \frac{1}{\bar{\kappa}}}{(1+\delta)R} - 2 \right].$$

In both cases, for all γ beyond a threshold depending only on (δ, η) , and uniformly over $\tau \in \mathcal{C}_\gamma$, $\frac{RJ(\tau)}{\log \gamma} \geq \frac{1-o_{\delta,\eta}(1)}{(1+\delta)(1+\eta)}$. Taking the infimum over $\tau \in \mathcal{C}_\gamma$ and then the liminf in γ ,

$$\liminf_{\gamma \rightarrow \infty} \inf_{\tau \in \mathcal{C}_\gamma} \frac{RJ(\tau)}{\log \gamma} \geq \frac{1}{(1+\delta)(1+\eta)}.$$

The left-hand side does not depend on (δ, η) ; letting $\delta \downarrow 0$ and $\eta \downarrow 0$ completes the proof. $\square$

Corollary 3.17 (Audit-rate form). *For $\alpha \in (0, 1]$, every detector whose benign false-alarm rate is at most α — that is, $\mathbb{E}_\infty[\tau] \geq 1/\alpha$ — satisfies*

$$J(\tau) \geq (1 - o(1)) \frac{\log(1/\alpha)}{R} \quad (\alpha \rightarrow 0).$$

Equivalently: achieving worst-case expected detection delay at most D forces an audit budget $\alpha \geq \exp(-(1+o(1))DR)$ as $D \rightarrow \infty$.

Proof. The first display is Theorem 3.16 with $\gamma = 1/\alpha$. For the second: if $J(\tau) \leq D$ and $\mathbb{E}_\infty[\tau] \geq 1/\alpha$, the first display gives $(1 - o(1)) \log(1/\alpha)/R \leq D$, i.e. $\log(1/\alpha) \leq (1 + o(1))DR$. $\square$

Remark 3.18 (What the proof used, and what it did not). It is worth taking inventory, both to certify the hypotheses and to expose the theorem’s modularity. The *benign* law entered only twice: through the factorisation (Lemma 2.7), and through the budget $\mathbb{E}_\infty[\tau] \geq \gamma$. Its stationarity and ergodicity played no role in the proof — they matter for defining R , for verifying Condition (U) in concrete models, and for §4. The *attack* law entered only through the single number R and through Condition (U). Consequently the proof is modular in exactly the sense anticipated in Remark 3.13: if the deviation bound $\sup_\nu \text{ess sup } \mathbb{P}_\nu(\Lambda_\nu^{\nu+n-1} \geq (1 + \delta)n\bar{R} \mid \mathcal{F}_{\nu-1}) \rightarrow 0$ is verified at *any* threshold rate $\bar{R} \geq R$, then running the identical argument with $c = (1 + \delta)n\bar{R}$ yields $J^*(\gamma) \geq (1 - o(1))(\log \gamma)/\bar{R}$ — a weaker but valid speed limit. Three further notes. First, the conclusion agrees to first order with the classical theory — Lorden (1971) for i.i.d. observations, Lai (1998) for dependent ones — but the proof is self-contained and its hypothesis is a fixed-endpoint deviation bound, weaker than the running-maximum (“*r*-quick”) conditions of the literature; the price is that our constants are asymptotic rather than explicit at finite γ . Second, the bound is tight: a CUSUM detector built on $\sum_t Z_t$ attains delay $(1 + o(1))(\log \gamma)/R$ (Theorems 4.7 and 4.8), using the lower-deviation twins recorded in Propositions 3.11 and 3.12, so the constant 1 in the theorem is exact. Third, J is an *expected* delay; the finite-window safety statement — that catching an attack of length L with miss probability δ forces $LR \gtrsim \log \gamma$ (Definition 2.16) — concerns the delay’s tail and is settled in §4.3 (Theorems 4.12 and 4.13).

4 The matching upper bound: CUSUM attains the speed limit

The lower bound of §3.3 says that no detector in $\mathcal{C}_\gamma$ beats delay $(\log \gamma)/R$. We now exhibit one that achieves it: Page’s cumulative-sum detector, run on the same log-likelihood increments Z_t that the whole theory is built from. The analysis splits along the familiar benign/attack axis, and each half needs strictly less than one might fear. The false-alarm side needs nothing beyond the factorisation: the likelihood ratio started at any time is a nonnegative martingale under the benign law, and a maximal (Ville) inequality calibrates the threshold. The delay side needs only the *lower*-deviation twin of Condition (U), strengthened from convergence to summability — because an expected first-passage time is a *sum* of tail probabilities, decay alone is not enough, but the exponential bounds already recorded in Propositions 3.11 and 3.12 give summability at once. Together with Theorem 3.16, this pins the optimal delay exactly.

4.1 The detector, and its false-alarm calibration

Definition 4.1 (CUSUM; Page 1954). The *cumulative-sum statistic* is $W_0 := 0$ and $W_t := \max(0, W_{t-1} + Z_t)$ for $t \geq 1$; for a threshold $b > 0$ the *CUSUM detector* is

$$\tau_b := \min\{t \geq 1 : W_t \geq b\}.$$

Since W_t is $\mathcal{F}_t$ -measurable, τ_b is a detector in the sense of Definition 2.13.

Lemma 4.2 (Running-maximum identity). *For every $t \geq 1$,*

$$W_t = \max\left(0, \max_{1 \leq \nu \leq t} \Lambda_\nu^t\right).$$

Consequently: (i) $\{\tau_b \leq N\} = \bigcup_{\nu=1}^N \{\exists t \in [\nu, N] : \Lambda_\nu^t \geq b\}$ for every N ; and (ii) on $\{\tau_b \geq \nu\}$ one has $\tau_b \leq \sigma_\nu := \min\{t \geq \nu : \Lambda_\nu^t \geq b\}$.

Proof. Induction on t . For $t = 1$: $W_1 = \max(0, Z_1) = \max(0, \Lambda_1^1)$. For the step, using the inductive identity and distributing the addition of Z_t over the maximum,

$$W_t = \max(0, W_{t-1} + Z_t) = \max\left(0, Z_t, \max_{\nu \leq t-1} \Lambda_\nu^{t-1} + Z_t\right) = \max\left(0, \Lambda_t^t, \max_{\nu \leq t-1} \Lambda_\nu^t\right),$$

which is the identity at t . For (i): if $W_t \geq b$ for some $t \leq N$ then $\Lambda_\nu^t \geq b$ for some $\nu \leq t$, and conversely $\Lambda_\nu^t \geq b$ with $\nu \leq t \leq N$ forces $W_t \geq b$, hence $\tau_b \leq t \leq N$. For (ii): suppose $\tau_b \geq \nu$ and $\sigma_\nu = t < \infty$; then $\Lambda_\nu^t \geq b$ with $t \geq \nu$, so $W_t \geq \Lambda_\nu^t \geq b$ by the identity, giving $\tau_b \leq t = \sigma_\nu$; if $\sigma_\nu = \infty$ the claim is vacuous. $\square$

Lemma 4.3 (Ville's inequality for the likelihood ratio). *For every $\nu \geq 1$ and $b > 0$, $\mathbb{P}_\infty(\exists t \geq \nu : \Lambda_\nu^t \geq b) \leq e^{-b}$.*

Proof. Set $M_{\nu-1} := 1$ and $M_t := e^{\Lambda_\nu^t}$ for $t \geq \nu$. Under $\mathbb{P}_\infty$, by Assumption 2.1,

$$\mathbb{E}_\infty[e^{Z_t} | \mathcal{F}_{t-1}] = \int \frac{f_1(y | S_{1:t-1})}{f_0(y | S_{1:t-1})} f_0(y | S_{1:t-1}) \mu(dy) = \int f_1(y | S_{1:t-1}) \mu(dy) = 1,$$

so $(M_t)_{t \geq \nu-1}$ is a nonnegative martingale with $\mathbb{E}_\infty[M_{\nu-1}] = 1$. Let $\sigma := \min\{t \geq \nu : M_t \geq e^b\}$. The stopped process $(M_{\sigma \wedge t})$ is again a martingale: writing $M_{\sigma \wedge t} = M_{\sigma \wedge (t-1)} + \mathbf{1}\{\sigma > t-1\}(M_t - M_{t-1})$ and noting $\{\sigma > t-1\} \in \mathcal{F}_{t-1}$, the conditional expectation given $\mathcal{F}_{t-1}$ is $M_{\sigma \wedge (t-1)}$. Hence for every finite horizon $T \geq \nu$,

$$1 = \mathbb{E}_\infty[M_{\sigma \wedge T}] \geq \mathbb{E}_\infty[M_\sigma \mathbf{1}\{\sigma \leq T\}] \geq e^b \mathbb{P}_\infty(\sigma \leq T),$$

the first inequality because $M \geq 0$ on $\{\sigma > T\}$. Letting $T \uparrow \infty$ gives $\mathbb{P}_\infty(\sigma < \infty) \leq e^{-b}$, and $\{\sigma < \infty\} = \{\exists t \geq \nu : \Lambda_\nu^t \geq b\}$. $\square$

Proposition 4.4 (False-alarm calibration). *For every $b \geq 2$, $\mathbb{E}_\infty[\tau_b] \geq e^b/8$. Consequently, for every $\gamma \geq 1$ the choice $b_\gamma := \log(8\gamma)$ puts $\tau_{b_\gamma} \in \mathcal{C}_\gamma$.*

Proof. Fix $N \in \mathbb{N}$. By Lemma 4.2(i), the union bound, and Lemma 4.3,

$$\mathbb{P}_\infty(\tau_b \leq N) \leq \sum_{\nu=1}^N \mathbb{P}_\infty(\exists t \geq \nu : \Lambda_\nu^t \geq b) \leq N e^{-b}.$$

Hence $\mathbb{E}_\infty[\tau_b] \geq \mathbb{E}_\infty[\tau_b \mathbf{1}\{\tau_b > N\}] \geq N(1 - N e^{-b})$. Choose $N = \lceil e^b/2 \rceil$, so that $N \geq e^b/2$ and $N e^{-b} \leq \frac{1}{2} + e^{-b} \leq \frac{1}{2} + e^{-2} < \frac{3}{4}$ for $b \geq 2$; then $\mathbb{E}_\infty[\tau_b] \geq \frac{e^b}{2} \cdot \frac{1}{4} = e^b/8$. The calibration claim follows since $b_\gamma = \log(8\gamma) \geq \log 8 > 2$ and $e^{b_\gamma}/8 = \gamma$. $\square$

4.2 Delay, and exactness of the speed limit

Why is CUSUM's worst case over pre-change histories benign? Because the statistic never goes below its reset value: $W_{\nu-1} \geq 0$, so the value carried *into* the attack can only help, and Lemma 4.2(ii) dominates the conditional delay, uniformly in the history, by the passage time of the fresh post-change walk Λ_ν to level b . This restart structure is the formal sense in which the

optimal monitor procrastinates: it forgets, waits at zero, and spends nothing until the evidence itself moves. What remains is to bound an *expected* first-passage time, and an expectation is a sum of tail probabilities — so the lower-deviation twin of Condition (U) is needed in summable, not merely vanishing, form.

Condition 4.5 ((L)). For $\delta \in (0, 1)$ and $n \geq 1$ set

$$\varepsilon'_n(\delta) := \sup_{\nu \geq 1} \mathbb{E}_\nu \left(\Lambda_\nu^{\nu+n-1} \leq (1-\delta)nR \mid \mathcal{F}_{\nu-1} \right).$$

Condition (L) holds if $\sum_{n \geq 1} \varepsilon'_n(\delta) < \infty$ for every $\delta \in (0, 1)$.

Remark 4.6 (Discharging (L)). Propositions 3.11 and 3.12 recorded *exponential* bounds on exactly these lower deviations — $\varepsilon'_n(\delta) \leq \exp(-\delta^2 R^2 n / 8 \tilde{B}^2)$ for $n \geq n_0(\delta)$ in the positive finite-state model, and $\varepsilon'_n(\delta) \leq \exp(-\delta^2 n R / 32)$ for $n \geq n_1(\delta)$ in the Gaussian AR(1) mean-shift model — so Condition (L) holds in both, with the same explicit, mixing-time constants, and Conditions (U) and (L) hold there simultaneously. For hidden-state models, (L) has the same status as (U): open, for the same filter-stability reason (Remark 3.13).

Theorem 4.7 (CUSUM delay). *Assume 2.1–2.4 with $R \in (0, \infty)$, and Condition (L). Then, for every $\delta \in (0, 1)$ and every $b > 0$,*

$$J(\tau_b) \leq n_b + r_\delta(n_b), \quad \text{where } n_b := \left\lceil \frac{b}{(1-\delta)R} \right\rceil, \quad r_\delta(m) := \sum_{n \geq m} \varepsilon'_n(\delta).$$

In particular $J(\tau_b) \leq (1 + o(1))b/R$ as $b \rightarrow \infty$.

Proof. Fix $\nu \geq 1$. By Lemma 4.2(ii), pointwise, $(\tau_b - \nu + 1)^+ \leq \sigma_\nu - \nu + 1$: on $\{\tau_b \geq \nu\}$ this is the domination $\tau_b \leq \sigma_\nu$, and on the complement the left side is 0 while the right side is at least 1 (as $\sigma_\nu \geq \nu$). Taking conditional expectations, $\mathbb{E}_\nu[(\tau_b - \nu + 1)^+ \mid \mathcal{F}_{\nu-1}] \leq \mathbb{E}_\nu[\sigma_\nu - \nu + 1 \mid \mathcal{F}_{\nu-1}]$ almost surely. The random variable $X := \sigma_\nu - \nu + 1$ takes values in $\{1, 2, \dots\} \cup \{\infty\}$, so by the conditional layer-cake formula (monotone convergence),

$$\mathbb{E}_\nu[X \mid \mathcal{F}_{\nu-1}] = \sum_{n \geq 0} \mathbb{P}_\nu(X > n \mid \mathcal{F}_{\nu-1}).$$

For $n \geq n_b$ we have $(1-\delta)nR \geq (1-\delta)n_bR \geq b$, and the event $\{X > n\}$ entails $\Lambda_\nu^t < b$ for all $t \in [\nu, \nu + n - 1]$, in particular $\Lambda_\nu^{\nu+n-1} < b \leq (1-\delta)nR$; hence $\mathbb{P}_\nu(X > n \mid \mathcal{F}_{\nu-1}) \leq \varepsilon'_n(\delta)$ a.s. Bounding the terms $n < n_b$ by 1,

$$\mathbb{E}_\nu[X \mid \mathcal{F}_{\nu-1}] \leq n_b + \sum_{n \geq n_b} \varepsilon'_n(\delta) = n_b + r_\delta(n_b) \quad \text{a.s.,}$$

uniformly in ν and in the history (the right-hand side is deterministic). Taking the essential supremum and the supremum over ν gives the display. For the asymptotic statement: $r_\delta(n_b) \rightarrow 0$ as $b \rightarrow \infty$ (tail of a convergent series, since $n_b \rightarrow \infty$), so $\limsup_b R J(\tau_b)/b \leq 1/(1-\delta)$ for every $\delta \in (0, 1)$; let $\delta \downarrow 0$. $\square$

Theorem 4.8 (Exactness of the speed limit). *Assume 2.1–2.4 with $R \in (0, \infty)$, and Conditions (U) and (L). Then*

$$J^*(\gamma) = (1 \pm o(1)) \frac{\log \gamma}{R} \quad (\gamma \rightarrow \infty),$$

the upper bound being attained by the CUSUM detector τ_{b_γ} with $b_\gamma = \log(8\gamma)$.

Proof. By Proposition 4.4, $\tau_{b_\gamma} \in \mathcal{C}_\gamma$, so $J^*(\gamma) \leq J(\tau_{b_\gamma}) \leq (1 + o(1))b_\gamma/R = (1 + o(1))(\log \gamma)/R$ by Theorem 4.7 (the constant $\log 8$ is absorbed by the $o(1)$). The matching lower bound is Theorem 3.16. $\square$

Corollary 4.9 (Exact speed limits, and the delay-inflation factor). *(i) For the positive finite-state model of §2.6 (with $R > 0$): $J^*(\gamma) = (1 \pm o(1))(\log \gamma)/R$, with R the explicit display of §2.6.*

(ii) For the Gaussian AR(1) mean-shift model: $J^(\gamma) = (1 \pm o(1))\frac{1+\phi}{1-\phi} \cdot \frac{\log \gamma}{d}$, where $d = \mu^2/(2\sigma^2)$ is the single-step divergence. The independence surrogate (Definition 2.12) for the same marginals is an i.i.d. model with rate d , whose exact optimal delay — by the same two theorems, trivially applicable in the i.i.d. case — is $(1 \pm o(1))(\log \gamma)/d$. Hence the surrogate misstates the true optimal delay by exactly the factor*

$$\frac{1 + \phi}{1 - \phi} :$$

an understatement (optimistic, unsafe) when $\phi > 0$, and an overstatement (conservative) when $\phi < 0$.

Proof. In both models Conditions (U) and (L) hold with exponential rates (Propositions 3.11 and 3.12; Remark 4.6), and the rate constants are those computed there; apply Theorem 4.8. For (ii), $R = d(1 - \phi)/(1 + \phi)$ by Proposition 3.12(i), so $(\log \gamma)/R = \frac{1+\phi}{1-\phi}(\log \gamma)/d$. For the surrogate: an i.i.d. model has memory 0 with $R = d$ (Proposition 2.10); its increments under attack are i.i.d. $N(d, 2d)$, so both deviation conditions hold by the Gaussian tail bound exactly as in Proposition 3.12(ii) with d in place of R , and Theorem 4.8 applies. $\square$

Remark 4.10 (Inventory, and what remains). The symmetry with Remark 3.18 is worth displaying. The false-alarm side of CUSUM (Proposition 4.4) used *only* the factorisation, through Ville’s inequality — no stationarity, no ergodicity, no deviation condition; it is the maximal-inequality form of the Chernoff bound that proved mutual singularity in Proposition 3.6. The delay side (Theorem 4.7) used *only* Condition (L) — and not Condition (U), which belongs entirely to the lower bound. So the exactness of the speed limit (Theorem 4.8) rests on precisely (U) + (L), the statement that the post-change log-likelihood ratio grows at rate exactly R , uniformly over launch times and histories. Three caveats. First, computing Z_t requires *both* predictive models: CUSUM as used here is the information-theoretic benchmark; a monitor that knows only the benign law and a finite or smoothly parametrised menu of attacks attains the same first-order limit by a max-CUSUM over the menu (§6.3). Second, J is the worst-case *expected* delay; the finite-window safety form — β_L small iff $LR \gtrsim \log \gamma$, the statement a safety case actually needs — requires non-asymptotic control of the delay’s tail and the false-alarm side jointly, and is settled in §4.3. Third, the restart structure that made the worst case tractable is also a design lesson: the optimal monitor is maximally forgetful, spending nothing until the likelihood itself accumulates — the same procrastination principle as the optimal policies of the selective-monitoring problem (Grigore & Kiefer 2018), now exact in the present model.

4.3 Finite-window safety: a sharp threshold

The expected delay J is the classical functional, but a safety case asks a sharper question (Definition 2.16): an attack runs for L steps and then *completes* — the backdoor ships, the exfiltration

finishes — so what matters is the probability of stopping *inside the window*, $\beta_L(\tau)$, not the mean delay. Markov's inequality converts J into a tail bound only at a ruinous loss (Remark 2.17), so we bound β_L directly. The reward for doing so is the cleanest statement in the paper: at budget γ the catchable/uncatchable boundary for attack length sits exactly at $L = (\log \gamma)/R$, with the miss probability tending to 0 on one side and to 1 on the other — a sharp threshold. Pleasingly, each side needs even less than the corresponding half of the expected-delay theory: achievability needs only the *vanishing* form of the lower-deviation condition (summability was needed only because an expectation is a sum of tails), and the converse needs only Condition (U).

Condition 4.11 ((L_0)). With $\varepsilon'_n(\delta)$ as in Condition 4.5: *Condition (L_0) holds* if $\varepsilon'_n(\delta) \rightarrow 0$ as $n \rightarrow \infty$, for every $\delta \in (0, 1)$. (Thus $(L) \Rightarrow (L_0)$; the exponential discharges of Remark 4.6 give both.)

Theorem 4.12 (Window achievability, non-asymptotic). *Assume 2.1–2.4 with $R \in (0, \infty)$. For every $\delta \in (0, 1)$, every threshold $b > 0$, and every L with $(1 - \delta)LR \geq b$, $\beta_L(\tau_b) \leq \varepsilon'_L(\delta)$.*

Proof. Fix ν . Pointwise, $\{\tau_b \geq \nu + L\} \subseteq \{\sigma_\nu \geq \nu + L\}$: on $\{\tau_b \geq \nu + L\}$ we have in particular $\tau_b \geq \nu$, whence $\sigma_\nu \geq \tau_b \geq \nu + L$ by Lemma 4.2(ii). The event $\{\sigma_\nu \geq \nu + L\} = \{\sigma_\nu - \nu + 1 > L\}$ entails $\Lambda_\nu^t < b$ for all $t \in [\nu, \nu + L - 1]$, in particular $\Lambda_\nu^{\nu+L-1} < b \leq (1 - \delta)LR$. Hence, almost surely, $\mathbb{P}_\nu(\tau_b \geq \nu + L \mid \mathcal{F}_{\nu-1}) \leq \mathbb{P}_\nu(\Lambda_\nu^{\nu+L-1} \leq (1 - \delta)LR \mid \mathcal{F}_{\nu-1}) \leq \varepsilon'_L(\delta)$. Taking the essential supremum and the supremum over ν gives the claim. $\square$

Theorem 4.13 (Window converse, non-asymptotic). *Assume 2.1–2.4 with $R \in (0, \infty)$. For every $\gamma \geq 1$, every $\tau \in \mathcal{C}_\gamma$, every $L \geq 1$, and every $\delta > 0$,*

$$\beta_L(\tau) \geq 1 - \varepsilon_L(\delta) - \frac{L e^{(1+\delta)LR}}{\gamma},$$

with $\varepsilon_L(\delta)$ the quantity of Condition 3.9 (no convergence hypothesis is used: the display is an identity-level consequence of the factorisation).

Proof. Write $\beta := \beta_L(\tau)$ and $c := (1 + \delta)LR$. If $1 - \beta - \varepsilon_L(\delta) \leq 0$ the display is trivial, so assume $p := (1 - \beta - \varepsilon_L(\delta))e^{-c} > 0$. By Definition 2.16, for every ν , $\mathbb{P}_\nu(\tau \geq \nu + L \mid \mathcal{F}_{\nu-1}) \leq \beta$ almost surely; since $\{\tau \geq \nu\} \in \mathcal{F}_{\nu-1}$, on that event $\mathbb{P}_\nu(A_\nu \mid \mathcal{F}_{\nu-1}) = 1 - \mathbb{P}_\nu(\tau \geq \nu + L \mid \mathcal{F}_{\nu-1}) \geq 1 - \beta$, where $A_\nu := \{\nu \leq \tau \leq \nu + L - 1\} \in \mathcal{F}_{\nu+L-1}$. Lemma 3.14 (with window $n = L$ and this c), together with $\mathbb{P}_\nu(\Lambda_\nu^{\nu+L-1} > c \mid \mathcal{F}_{\nu-1}) \leq \varepsilon_L(\delta)$, gives almost surely on $\{\tau \geq \nu\}$

$$\mathbb{P}_\infty(A_\nu \mid \mathcal{F}_{\nu-1}) \geq (1 - \beta - \varepsilon_L(\delta))e^{-c} = p, \quad \text{hence} \quad \mathbb{P}_\infty(\tau \geq \nu + L \mid \mathcal{F}_{\nu-1}) \leq 1 - p.$$

Applying this at $\nu = kL + 1$ for $k = 0, 1, 2, \dots$ (the countably many null sets union into one) and invoking the chaining Lemma 3.15 with block length L yields $\mathbb{E}_\infty[\tau] \leq L/p$. The budget $\gamma \leq \mathbb{E}_\infty[\tau]$ then rearranges to $1 - \beta - \varepsilon_L(\delta) \leq L e^{(1+\delta)LR}/\gamma$, which is the display. $\square$

Theorem 4.14 (Sharp threshold for catchable attacks). *Assume 2.1–2.4 with $R \in (0, \infty)$, Condition (U), and Condition (L_0) . Let $L_\gamma \rightarrow \infty$ as $\gamma \rightarrow \infty$.*

- (i) (Catch.) *If $\liminf_\gamma L_\gamma R / \log \gamma > 1$, then the CUSUM detector τ_{b_γ} with $b_\gamma = \log(8\gamma)$ lies in $\mathcal{C}_\gamma$ and $\beta_{L_\gamma}(\tau_{b_\gamma}) \rightarrow 0$: the attack is caught within its window, within budget, with probability tending to one, uniformly over launch times and pre-change histories.*

- (ii) (Miss.) If $\limsup_{\gamma} L_{\gamma}R/\log \gamma < 1$, then for every choice of detectors $\tau_{\gamma} \in \mathcal{C}_{\gamma}$, $\beta_{L_{\gamma}}(\tau_{\gamma}) \rightarrow 1$: the attack is missed with probability tending to one, whatever the detector.

Equivalently, in audit-rate form: at audit rate α , an attack of length L is catchable iff $LR \geq (1 \pm o(1)) \log(1/\alpha)$.

Proof. (i) Write $\liminf L_{\gamma}R/\log \gamma = 1 + c_0$ with $c_0 > 0$, so that $L_{\gamma}R \geq (1 + \frac{c_0}{2}) \log \gamma$ for all large γ , and pick $\delta \in (0, 1)$ with $(1 - \delta)(1 + \frac{c_0}{2}) > 1$. Then for γ large, $(1 - \delta)L_{\gamma}R \geq (1 - \delta)(1 + \frac{c_0}{2}) \log \gamma \geq \log \gamma + \log 8 = b_{\gamma}$, so Theorem 4.12 applies and $\beta_{L_{\gamma}}(\tau_{b_{\gamma}}) \leq \varepsilon'_{L_{\gamma}}(\delta) \rightarrow 0$ by (L₀), since $L_{\gamma} \rightarrow \infty$. Membership in $\mathcal{C}_{\gamma}$ is Proposition 4.4.

(ii) Write $\limsup L_{\gamma}R/\log \gamma = 1 - c_0$ with $c_0 \in (0, 1]$, and pick $\delta > 0$ with $(1 + \delta)(1 - \frac{c_0}{2}) \leq 1 - \delta$ (possible: as $\delta \downarrow 0$ the left side tends to $1 - \frac{c_0}{2} < 1$). For γ large, $L_{\gamma}R \leq (1 - \frac{c_0}{2}) \log \gamma$, hence $(1 + \delta)L_{\gamma}R \leq (1 - \delta) \log \gamma$ and

$$\frac{L_{\gamma}e^{(1+\delta)L_{\gamma}R}}{\gamma} \leq L_{\gamma}\gamma^{-\delta} \leq \frac{\log \gamma}{R}\gamma^{-\delta} \rightarrow 0.$$

Theorem 4.13 then gives, uniformly over $\tau_{\gamma} \in \mathcal{C}_{\gamma}$, $\beta_{L_{\gamma}}(\tau_{\gamma}) \geq 1 - \varepsilon_{L_{\gamma}}(\delta) - o(1) \rightarrow 1$, using Condition (U) and $L_{\gamma} \rightarrow \infty$ for the middle term. The audit-rate form is the substitution $\gamma = 1/\alpha$. $\square$

Corollary 4.15 (Explicit safety certificates). *Fix $\delta \in (0, 1)$ and a budget $\gamma \geq 1$, and run CUSUM at threshold $b_{\gamma} = \log(8\gamma)$.*

- (i) (Positive finite-state chains, §2.6.) If $L \geq \max\{n_0(\delta), \frac{\log(8\gamma)}{(1-\delta)R}\}$, then $\tau_{b_{\gamma}}$ is $(L, e^{-\delta^2 R^2 L/(8\tilde{B}^2)})$ -safe at budget γ , with $n_0(\delta), \tilde{B}$ the explicit constants of Proposition 3.11.
- (ii) (Gaussian AR(1) mean shift.) If $L \geq \max\{n_1(\delta), \frac{\log(8\gamma)}{(1-\delta)R}\}$ with $R = d\frac{1-\phi}{1+\phi}$, then $\tau_{b_{\gamma}}$ is $(L, e^{-\delta^2 LR/32})$ -safe at budget γ , with $n_1(\delta)$ as in Proposition 3.12.

In both cases the certificate is fully non-asymptotic: every constant is computable from the model.

Proof. Proposition 4.4 gives $\tau_{b_{\gamma}} \in \mathcal{C}_{\gamma}$. The length hypothesis gives $(1 - \delta)LR \geq b_{\gamma}$, so Theorem 4.12 bounds $\beta_L(\tau_{b_{\gamma}})$ by $\varepsilon'_L(\delta)$, which the cited propositions bound by the stated exponentials for L beyond their thresholds. This is (L, β) -safety in the sense of Definition 2.16. $\square$

Remark 4.16 (Inventory; and the safety-case reading). The split of hypotheses is again exact: the converse (Theorem 4.13) is an unconditional inequality whose asymptotic force comes from Condition (U) alone; achievability (Theorem 4.12) uses the lower-deviation condition in its weakest, merely-vanishing form (L₀) — summability was an artefact of bounding an *expectation*, and the window form, which is what safety actually needs, is cheaper. The safety-case reading of Theorem 4.14 deserves stating in words: *at audit rate α , the monitor's protection extends exactly to attacks carrying total log-likelihood evidence $LR \gtrsim \log(1/\alpha)$; an adversary who can complete its task with less total evidence — fewer steps, or a lower per-step rate R , the quantity it controls through pacing and hiding (Proposition 2.19) — is provably uncatchable at that budget, by any detector whatsoever.* The independence surrogate misjudges this boundary by exactly the factor d/R in the length coordinate — for the AR(1) model, the factor $(1 + \phi)/(1 - \phi)$ of Corollary 4.9(ii) — which is the quantitative content of the claim that correlated scores break independence-based safety cases. §6 develops this reading: the d -versus- R soundness comparison (§6.1), the attacker's objective $\min LR$ over its feasible strategies (§6.2), and the estimation of these quantities on deployed monitors.

5 Closed forms

The development so far ran on two abstract quantities: the single-step divergence d that an analyst reads off score histograms, and the rate R that actually governs detection. This section computes them. The two-state computations below resolve the sign of $R - d$ left open in the proof of Proposition 2.10: it goes both ways, and each direction is realised by a model simple enough to carry exact formulas through the entire theory — both examples have strictly positive kernels, so Conditions (U) and (L) hold with the explicit constants of Proposition 3.11, and Theorems 4.8 and 4.14 apply verbatim. The Gaussian and hidden-state models follow in §§5.2–5.4.

5.1 Two-state chains

Throughout this subsection $\mathcal{S} = \{0, 1\}$ (read: low and high suspicion), and we write

$$D_2(p \parallel q) := p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q} \quad (p, q \in (0, 1))$$

for the binary divergence. A two-state kernel is determined by its exit probabilities $a := Q(0, 1)$ and $b := Q(1, 0)$; positivity means $a, b \in (0, 1)$, and the stationary law is $\pi = (\frac{b}{a+b}, \frac{a}{a+b})$.

Proposition 5.1 (General two-state closed forms). *Let the benign and attack kernels have exit parameters (a_0, b_0) and (a_1, b_1) in $(0, 1)$. Then*

$$R = \frac{b_1}{a_1 + b_1} D_2(a_1 \parallel a_0) + \frac{a_1}{a_1 + b_1} D_2(b_1 \parallel b_0), \quad d = D_2\left(\frac{a_1}{a_1 + b_1} \parallel \frac{a_0}{a_0 + b_0}\right).$$

Proof. Immediate from the displays of §2.6: R is the π_1 -average of the row divergences, and each row of a two-state kernel is a two-point law determined by its exit probability; d is the binary divergence of the stationary laws, determined by their masses at state 1. $\square$

Example A: a persistent marginal shift, and the discrete delay-inflation factor. The cleanest one-parameter family separating R from d is the *lazy resampler*. Fix a refresh rate $\varepsilon \in (0, 1]$ and target marginals $\pi_0 = (\frac{1}{2}, \frac{1}{2})$ (benign) and $\pi_1 = (1 - w, w)$ (attack), $w \in (\frac{1}{2}, 1)$. Both regimes use the kernel $Q_b := (1 - \varepsilon)I + \varepsilon K_b$, $K_b(i, j) := \pi_b(j)$: at each step the score is refreshed from the regime's marginal with probability ε and otherwise frozen. Both chains are positive, both have stationary law π_b , and both move at the *same* speed: the attack shifts the marginal, not the tempo. In the exit parametrisation, $a_1 = \varepsilon w$, $b_1 = \varepsilon(1 - w)$, and $a_0 = b_0 = \varepsilon/2$.

Proposition 5.2 (Lazy resampler). *For the family above:*

- (i) *exactly, $R(\varepsilon) = (1 - w) D_2(\varepsilon w \parallel \frac{\varepsilon}{2}) + w D_2(\varepsilon(1 - w) \parallel \frac{\varepsilon}{2})$ and $d = D_2(w \parallel \frac{1}{2})$, with d independent of ε , and $R(1) = d$ (the memoryless case);*
- (ii) *as $\varepsilon \downarrow 0$, $R(\varepsilon) = C(w)\varepsilon + O(\varepsilon^2)$ with $C(w) := w(1 - w) \log(4w(1 - w)) + 2(w - \frac{1}{2})^2 > 0$, so $R(\varepsilon) \asymp \varepsilon$ while d stays fixed: persistence starves the rate without touching the histograms;*
- (iii) *moreover $R(\varepsilon) \leq \varepsilon d$ for every $\varepsilon \in (0, 1]$, with equality at $\varepsilon = 1$; hence, for each fixed ε , the exact optimal delay (Theorem 4.8, via Proposition 3.11) satisfies*

$$J^*(\gamma) = (1 \pm o(1)) \frac{\log \gamma}{R(\varepsilon)} \geq (1 \pm o(1)) \frac{1}{\varepsilon} \cdot \frac{\log \gamma}{d} :$$

the independence surrogate, whose own optimal delay is $(1 \pm o(1))(\log \gamma)/d$, understates the truth by a factor of at least the inverse refresh rate $1/\varepsilon$.

Proof. (i) Insert the exit parameters into Proposition 5.1; the marginals are π_b by construction, giving d . At $\varepsilon = 1$ the kernels are K_b , i.e. the scores are i.i.d., and $R = d$ is Proposition 2.10; alternatively, the symmetry $D_2(p\|\frac{1}{2}) = D_2(1-p\|\frac{1}{2})$ collapses the formula for $R(1)$ to $D_2(w\|\frac{1}{2})$.

(ii) For fixed $p, q \in (0, 1)$, expanding $\log \frac{1-\varepsilon p}{1-\varepsilon q} = \varepsilon(q-p) + O(\varepsilon^2)$ gives $D_2(\varepsilon p \|\varepsilon q) = \varepsilon(p \log \frac{p}{q} - p + q) + O(\varepsilon^2)$. Applying this to both terms of (i) (with $(p, q) = (w, \frac{1}{2})$ and $((1-w), \frac{1}{2})$) and collecting,

$$R(\varepsilon) = \varepsilon \left[\underbrace{w(1-w) \log(4w(1-w))}_{\text{log terms}} + \underbrace{(\frac{1}{2}-w)(1-2w)}_{\text{linear terms}} \right] + O(\varepsilon^2),$$

and $(\frac{1}{2}-w)(1-2w) = 2(w - \frac{1}{2})^2$, giving $C(w)$ as displayed. For positivity, substitute $x := 4(w - \frac{1}{2})^2 \in (0, 1)$, so that $4w(1-w) = 1-x$: $C = \frac{1}{4}(1-x) \log(1-x) + \frac{x}{2} := \varphi(x)$, with $\varphi(0) = 0$ and $\varphi'(x) = \frac{1}{4}(1 - \log(1-x)) > 0$ on $(0, 1)$, so φ is strictly increasing and $C > 0$ for $w \neq \frac{1}{2}$.

(iii) Couple in the refresh structure. Let $\xi = (\xi_t)_{t \geq 2}$ be i.i.d. Bernoulli(ε) refresh indicators — the same law under both regimes — and let $X_0, X_1, X_2, \dots$ be i.i.d. with law π_b , independent of ξ . Setting $S_1 := X_0$ and, for $t \geq 2$, $S_t := S_{t-1}$ if $\xi_t = 0$ and $S_t := X_{N(t)}$ if $\xi_t = 1$ (with $N(t)$ the number of refreshes up to t) realises exactly the law of the lazy resampler in regime b . Now argue as in Proposition 2.19. The score block $S_{1:n}$ is a deterministic function of the latent pair $(\xi_{2:n}, X_{0:N(n)})$, whose attack-to-benign divergence is, by the chain rule, $0 + \mathbb{E}[N(n) + 1] D(\pi_1 \|\pi_0) = (1 + \varepsilon(n-1))d$. Since divergence cannot increase under a function, the data-processing inequality gives $D(P_1^{(n)} \| P_0^{(n)}) \leq (1 + \varepsilon(n-1))d$ for the scores; dividing by n and letting $n \rightarrow \infty$ yields $R(\varepsilon) \leq \varepsilon d$. The delay display is Theorem 4.8, applicable by Proposition 3.11 since both kernels are strictly positive; the surrogate's delay is the i.i.d. case of the same theorem, as in Corollary 4.9(ii). $\square$

Remark 5.3 (The mixing-time price, made concrete). Example A is a concrete instance of the principle of §2.1: *long gaps cost mixing time, not rate*. The smallest entry of Q_1 is $q_{\min} = \varepsilon(1-w)$, so the contraction parameter of Proposition 3.11 is $\rho = 1 - 2\varepsilon(1-w)$ and the constants of the deviation bounds — hence the thresholds $n_0(\delta)$ in the safety certificates of Corollary 4.15 — scale like $1/(1-\rho) \asymp 1/\varepsilon$. Consequently the $o(1)$ in the optimal-delay statement, taken as $\gamma \rightarrow \infty$ at fixed ε , is *not* uniform in ε : the two limits do not commute, and statements about $\varepsilon \downarrow 0$ concern the leading constant $1/R(\varepsilon)$ only. This is exactly the non-uniformity anticipated in Remark 2.18, now with every constant explicit. Note also that part (iii) of Proposition 5.2 anticipates the hidden-state results of §5.4: the refresh epochs are invisible in the score stream, and $\varepsilon d - R(\varepsilon) > 0$ for $\varepsilon < 1$ is an instance of the information deficit of Proposition 2.19.

Example B: a correlation-only attack, invisible to the surrogate.

Proposition 5.4 (Correlation-only attack). *Let the benign law be i.i.d. fair flips, $Q_0(i, j) = \frac{1}{2}$, and the attack the symmetric persistent chain $Q_1 = \begin{pmatrix} s & 1-s \\ 1-s & s \end{pmatrix}$ with $s \in (0, 1)$ and $s \neq \frac{1}{2}$. Both stationary laws are $(\frac{1}{2}, \frac{1}{2})$, so*

$$d = 0, \quad \text{while} \quad R = \log 2 - H(s) > 0, \quad H(s) := -s \log s - (1-s) \log(1-s).$$

Consequently, by Theorems 4.8 and 4.14 (via Proposition 3.11; both kernels are positive), the true optimal delay is finite, $J^(\gamma) = (1 \pm o(1))(\log \gamma)/(\log 2 - H(s))$, and an attack of length $L \geq (1+c)(\log \gamma)/(\log 2 - H(s))$ is caught with probability tending to one — whereas under the independence surrogate the attack and benign laws coincide, so the surrogate predicts the attack is undetectable at any budget.*

Proof. The marginals are equal by symmetry of Q_1 (its stationary law is uniform), giving $d = 0$. Each row divergence is $D((s, 1-s) \parallel (\frac{1}{2}, \frac{1}{2})) = s \log 2s + (1-s) \log 2(1-s) = \log 2 - H(s)$, and R is the average of the two identical row divergences. Positivity of R for $s \neq \frac{1}{2}$ is strict concavity of H with maximum $\log 2$ at $\frac{1}{2}$. The delay and window statements are the cited theorems; for the surrogate, Definition 2.12 replaces both laws by i.i.d. sequences with the (equal) marginals, after which the change-point family of Definition 2.2 is constant in ν and no test distinguishes any $\mathbb{P}_\nu$ from $\mathbb{P}_\infty$. $\square$

Remark 5.5 (Reading Example B). The rate $\log 2 - H(s)$ is the *entropy deficit* of the attack chain relative to the fair coin: what betrays the attacker is not where its scores sit but how little they shuffle. Two corners are instructive. As $s \rightarrow 1$ the rate tends to $\log 2$: a score that *freezes* is maximally conspicuous against i.i.d. flips — though the mixing constants degrade simultaneously ($q_{\min} = 1-s \rightarrow 0$), so the certificates of Corollary 4.15 hold per fixed s , with thresholds growing like $1/(1-s)$, the same non-uniformity as in Remark 5.3. Jointly with Example A this settles the sign of $R - d$ left open in Proposition 2.10: it is genuinely model-dependent — Example A gives $R < d$ for small ε (indeed $R \leq \varepsilon d$), Example B gives $d = 0 < R$. For a safety case the moral is asymmetric and worth stating plainly: a monitor calibrated on histograms alone can be *exponentially over-credited* (Example A) or can *entirely miss a detectable attack* (Example B); only the rate sees both.

5.2 The Gaussian mean shift

The model of Proposition 3.12: benign and attack are stationary AR(1) processes with common autocorrelation $\phi \in (-1, 1)$ and innovation variance σ_ε^2 , differing only in stationary mean (0 versus μ); $\sigma^2 = \sigma_\varepsilon^2/(1-\phi^2)$ is the common stationary variance. We collect its consequences, the cleanest closed form in the paper.

Proposition 5.6 (Gaussian mean shift: exact speed limit and inflation factor). *With $d = \mu^2/(2\sigma^2)$:*

- (i) $R = d \frac{1-\phi}{1+\phi}$;
- (ii) Conditions (U) and (L) hold (Proposition 3.12; Remark 4.6), so by Theorem 4.8, $J^*(\gamma) = (1 \pm o(1))^{\frac{1+\phi}{1-\phi}} \cdot \frac{\log \gamma}{d}$ as $\gamma \rightarrow \infty$;
- (iii) against the independence surrogate, whose optimal delay is $(1 \pm o(1))(\log \gamma)/d$, the true optimum is inflated by exactly $\frac{1+\phi}{1-\phi}$: a factor > 1 (the surrogate is optimistic, unsafe) for $\phi > 0$, diverging as $\phi \uparrow 1$; a factor < 1 (conservative) for $\phi < 0$, vanishing as $\phi \downarrow -1$.

Proof. (i) and (ii) are Proposition 3.12 and Corollary 4.9(ii); (iii) is the arithmetic of $\frac{1+\phi}{1-\phi}$, increasing on $(-1, 1)$ from 0 to $+\infty$ with value 1 at $\phi = 0$. $\square$

Remark 5.7 (Why the mean shift is analytically benign). The decisive feature is Proposition 3.12(i): the increment $Z_t = \frac{a}{\sigma_\varepsilon^2}(\frac{a}{2} + \varepsilon_t)$, with $a = (1-\phi)\mu$, is a function of the innovation ε_t alone — the previous state S_{t-1} cancels, because a pure mean shift moves S_t and ϕS_{t-1} in lockstep. Hence the post-change increments are i.i.d. and independent of the pre-change history, and the essential supremum over histories in Condition (U) is vacuous. The next subsection shows that this is a property of the mean shift, not of Gaussian AR(1) attacks in general: when the increment genuinely depends on an unbounded state, the worst-case-over-histories formalism of Definition 2.14 can break, even though the rate law does not.

5.3 A cautionary correlation-only model

We now give the Gaussian analogue of Example B of §5.1 — an attack that shifts no marginal, only the correlation — and use it to expose a limitation of the worst-case formalism on unbounded state spaces.

Proposition 5.8 (Correlation-only attack: rate, and $d = 0$). *Let the benign law be i.i.d. $N(0, \sigma^2)$ and the attack the stationary $AR(1)$ with the same marginal,*

$$f_0(\cdot \mid x_{1:t-1}) = N(0, \sigma^2), \quad f_1(y \mid x_{1:t-1}) = N(\phi x_{t-1}, \sigma^2(1 - \phi^2)), \quad \phi \in (-1, 1) \setminus \{0\}.$$

Both stationary marginals are $N(0, \sigma^2)$, so $d = 0$; the model has memory 1 with integrable increments, so Theorem 3.2 applies and $R = -\frac{1}{2} \log(1 - \phi^2) > 0$.

Proof. Equal marginals give $d = 0$. For $t \geq 2$ the increment is, writing $u := y - \phi x_{t-1}$,

$$Z_t = \log \frac{f_1(S_t \mid S_{t-1})}{f_0(S_t \mid S_{t-1})} = -\frac{1}{2} \log(1 - \phi^2) - \frac{(S_t - \phi S_{t-1})^2}{2\sigma^2(1 - \phi^2)} + \frac{S_t^2}{2\sigma^2},$$

and $Z_1 = 0$ (equal initial densities). By Theorem 3.2, $R = \mathbb{E}_{\mathbb{P}_{\text{att}}}[\ell_1(S_1, S_2)] = \mathbb{E}_{\mathbb{P}_{\text{att}}}[D(f_1(\cdot \mid S_1) \parallel f_0(\cdot \mid S_1))]$. The conditional divergence of $N(\phi x, \sigma^2(1 - \phi^2))$ from $N(0, \sigma^2)$ is the standard Gaussian formula

$$D(N(\phi x, \sigma^2(1 - \phi^2)) \parallel N(0, \sigma^2)) = -\frac{1}{2} \log(1 - \phi^2) - \frac{\phi^2}{2} + \frac{\phi^2 x^2}{2\sigma^2},$$

and averaging over the stationary $S_1 \sim N(0, \sigma^2)$ (so $\mathbb{E}[S_1^2] = \sigma^2$) cancels the last two terms, leaving $R = -\frac{1}{2} \log(1 - \phi^2)$. Strict positivity holds for $\phi \neq 0$ since $1 - \phi^2 < 1$. Integrability is clear (Z_t is quadratic in jointly Gaussian variables). $\square$

Proposition 5.9 (Condition (U) fails on unbounded histories). *For the model of Proposition 5.8, $\varepsilon_n(\delta) = 1$ for every $\delta > 0$ and every $n \geq 1$. Hence Condition (U) does not hold, and the lower bound (Theorem 3.16) and the miss half of the threshold (Theorem 4.14(ii)) are not established by the methods of this paper for this model.*

Proof. Fix $\nu \geq 2$ and condition on $\mathcal{F}_{\nu-1}$. Under $\mathbb{P}_\nu$ the post-change states obey $S_{\nu-1+k} = \phi^k S_{\nu-1} + W_k$, where $W_k := \sum_{j=1}^k \phi^{k-j} \varepsilon_{\nu-1+j}$ is built from the post-change innovations and is independent of $S_{\nu-1}$. The increments are, for $t = \nu, \dots, \nu + n - 1$, $Z_t = -\frac{1}{2} \log(1 - \phi^2) - \frac{\varepsilon_t^2}{2\sigma^2(1 - \phi^2)} + \frac{S_t^2}{2\sigma^2}$, since $S_t - \phi S_{t-1} = \varepsilon_t$ post-change. The innovation terms do not involve $S_{\nu-1}$; the only dependence on the history is through the squared states. Writing $S_{\nu-1} = x$,

$$\Lambda_\nu^{\nu+n-1} = \underbrace{\left[-\frac{n}{2} \log(1 - \phi^2) - \sum_{t=\nu}^{\nu+n-1} \frac{\varepsilon_t^2}{2\sigma^2(1 - \phi^2)} \right]}_{\text{independent of } x} + \frac{1}{2\sigma^2} \sum_{k=0}^{n-1} (\phi^{k+1} x + W_{k+1})^2.$$

The final sum is a quadratic in x with positive leading coefficient $\frac{1}{2\sigma^2} \sum_{k=0}^{n-1} \phi^{2(k+1)} > 0$, so, holding the innovations fixed, $\Lambda_\nu^{\nu+n-1} \rightarrow +\infty$ as $|x| \rightarrow \infty$; consequently $g(x) := \mathbb{P}_\nu(\Lambda_\nu^{\nu+n-1} \geq (1 + \delta)nR \mid S_{\nu-1} = x) \rightarrow 1$ as $|x| \rightarrow \infty$. Under $\mathbb{P}_\nu$ the law of $S_{\nu-1}$ is the benign marginal $N(0, \sigma^2)$ (Lemma 2.6), of full support $\mathbb{R}$, and the conditional probability is a function of $S_{\nu-1}$ alone by the Markov property; hence $\text{ess sup } g = \sup_{x \in \mathbb{R}} g(x) = 1$, giving $\varepsilon_n(\delta) = 1$. $\square$

Remark 5.10 (What breaks, what survives, and the needed relaxation). The failure is sharp but narrow, and its anatomy is instructive.

What survives. The rate law (Theorem 3.2) is untouched: $R = -\frac{1}{2} \log(1 - \phi^2)$ exists as stated. So is mutual singularity (Proposition 3.6), which used only the almost-sure convergence, not (U). Most importantly, the *lower-deviation* conditions (L) and (L₀) *do* hold, so the monitor still works. The reason is an asymmetry: the offending histories are exactly those that make Λ large, never small. Conditioning on $S_{\nu-1} = x$, the calculation in the proof gives

$$\mathbb{E}_\nu[\Lambda_\nu^{\nu+n-1} \mid S_{\nu-1} = x] = nR + \frac{c_n}{2} \left(\frac{x^2}{\sigma^2} - 1 \right), \quad c_n := \frac{\phi^2(1 - \phi^{2n})}{1 - \phi^2} \in \left(0, \frac{\phi^2}{1 - \phi^2} \right),$$

so the conditional mean is $\geq nR - \frac{\phi^2}{2(1-\phi^2)}$, minimised at $x = 0$ and only increased by extreme histories. A lower deviation $\{\Lambda \leq (1 - \delta)nR\}$ is therefore a genuine large-deviation event for a quadratic form in Gaussians, with sub-exponential decay in n that is uniform over the history; hence $\varepsilon'_n(\delta) \rightarrow 0$ summably and Conditions (L), (L₀) hold. Consequently the achievability results — the CUSUM delay bound (Theorem 4.7) and window achievability (Theorem 4.12) — apply verbatim: *a CUSUM monitor on Z_t catches the correlation-only attack within delay $(1 + o(1))(\log \gamma)/R$ and within window $L \gtrsim (\log \gamma)/R$* , exactly the discrete conclusion of Proposition 5.4. What is lost is only the matching *lower* bound: we cannot certify, by these methods, that no detector does better.

Why the loss is an artefact, not a phenomenon. The histories that defeat (U) are those ending in an extreme score $|S_{\nu-1}| \gg \sigma$. After such a history the log-likelihood ratio jumps up immediately ($Z_\nu \approx S_\nu^2/2\sigma^2$ is huge), so the attack is *trivially* detected — a CUSUM monitor fires at once. These are the easiest cases for the monitor, and they cannot help any detector evade; they should play no role in a lower bound on delay. The change-of-measure argument nonetheless trips on them, because it pays e^Λ to move to the benign measure and that cost is unbounded precisely where Λ is large. The pessimism is in the proof technique — and, upstream, in the *worst-case-over-all-histories* essential supremum of Definition 2.14, which grants the adversary a pre-change history of vanishing benign probability.

The needed relaxation. The fix is to measure delay against a *typical* pre-change history rather than the absolute worst one: replace ess sup over $\mathcal{F}_{\nu-1}$ by the essential supremum over a set of benign probability at least $1 - \eta$, and send $\eta \downarrow 0$ after $\gamma \rightarrow \infty$. On any such typical set the states are $O(\sigma\sqrt{\log(1/\eta)})$, the increments are controlled, and (U) is recovered with n -dependence degraded only by a $\log(1/\eta)$ factor; the lower bound then goes through with R unchanged. Carrying this out requires re-opening Definition 2.14, which we do not undertake here; it is the one structural revision the unbounded-state case demands, and note that it leaves every result for bounded-increment models (§2.6, §5.1) and for the mean-shift model (§5.2, where increments are history-free) exactly as stated. For this paper, then: the correlation-only Gaussian attack is provably *caught* at the rate $R = -\frac{1}{2} \log(1 - \phi^2)$, and its provable *un-catchability* below that rate awaits the typical-set formulation.

5.4 Hidden state: monotonicity, the deficit identity, and a defender's bound

Proposition 2.19 treated one hidden-state model, the renewal attacker, and found $R = \kappa/\bar{g} - \iota$ with an information deficit $\iota \geq 0$: hiding the malicious epochs behind the score stream strictly lowers the rate, in the attacker's favour. This subsection extracts the general principle behind that calculation, and then — for the first time in the paper — turns it to the defender's advantage,

by bounding how much hiding can buy. Everything here is a statement about the *value* of R ; converting it to a delay guarantee still requires the deviation conditions of §3.2, which for hidden-state models remain open (Remark 3.13). We work with natural logarithms throughout, so divergences and entropies are in nats.

Proposition 5.11 (Monotonicity under coarsening: hiding helps the attacker). *Suppose the observer sees not (S_t) but a coarsening (Y_t) , where $Y_{1:n} = \Psi_n(S_{1:n})$ for measurable maps Ψ_n that are the same under both hypotheses (for instance a fixed per-step lumping $Y_t = \psi(S_t)$). If the coarsened process has a divergence rate R^Y , then $R^Y \leq R$.*

Proof. The coarsened laws are pushforwards, $\mathbb{P}_b^{Y,(n)} = \mathbb{P}_b^{(n)} \circ \Psi_n^{-1}$ for $b \in \{\text{att}, \infty\}$. The data-processing inequality for relative entropy gives $D(\mathbb{P}_{\text{att}}^{Y,(n)} \parallel \mathbb{P}_{\infty}^{Y,(n)}) \leq D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_{\infty}^{(n)})$; divide by n and let $n \rightarrow \infty$ (Definition 2.9). $\square$

The oracle bound $R \leq \kappa/\bar{g}$ of Proposition 2.19 is the instance in which the observer drops the latent mark: observing S_t alone is a coarsening of observing the pair (X_t, S_t) , whose rate is the oracle rate. The next proposition makes the gap exact, in general.

Proposition 5.12 (General information-deficit identity). *Let $X = (X_t)$ be a stationary ergodic latent process with law Q on a countable alphabet, the same under both hypotheses. Suppose that under the benign law the scores are i.i.d. q_0 , independent of X , while under the attack law the scores are conditionally independent given X with $S_t \sim a(\cdot \mid X_t)$, $a(\cdot \mid x) \ll q_0$. Write $\kappa(x) := D(a(\cdot \mid x) \parallel q_0)$ and $\bar{\kappa} := \mathbb{E}_Q[\kappa(X_0)] < \infty$ (the oracle rate). Then*

$$R = \bar{\kappa} - \iota, \quad \iota := \lim_{n \rightarrow \infty} \frac{1}{n} I_{\text{att}}(X_{1:n}; S_{1:n}) \in [0, \bar{\kappa}],$$

and the limit exists. (The renewal model of §2.7 is the case $X_t \in \{0, 1\}$ with $a(\cdot \mid 1) = q_1$ and $a(\cdot \mid 0) = q_0$, giving $\bar{\kappa} = \kappa/\bar{g}$.)

Proof. Form the pair laws on $(X_{1:n}, S_{1:n})$ as in Proposition 2.19: both use $X \sim Q$, the benign pair makes S i.i.d. q_0 independent of X , the attack pair uses $a(\cdot \mid X_t)$ conditionally independently. Chain-rule the relative entropy of the pair laws two ways. Decomposing as X then $S \mid X$, the X -marginals agree and the conditional term is $\mathbb{E}_Q[\sum_{t \leq n} D(a(\cdot \mid X_t) \parallel q_0)] = n\bar{\kappa}$ by stationarity. Decomposing as S then $X \mid S$, and using that under the benign pair law X is independent of S (so its conditional equals the prior Q), the term is $D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_{\infty}^{(n)}) + I_{\text{att}}(X_{1:n}; S_{1:n})$. Equate:

$$\frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_{\infty}^{(n)}) = \bar{\kappa} - \frac{1}{n} I_{\text{att}}(X_{1:n}; S_{1:n}).$$

The left side converges to R (Definition 2.9; for this hidden-state model the existence of the rate is the cited Shannon–McMillan–Breiman theorem, Remark 3.8), so the right side converges, giving $\iota := \bar{\kappa} - R$; nonnegativity of mutual information gives $\iota \geq 0$, and $R \geq 0$ gives $\iota \leq \bar{\kappa}$. $\square$

So far this only re-expresses the attacker’s advantage. The defender’s bound comes from observing that the deficit cannot exceed the randomness the attacker spends on concealment.

Proposition 5.13 (Defender’s lower bound on the rate). *Under the hypotheses of Proposition 5.12, let $h(X) := \lim_n \frac{1}{n} H(X_{1:n})$ be the entropy rate of the latent process. Then $\iota \leq h(X)$, hence*

$$R \geq \bar{\kappa} - h(X).$$

For the renewal attacker, with gap law G of mean $\bar{g}$ and entropy $H(G)$, the latent entropy rate is $h(X) = H(G)/\bar{g}$, so $R \geq \frac{\kappa - H(G)}{\bar{g}}$.

Proof. For each n , $I_{\text{att}}(X_{1:n}; S_{1:n}) \leq H(X_{1:n})$ since X is discrete (mutual information is at most the entropy of either argument). Divide by n and pass to the limit: $\iota \leq h(X)$, the entropy rate existing by stationarity (subadditivity). Substitute into Proposition 5.12. For the renewal specialisation, $\bar{\kappa} = \kappa/\bar{g}$, and the entropy rate of a stationary renewal process with gap law G is $h(X) = H(G)/\bar{g}$ (a standard fact; heuristically, $n/\bar{g}$ renewals each cost $H(G)$ nats — we use only the inequality $h(X) \leq H(G)/\bar{g}$). $\square$

Lemma 5.14 (Timing entropy is at most logarithmic in the gap). *Any probability law G on $\{1, 2, \dots\}$ with mean $\bar{g}$ satisfies $H(G) \leq \log(e\bar{g})$.*

Proof. Among laws on $\{1, 2, \dots\}$ with mean $\bar{g}$, entropy is maximised by the geometric law $G_{\star}(k) = (1-p)^{k-1}p$ with $p = 1/\bar{g}$ (the maximum-entropy law under a mean constraint is exponential in k ; equivalently, $D(G\|G_{\star}) \geq 0$ expands to $H(G) \leq H(G_{\star})$ whenever $\mathbb{E}_G[K] = \mathbb{E}_{G_{\star}}[K]$). Its entropy is $H(G_{\star}) = H_b(p)/p$ with $H_b(p) = -p \log p - (1-p) \log(1-p)$. Now

$$H(G_{\star}) = \frac{-p \log p - (1-p) \log(1-p)}{p} = \log \frac{1}{p} + \frac{1-p}{p} (-\log(1-p)) \leq \log \bar{g} + \frac{1-p}{p} \cdot \frac{p}{1-p} = \log \bar{g} + 1,$$

using $-\log(1-p) \leq p/(1-p)$ and $1/p = \bar{g}$. Hence $H(G) \leq \log \bar{g} + 1 = \log(e\bar{g})$. $\square$

Corollary 5.15 (Guaranteed catchability of any pacing). *For the renewal attacker with per-event evidence κ and mean gap $\bar{g}$,*

$$R \geq \frac{\kappa - \log(e\bar{g})}{\bar{g}},$$

regardless of how the attacker randomises its inter-event gaps. In particular, whenever $\kappa > \log(e\bar{g})$ the rate is bounded below by a positive constant determined by $(\kappa, \bar{g})$ alone: the speed limit is finite and the attack is catchable in principle, no matter the pacing law G .

Proof. Combine Proposition 5.13 with Lemma 5.14. $\square$

Remark 5.16 (Reading the bracket). Propositions 5.12 and 5.13 bracket the hidden rate, $\bar{\kappa} - h(X) \leq R \leq \bar{\kappa}$, between the oracle rate and the oracle rate minus the latent’s entropy rate. The width of the bracket is the attacker’s maximal hiding gain, and it is exactly the timing entropy it can afford to spend: a near-deterministic schedule ($h(X) \approx 0$) barely hides ($R \approx \bar{\kappa}$, the oracle rate), while randomised pacing hides more — but never more than $h(X)$, and $h(X)$ is itself capped at $\log(e\bar{g})/\bar{g}$ by Lemma 5.14. The resulting picture is a genuine tension for the attacker, and the first quantitative *good* news for the monitor in this paper: to dilute its per-event footprint the attacker must lengthen $\bar{g}$, but the concealment that buys (the deficit) grows only like $\log \bar{g}$, while the evidence it must dilute is κ per event. Corollary 5.15 is the clean consequence — an attacker whose every malicious step is even modestly conspicuous (κ a few nats) cannot hide behind *any* pacing once $\log(e\bar{g})$ falls below κ . Two caveats. First, these are rate-level statements; a delay or safety guarantee needs Conditions (U)/(L) for the renewal model, which are open (Remark 3.13), so Corollary 5.15 currently certifies a positive *rate*, not yet a finite *delay*. Second, the bound $R \geq \bar{\kappa} - h(X)$ can be loose: it is tight only when the scores reveal essentially all of the latent’s entropy, and an attacker whose emissions $a(\cdot | x)$ barely depend on x leaks little, making $\iota \ll h(X)$ and $R \ll \bar{\kappa}$ — the regime where Proposition 5.11’s coarsening bites hardest. The two bounds are complementary: monotonicity caps R from above by how much the attacker hides, the deficit bound floors it from below by how little hiding the timing randomness permits.

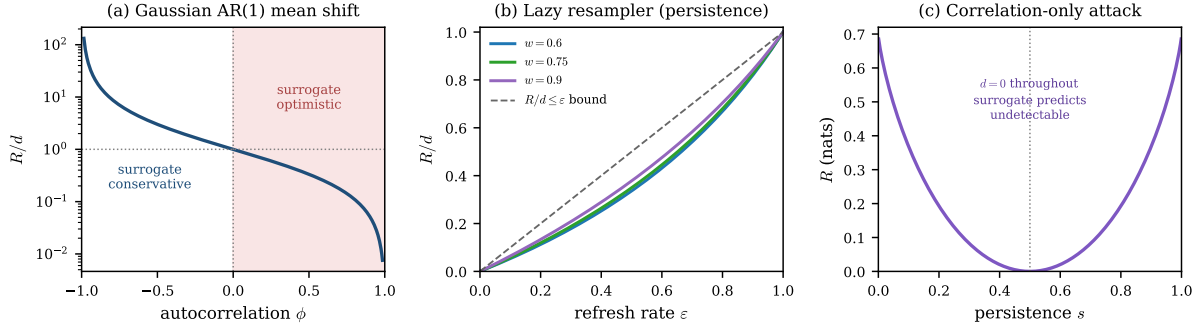


Figure 1: The three closed-form phenomena, all in nats. **(a)** Gaussian AR(1) mean shift (Proposition 5.6): the ratio $R/d = (1 - \phi)/(1 + \phi)$ against the autocorrelation ϕ , log scale. Positive correlation ($\phi > 0$, shaded) makes $R < d$, so the independence-based prediction of the optimal delay is optimistic by the factor $(1 + \phi)/(1 - \phi)$; negative correlation makes it conservative. **(b)** Lazy resampler (Proposition 5.2): R/d against the refresh rate ε , for marginal-shift parameters $w \in \{0.6, 0.75, 0.9\}$. Every curve lies on or below the bound $R/d \leq \varepsilon$ (dashed) and meets 1 at $\varepsilon = 1$ (the memoryless case): persistence starves the rate while leaving d fixed. **(c)** Correlation-only two-state attack (Proposition 5.4): $R = \log 2 - H(s)$ against the persistence s , with $d = 0$ throughout. The rate vanishes only at $s = \frac{1}{2}$ (the benign law) and is positive everywhere else, though the independence surrogate predicts the attack is undetectable.

5.5 Panorama: the closed forms and the attacker's levers

Figure 1 collects the three qualitative phenomena the closed forms isolate; Table 1 collects the formulas themselves. Read together they say one thing in three registers: the single-step divergence d that a histogram reports and the rate R that governs detection are different quantities, and an adversary controls their ratio.

Remark 5.17 (The attacker's three levers). The closed forms expose exactly three ways an adversary lowers the rate R — and, by the threshold of Theorem 4.14, the total evidence LR it must avoid emitting. Each carries a formula or a bound, and each respects a limit the defender can name.

- (1) *Persistence / correlation*. Spreading evidence across temporally correlated steps lowers R below d without altering the score histogram: by the factor $(1 - \phi)/(1 + \phi)$ for the AR(1) mean shift, and down to $R \leq \varepsilon d$ for the lazy resampler at refresh rate ε . The limit: this lever cannot create undetectability, only dilution — $R > 0$ whenever the conditional laws genuinely differ (Proposition 3.4), and the cost is paid in mixing time, not rate (Remark 5.3).
- (2) *Pacing*. Inserting gaps between malicious steps lowers the oracle rate to $\bar{\kappa} = \kappa/\bar{g}$, vanishing as the mean gap $\bar{g} \rightarrow \infty$. The limit: an infinite-mean gap is required to reach $R = 0$ (Remark 2.20), and short of that, randomising the pacing buys only $\log \bar{g}$ of concealment against κ of evidence per event, so $\kappa > \log(e\bar{g})$ guarantees catchability whatever the pacing law (Corollary 5.15).
- (3) *Hiding*. Routing malice through a latent state the monitor cannot observe lowers R below the oracle rate by the deficit ι , the information the scores leak about the latent. The

Model	Single-step d	Rate R	Reference
i.i.d. (memoryless)	$D(\mu_1 \parallel \mu_0)$	$= d$	Prop. 2.10
Two-state Markov (general)	$D_2(\frac{a_1}{a_1+b_1} \parallel \frac{a_0}{a_0+b_0})$	$\frac{b_1}{a_1+b_1} D_2(a_1 \parallel a_0) + \frac{a_1}{a_1+b_1} D_2(b_1 \parallel b_0)$	Prop. 5.1
Lazy resampler (persistence)	$D_2(w \parallel \frac{1}{2})$, fixed in ε	$\leq \varepsilon d; = C(w)\varepsilon + O(\varepsilon^2)$	Prop. 5.2
Correlation-only (two-state)	0	$\log 2 - H(s)$	Prop. 5.4
Gaussian AR(1) mean shift	$\mu^2/2\sigma^2$	$d(1-\phi)/(1+\phi)$	Prop. 5.6
Gaussian correlation-only	0	$-\frac{1}{2} \log(1-\phi^2)$	Prop. 5.8
Renewal / hidden (mean gap $\bar{g}$)	—	$\bar{\kappa} - \iota; \frac{\kappa - H(G)}{\bar{g}} \leq R \leq \frac{\kappa}{\bar{g}}$	Props. 2.19, 5.12, 5.

Table 1: Closed forms for the single-step divergence d and the rate R . Here D_2 is the binary divergence, H the binary entropy, D the divergence between marginals; (a_b, b_b) are two-state exit probabilities, w a marginal-shift parameter, ε a refresh rate, s a persistence, ϕ an autocorrelation, and for the renewal model κ is the per-event divergence, $\bar{g}$ the mean gap, $H(G)$ the gap entropy, $\bar{\kappa} = \kappa/\bar{g}$ the oracle rate, ι the information deficit. The renewal row has no single marginal divergence: a stationary renewal attacker can leave the score marginal unchanged, so d is not the relevant invariant.

limit: the deficit cannot exceed the latent’s entropy rate, $\iota \leq h(X)$, so $R \geq \bar{\kappa} - h(X)$ (Proposition 5.13); a correlation-only attack ($d = 0$) is the pure case, detectable at rate R despite leaving the marginal untouched.

A single quantity absorbs all three. Writing $\mathcal{A}$ for the class of attack laws that complete the adversary’s task within the deployment, the attacker’s problem is

$$\min_{P_1 \in \mathcal{A}} L(P_1) R(P_1),$$

the smallest total evidence any feasible strategy must emit; the monitor is safe at budget γ against the whole class precisely when this minimum exceeds $\log \gamma$ (Theorem 4.14). The independence surrogate evaluates the objective with d in place of R , and the closed forms above are the exact discrepancies it incurs — understating LR by $(1+\phi)/(1-\phi)$ for the AR(1) attacker, by $1/\varepsilon$ for the lazy resampler, and unboundedly (predicting safety where none exists) for any correlation-only or paced attacker with $d \approx 0 < R$. This objective is made precise for the pacing lever in §6.2; characterising $\mathcal{A}$ for a deployed monitor, and estimating the resulting $\min LR$ from data, is the empirical question the theory hands to measurement.

6 The safety case

The threshold theorem (Theorem 4.14) draws a line: at audit rate $\alpha = 1/\gamma$, an attack is caught precisely when the total evidence it emits, LR , exceeds $\log \gamma$. A *safety case* is an argument that a deployment sits on the safe side of that line — that any attack capable of completing its task is caught before it does. This section asks what such an argument must contain to be valid, identifies the failure mode of the argument an analyst is most likely to make, and sets out how the quantities it needs are estimated. We begin with the analyst’s natural argument: treat the scores as independent and reason from their histograms.

6.1 The independence safety case, and when it is sound

An analyst who has measured the benign and attack score marginals $\mu_\infty, \mu_{\text{att}}$, and who models the stream as independent, computes the single-step divergence $d = D(\mu_{\text{att}} \parallel \mu_\infty)$ and applies the detection theory valid for i.i.d. observations — which is the present theory with R replaced by d (Proposition 2.10). That analyst predicts an attack of length L is caught exactly when it carries enough evidence at the rate they believe in.

Definition 6.1 (The independence safety case). At budget γ , the *independence safety case* predicts that an attack of length L is *caught* when $Ld \geq \log \gamma$, and missed otherwise. A deployment is certified safe when every attack long enough to complete its task is predicted caught.

The truth, by Theorem 4.14, replaces d by R : an attack is caught when $LR > (1 + o(1)) \log \gamma$ and uncatchable — missed with probability tending to one by *every* budget-compliant detector — when $LR < (1 - o(1)) \log \gamma$. The safety case is hazardous when it predicts “caught” for an attack that is in fact uncatchable, for that is a false assurance of safety.

Definition 6.2 (Soundness). The independence safety case is *sound* if it never predicts “caught” for an uncatchable attack: whenever $Ld \geq \log \gamma$, the attack satisfies $LR \geq (1 - o(1)) \log \gamma$.

Theorem 6.3 (Soundness characterisation). *Fix a model satisfying Assumptions 2.1–2.4 with $R \in (0, \infty)$ and Conditions (U) and (L). The independence safety case is sound if and only if $d \leq R$. More precisely, considering attack lengths $L = L_\gamma$ that grow with γ :*

- (i) ($d < R$: sound, but conservative.) *Every attack predicted caught is caught. The case is strictly cautious: it predicts “missed” — and so raises a threat — for every length in the band*

$$(1 + o(1)) \frac{\log \gamma}{R} \leq L_\gamma < \frac{\log \gamma}{d},$$

on which the attack is in fact caught; the band has multiplicative width $(1 - o(1))R/d$.

- (ii) ($d = R$: exact.) *The case coincides with the truth up to the $o(1)$ in the threshold.*

- (iii) ($d > R$: unsound.) *The case predicts “caught” for genuinely uncatchable attacks: for every length in the band*

$$\frac{\log \gamma}{d} \leq L_\gamma < (1 - o(1)) \frac{\log \gamma}{R},$$

the attack is missed with probability tending to one by every budget-compliant detector, yet is certified safe. The dangerous band has multiplicative width $(1 - o(1))d/R$.

Proof. Theorem 4.14 gives the dichotomy in terms of $\liminf$ and $\limsup$ of $L_\gamma R / \log \gamma$: the attack is caught (by the CUSUM detector $\tau_{b_\gamma} \in \mathcal{C}_\gamma$) when this exceeds 1, and uncatchable by any $\tau \in \mathcal{C}_\gamma$ when it is below 1.

Soundness when $d \leq R$. If the case predicts caught, $L_\gamma d \geq \log \gamma$, i.e. $L_\gamma \geq (\log \gamma)/d$, so $\frac{L_\gamma R}{\log \gamma} \geq \frac{R}{d} \geq 1$. When $d < R$ the ratio is bounded below by the constant $R/d > 1$, so $\liminf L_\gamma R / \log \gamma \geq R/d > 1$ and the attack is caught (Theorem 4.14(i)); when $d = R$ the prediction is $L_\gamma R \geq \log \gamma$, matching the truth up to $o(1)$. In neither case is an uncatchable attack predicted caught: sound. The conservative band of (i) is the set of lengths predicted missed ($L_\gamma < (\log \gamma)/d$) yet caught ($L_\gamma > (1 + o(1))(\log \gamma)/R$); it is non-empty for large γ precisely when $d < R$, with the stated width.

Unsoundness when $d > R$. Choose any $\eta \in (0, 1)$ with $(1-\eta)/R > 1/d$ — possible since $d > R$ — and lengths L_γ with $(\log \gamma)/d \leq L_\gamma < (1-\eta)(\log \gamma)/R$, a non-empty range for all large γ . The left inequality makes the case predict caught; the right gives $\limsup L_\gamma R / \log \gamma \leq 1-\eta < 1$, so by Theorem 4.14(ii) every $\tau \in \mathcal{C}_\gamma$ misses with probability tending to one. Thus uncatchable attacks are certified safe: unsound. The dangerous band is $[(\log \gamma)/d, (1-o(1))(\log \gamma)/R]$, of multiplicative width $(1-o(1))d/R$. $\square$

Corollary 6.4 (The sign of $d - R$ across the models). *(i) Gaussian AR(1) mean shift. $d/R = (1+\phi)/(1-\phi)$, so the independence safety case is unsound iff $\phi > 0$, with a dangerous band of lengths of width $(1+\phi)/(1-\phi)$ (for instance $9\times$ at $\phi = 0.8$); it is conservative for $\phi < 0$ and exact at $\phi = 0$.*

(ii) Lazy resampler. $d/R \geq 1/\varepsilon$ (Proposition 5.2), so the case is unsound with a dangerous band of width at least the inverse refresh rate $1/\varepsilon$ — arbitrarily large as persistence increases.

(iii) Correlation-only attacks ($d = 0 < R$). The case predicts “caught” only when $0 \geq \log \gamma$, i.e. never: it declares the attack undetectable at every budget. It is therefore sound (it certifies no catch, hence no false safety) but maximally blind — the attack is in fact caught for all $L \geq (1+o(1))(\log \gamma)/R$. For the Gaussian correlation-only model this catch direction rests only on Condition (L), which holds there even though (U) does not (Remark 5.10); the unsound direction is moot, as $d = 0$ never predicts catch.

Proof. Substitute the closed forms of §§5.1–5.3 into Theorem 6.3; for (iii), the catch claim is Theorem 4.12 (achievability), which needs only (L). $\square$

Remark 6.5 (The asymmetry, and the one bit a safety case must establish). The two error directions are not equally costly. When $d < R$ the independence case errs conservatively: it raises threats the monitor would in fact have caught, wasting effort but never granting false safety. When $d > R$ it errs dangerously, certifying as safe attacks no budget-compliant detector can catch. The entire hazard is thus carried by the sign of $d - R$, and that sign is exactly what a histogram cannot see: d is read from the marginals, while R requires the temporal structure. A safety case resting on d alone is therefore not merely imprecise but potentially unsound, and the gap is no small correction — by Corollary 6.4 it is the unbounded factor $(1+\phi)/(1-\phi)$ or $1/\varepsilon$ in the length an attacker may safely run. Establishing $d \leq R$, or estimating R and the sign of $d - R$ directly, is the minimal content a sound independence-based safety case must add; §6.4 takes up how, and warns that the estimate must itself be made in the safe direction.

6.2 The adversarial objective

The soundness theorem compares d and R for a single attack. A safety case must quantify over the attacker’s options: not one strategy but every strategy that completes the adversary’s task. We make this precise for the pacing lever, where the closed forms of §5 combine into a single accounting.

Fix the marked renewal model of §2.7: malicious events occur at the epochs of a stationary renewal process with gap law G of mean $\bar{g}$, and each malicious event emits a score of divergence $\kappa = D(q_1 \| q_0)$ from the benign law, the benign steps emitting q_0 . The adversary’s task requires a fixed number M of malicious events; the pacing law G is the attacker’s to choose, but κ — the irreducible footprint of a single malicious action — is fixed by the action, not the schedule.

Completing M events at mean gap $\bar{g}$ takes a window of length $L = (1 + o(1))M\bar{g}$ (renewal law of large numbers). The attacker's evasion problem is to spend as little total evidence as possible:

$$\min_G LR \quad \text{subject to completing the } M\text{-event task,}$$

since by the threshold theorem the deployment is safe against a pacing exactly when its LR exceeds $\log \gamma$.

Definition 6.6 (Task-completing pacing class). For $M \geq 1$ and $\kappa > 0$, let $\mathcal{A}_{M,\kappa}$ be the family of stationary marked-renewal attack laws with per-event divergence κ and gap law G ranging over all distributions on $\{1, 2, \dots\}$ of finite mean, each evaluated over the task window $L(G) = (1 + o(1))M\bar{g}(G)$.

Theorem 6.7 (Evidence per task). *For every G , with $\bar{g} = \bar{g}(G)$, $L = L(G)$, deficit ι and observable rate $R = \bar{\kappa} - \iota$ (Proposition 5.12, $\bar{\kappa} = \kappa/\bar{g}$):*

- (i) *the oracle total evidence over the task is $L\bar{\kappa} = (1 + o(1))M\kappa$, independent of the pacing G ;*
- (ii) *the observable total evidence satisfies the identity $LR = (1 + o(1))(M\kappa - L\iota)$;*
- (iii) *it is bounded below, independently of G except through the mean gap, by*

$$LR \geq (1 + o(1))M(\kappa - H(G)) \geq (1 + o(1))M(\kappa - \log(e\bar{g})).$$

Proof. (i) The oracle reads the marks; over L steps the task contributes M malicious events each of expected log-likelihood κ and benign steps contribute 0, so the oracle evidence is $M\kappa$; equivalently $L\bar{\kappa} = M\bar{g} \cdot (\kappa/\bar{g}) = M\kappa$. (ii) $LR = L\bar{\kappa} - L\iota = M\kappa - L\iota$ by Proposition 5.12. (iii) Recall $L = (1 + o(1))M\bar{g}$ by the renewal law of large numbers, so the $(1 + o(1))$ factors below all refer to this single ratio. By Proposition 5.13, $\iota \leq h(X) = H(G)/\bar{g}$, hence $L\iota \leq LH(G)/\bar{g} = (1 + o(1))MH(G)$. Substituting into the identity (ii),

$$LR = (1 + o(1))(M\kappa - L\iota) \geq (1 + o(1))(M\kappa - MH(G)) = (1 + o(1))M(\kappa - H(G)),$$

and since $H(G) \leq \log(e\bar{g})$ by Lemma 5.14, the right-hand side is at least $(1 + o(1))M(\kappa - \log(e\bar{g}))$. $\square$

The reading is sharp. *Pacing is evidence-neutral to the oracle:* spreading the M events over a longer window changes neither the oracle's total evidence $M\kappa$ nor, in the noiseless-mark limit, the observable's. The only way pacing helps the attacker is through the deficit $L\iota$ — the information the schedule's randomness withholds — and that is capped at $MH(G)$, the timing entropy per event. Conspicuousness κ and hiding are the levers that move total evidence; pacing alone is not.

Corollary 6.8 (Class safety certificate). *Suppose the achievability condition (L) holds for the attack model (Remark 4.6; for hidden-state pacings this is the open hypothesis of Remark 5.10). Then the deployment is safe — every task-completing attack is caught with probability tending to one — against the entire sub-class of $\mathcal{A}_{M,\kappa}$ with mean gap $\bar{g} \leq \bar{g}_\star$, where*

$$\bar{g}_\star = \exp\left(\kappa - 1 - \frac{\log \gamma}{M}\right),$$

equivalently whenever $M(\kappa - \log(e\bar{g})) > \log \gamma$. The certified-safe range of mean gaps grows exponentially in the per-event footprint κ and shrinks with the per-event budget demand $(\log \gamma)/M$. Two senses of “safe against the class” should be kept apart. The evidence bound (Theorem 6.7) is unconditional information theory and holds across the whole sub-class at once. Converting it to a caught-with-high-probability guarantee requires a detector: against any single known pacing this is the CUSUM of §4, and a single detector that is uniformly near-optimal across the sub-class is supplied by the max-CUSUM of §6.3 when the pacings are enumerated or smoothly parametrised. For a fully nonparametric pacing class the uniform detector is the open problem of Remark 6.14; the bound above should then be read as an oracle (known-pacing) certificate, achievable pacing by pacing.

Proof. By Theorem 6.7(iii), $\bar{g} \leq \bar{g}_\star$ gives $LR \geq (1 + o(1))M(\kappa - \log(e\bar{g})) > \log \gamma$ for large γ , so the attack is caught by Theorem 4.12; the displayed $\bar{g}_\star$ solves $M(\kappa - \log(e\bar{g})) = \log \gamma$, since $\log(e\bar{g}) = 1 + \log \bar{g}$. $\square$

Corollary 6.9 (Minimax delay against the class). *On any sub-class of $\mathcal{A}_{M,\kappa}$ for which Condition (U) holds, every detector $\tau \in \mathcal{C}_\gamma$ has worst-case delay*

$$\sup_{P_1} J(\tau; P_1) \geq (1 - o(1)) \frac{\log \gamma}{\inf_{P_1} R(P_1)},$$

the supremum and infimum over the sub-class; the attacker’s best evasion is to minimise R , i.e. to pace and hide up to the limits of Theorem 6.7.

Proof. The per-instance lower bound (Theorem 3.16) gives $J(\tau; P_1) \geq (1 - o(1))(\log \gamma)/R(P_1)$ for each admissible P_1 ; take the supremum over P_1 on the left and the infimum of R on the right. $\square$

Remark 6.10 (The pacing cliff, and what the certificate does not cover). Corollary 6.8 certifies safety against pacings no slower than $\bar{g}_\star$, and no further: as $\bar{g} \rightarrow \infty$ the floor $M(\kappa - \log(e\bar{g}))$ falls through zero and the certificate goes vacuous, consistent with the cliff of Remark 2.20: an infinite-mean gap drives the rate $R \rightarrow 0$, the speed limit degenerates, and the rate-based certificate correctly declines to cover that regime. What this expresses is a limit of the *rate* framework, not a proof that no detector can act: an adversary willing to spend unboundedly many benign steps per malicious one drives the long-run evidence rate to zero, so any guarantee at that boundary must come from a finer, non-rate analysis of the individual events. Within the covered regime, the existence of a worst-case pacing (a minimiser of LR over $\bar{g} \leq \bar{g}_\star$) follows from continuity of $\bar{g} \mapsto M(\kappa - H(G))$ -type bounds on the compact range of admissible mean gaps; we need only the infimum, which the displayed bound controls. The whole accounting is unconditional information theory — Theorem 6.7 uses no deviation condition — and only the conversion to a catch or delay guarantee (Corollaries 6.8, 6.9) inherits the hidden-state status of (U) and (L). For an attacker whose pacing is itself finite-state, both hold (Proposition 3.11) and the certificate is unconditional.

6.3 Detection without knowing the attack

Every detector to this point computes its increment $Z_t = \log f_1(S_t \mid S_{1:t-1})/f_0(S_t \mid S_{1:t-1})$ from *both* predictive models, the benign f_0 and the attack f_1 . A deployed monitor knows f_0 — it can be measured from honest traffic — but not which attack is underway: the adversary

draws its law from a class $\mathcal{A} = \{P_\theta : \theta \in \Theta\}$, and θ is hidden. This is the gap noted after Theorem 4.7 (Remark 3.18). The minimax *lower* bound is already in hand — by Corollary 6.9, no detector beats worst-case delay $(1 - o(1))(\log \gamma)/R_-$, where $R_- = \inf_{\theta \in \Theta} R_\theta$. The question is achievability: can a single detector, ignorant of θ , attain the per-attack speed limit $(\log \gamma)/R_\theta$ for every θ at once? For finite and for smoothly parametric classes, yes, at a price that is provably negligible.

Definition 6.11 (Max-CUSUM). For a finite collection $\{f_\theta : \theta \in \Theta_0\}$ of candidate attack predictives, run one CUSUM per candidate on its own increment $Z_t^{(\theta)} = \log f_\theta(S_t | S_{1:t-1})/f_0(S_t | S_{1:t-1})$, with running statistic $W_t^{(\theta)}$ and first-passage time $\tau^{(\theta)} = \inf\{t : W_t^{(\theta)} \geq b\}$. The *max-CUSUM* alarms at $\tau^{\max} = \min_{\theta \in \Theta_0} \tau^{(\theta)}$ — the first time any candidate's statistic crosses b . (Equivalently it thresholds $\max_\theta W_t^{(\theta)}$, the generalised-likelihood statistic over Θ_0 .)

Theorem 6.12 (Robust detection, finite class). *Let $\mathcal{A} = \{P_1, \dots, P_K\}$ be finite, each P_k satisfying Condition (L) with rate $R_k \in (0, \infty)$. Run the max-CUSUM with threshold $b = b_{\gamma, K} := \log(8K\gamma)$. Then*

- (i) *the false-alarm budget holds: $\mathbb{E}_\infty[\tau_b^{\max}] \geq \gamma$; and*
- (ii) *for every k , the worst-case delay under P_k satisfies $J(\tau_b^{\max}; P_k) \leq (1 + o(1))(\log \gamma)/R_k$ as $\gamma \rightarrow \infty$.*

The max-CUSUM is therefore simultaneously first-order optimal against every member of the class: it meets the per-attack speed limit for whichever attack is active, and the minimax bound $(\log \gamma)/R_-$ of Corollary 6.9, paying only the additive $\log K$ in the threshold — the price of not knowing θ , negligible against $\log \gamma$.

Proof. (i) Under $\mathbb{P}_\infty$ each candidate statistic is a benign CUSUM, so Proposition 4.4's Ville estimate gives $\mathbb{P}_\infty(\tau_b^{(k)} \leq n) \leq n e^{-b}$ for each k . A union bound over the K candidates gives $\mathbb{P}_\infty(\tau_b^{\max} \leq n) \leq K n e^{-b}$. With $N = \lceil e^b/(2K) \rceil$ and $b \geq \log(4K)$, $K N e^{-b} \leq \frac{1}{2} + K e^{-b} \leq \frac{3}{4}$, so $\mathbb{E}_\infty[\tau_b^{\max}] \geq N \mathbb{P}_\infty(\tau_b^{\max} > N) \geq N/4 \geq e^b/(8K)$. Setting $b = \log(8K\gamma)$ yields $\mathbb{E}_\infty[\tau_b^{\max}] \geq \gamma$. (ii) Under P_k the k -th candidate is the correctly specified CUSUM for the active attack, so Theorem 4.7 applies to it with threshold $b = \log(8K\gamma) = (1 + o(1)) \log \gamma$ (as K is fixed), giving $J(\tau_b^{(k)}; P_k) \leq (1 + o(1))b/R_k = (1 + o(1))(\log \gamma)/R_k$. Since $\tau_b^{\max} \leq \tau_b^{(k)}$ pathwise, the max-CUSUM alarms no later, so $J(\tau_b^{\max}; P_k) \leq J(\tau_b^{(k)}; P_k)$. $\square$

The same device covers a parametric family by discretisation, at the cost of a covering argument; the increment of a slightly-wrong candidate still carries almost the full rate, because a mismatch $\theta' \approx \theta$ costs only the second-order D -rate between near neighbours.

Theorem 6.13 (Robust detection, parametric class). *Let $\Theta \subset \mathbb{R}^p$ be compact, with $\theta \mapsto R_\theta$ continuous, $R_- = \inf_\Theta R_\theta > 0$, Condition (L) uniform over Θ , and the local-mismatch bound*

$$\sup_{\theta \in \Theta} \sup_{\theta' : |\theta' - \theta| \leq \delta} (R_\theta - \mathbb{E}_{P_{\theta'}}[Z_0^{(\theta)}]) \leq C \delta^2$$

for a constant C (the mismatched increment loses only the divergence rate between θ and θ'). Take a δ_γ -net $\Theta_{\delta_\gamma} \subset \Theta$, of cardinality $|\Theta_{\delta_\gamma}| \leq (C'/\delta_\gamma)^p$, with $\delta_\gamma \rightarrow 0$ slowly enough that $p \log(1/\delta_\gamma) =$

$o(\log \gamma)$ (e.g. $\delta_\gamma = 1/\log \gamma$), and run the max-CUSUM over Θ_{δ_γ} with $b = \log(8|\Theta_{\delta_\gamma}|\gamma)$. Then $\mathbb{E}_\infty[\tau_b^{\max}] \geq \gamma$, and

$$\sup_{\theta \in \Theta} \left(J(\tau_b^{\max}; P_\theta) - (1 + o(1)) \frac{\log \gamma}{R_\theta} \right) \leq 0,$$

so the detector is first-order minimax optimal, uniformly over the family, without knowing θ .

Proof sketch. False alarm is Theorem 6.12(i) with $K = |\Theta_{\delta_\gamma}|$; the threshold is $b = \log \gamma + p \log(1/\delta_\gamma) + O(1) = (1 + o(1)) \log \gamma$ by the choice of δ_γ . For the delay, fix the active θ and its net neighbour θ' , $|\theta - \theta'| \leq \delta_\gamma$. The θ' -candidate's increment has P_θ -mean $\mathbb{E}_{P_\theta}[Z_0^{(\theta')}] \geq R_\theta - C\delta_\gamma^2 = (1 - o(1))R_\theta$; uniform Condition (L) supplies the lower-deviation control for this mismatched increment, so Theorem 4.7 gives $J(\tau_b^{(\theta')}; P_\theta) \leq (1 + o(1))b/((1 - o(1))R_\theta) = (1 + o(1))(\log \gamma)/R_\theta$, and $\tau^{\max} \leq \tau_b^{(\theta')}$. This is the window-limited generalised-likelihood construction of the sequential-detection literature (Lai 1998; Xie & Moustakides 2022), specialised to a benign reference and a uniform rate floor. $\square$

Remark 6.14 (What this closes, and what stays open). For an attack class that is finite or a smooth finite-dimensional family, a single deployable monitor — one that knows only the benign law and the menu of possible attacks — attains the speed limit of §4 for whichever attack is run, closing the gap left after Theorem 4.7. The cost of ignorance is an additive $\log K$, or $p \log(1/\delta_\gamma)$, in the detection threshold: real, but lower-order, so the first-order limit is unchanged. The mixture (Bayesian) CUSUM, thresholding $\log \int f_\theta dw(\theta)/f_0$ against a prior w on Θ , is an alternative with the same first-order behaviour and is sometimes more convenient. What does *not* follow is the general nonparametric case: when $\mathcal{A}$ has infinite metric entropy — say, all stationary laws with rate at least R_- — no fixed net is uniformly fine and the covering penalty need not be $o(\log \gamma)$, so a single detector uniformly first-order optimal over the whole class need not exist. The minimax lower bound of Corollary 6.9 still holds there, bounding every detector; matching it over rich classes is the open problem. The practical reading is mild: a monitoring deployment that can enumerate, or smoothly parametrise, the attack strategies it means to defend against pays essentially nothing for not knowing which is live, and the empirical programme of §6.5 can fit that parametrisation from red-team data.

6.4 Estimating the rate, and a conservative certificate

To apply any of the foregoing one must estimate the rate R from a sample of benign and attack score traces. The single-step divergence d is read directly from the marginals and poses no difficulty; R is the delicate quantity, because it lives in the temporal structure, and the natural estimator — fit a model to the scores and compute its rate — carries a bias whose direction is exactly the one the safety case cares about.

Proposition 6.15 (The order- k estimator, and its endpoints). *Let $R^{[k]}$ denote the divergence rate of the order- k Markov approximation of the score process, $R^{[k]} = \mathbb{E}_{\mathbb{P}_{\text{att}}} [D(P_1(S_0 | S_{-k:-1}) \| P_0(S_0 | S_{-k:-1}))]$, and assume the per-step log-likelihood ratio is bounded, $|\log P_1(S_0 | S_{-\infty:-1})/P_0(S_0 | S_{-\infty:-1})| \leq B$ almost surely (automatic on a finite score alphabet with positive kernels; in general an explicit uniform-integrability hypothesis). Then $R^{[0]} = d$ and $R^{[k]} \rightarrow R$ as $k \rightarrow \infty$. Consequently the bias of a fixed-order fit, relative to the truth, begins at $R^{[0]} - R = d - R$ and closes only in the limit.*

Proof. At $k = 0$ the conditioning is empty and the inner divergence is $D(\mu_{\text{att}}\|\mu_{\infty}) = d$. As $k \rightarrow \infty$, $P_b(S_0 | S_{-k:-1}) \rightarrow P_b(S_0 | S_{-\infty:-1})$ (martingale convergence of the predictive densities), and the inner divergence converges pointwise to the increment whose stationary mean is R (Assumption 2.4); the bound B makes the order- k integrands uniformly bounded, so dominated convergence gives convergence of the expectation. $\square$

Remark 6.16 (The bias is signed like $d - R$, and is dangerous exactly when the safety case is). The estimator's hazard therefore coincides with the soundness hazard of §6.1. Two regimes, by the sign of $d - R$.

Overestimation ($d > R$). When benign and attack share temporal redundancy and the attack shifts the marginal — positive autocorrelation common to both laws, as in the AR(1) mean shift or the lazy resampler — consecutive scores are partly redundant, so the rate is below the marginal divergence ($R < d$, indeed $R \leq \varepsilon d$ for the resampler, Proposition 5.2). A short-memory fit misses the redundancy and returns $R^{[0]} = d > R$: it *overestimates* the rate, underestimates the delay, and is biased in the same dangerous direction as the independence surrogate it was meant to improve on.

Underestimation ($d < R$). When the attack instead introduces structure absent under the benign law — clustered renewal events, or a correlation-only shift with $d = 0$ — the marginal understates the rate, and the order- k estimates rise toward R from below. For a memory-one attack the rise is a single step, the order-one fit being exact: a persistent two-state attack against an i.i.d. benign stream gives $R^{[0]} = d \approx 0.08$ rising to $R \approx 0.20$ at order one, and a correlation-only attack gives $R^{[0]} = 0$ rising to the full $R = \log 2 - H(s)$, likewise at order one. A gradual climb across several orders is instead the signature of longer memory — a renewal or hidden-Markov attack, where each added order of fit recovers a further increment of the rate. Underestimation is the safe direction: it raises false threats, never false assurances.

The moral matches Remark 6.5: the dangerous regime is $d > R$, and a sound certificate cannot rest on a point estimate from a fixed low order, since exactly there the estimate is anti-conservative.

Corollary 6.17 (A conservative certificate from the deficit bound). *Because overestimating R is the unsound direction, a safety certificate should use a guaranteed lower bound on R . For the renewal model the deficit bound supplies one that needs no temporal fit: by Corollary 5.15, $R \geq (\kappa - \log(e\bar{g}))/\bar{g}$, a function only of the per-event divergence κ and the mean gap $\bar{g}$. If $\hat{\kappa}_{\text{lo}}$ is a lower confidence bound on κ (estimated from the score histogram on flagged events) and $\hat{g}_{\text{hi}}$ an upper confidence bound on the mean gap (estimated from the event rate), then, with the same confidence,*

$$R \geq \frac{\hat{\kappa}_{\text{lo}} - \log(e\hat{g}_{\text{hi}})}{\hat{g}_{\text{hi}}} := \hat{R}_{\text{lo}},$$

the right side being increasing in κ and decreasing in $\bar{g}$ throughout the catchable regime $\bar{g} < e^{\kappa-1}$, so that both confidence directions are conservative. The certificate of Corollary 6.8 then holds with $\hat{R}_{\text{lo}}$ in place of R . This trades the hard estimation of a rate for the easy estimation of two scalars, in the safe direction by construction.

Proof. Monotonicity of $g \mapsto (\kappa - 1 - \log g)/g$ was shown in Remark 6.10; substitute the confidence bounds and apply Corollary 5.15. $\square$

6.5 The empirical programme

The theory reduces a safety case to a small number of measurable quantities and hands the measurement to a deployment study. In order of logical priority:

- (1) *The feasibility precondition.* Does the monitoring pipeline expose a *per-action* score, or only an aggregate verdict? The entire theory is about the temporal stream (S_t) ; if a deployment collapses scores to a single end-of-episode number, R is undefined and the speed limit does not apply. Establishing that per-action scores are available, and logging them, is the first task; standard agentic-control stacks (ControlArena over Inspect) already expose and preserve a per-action monitor score, so the precondition is satisfiable, and the open question is whether a given production pipeline retains it.
- (2) *The sign of $d - R$.* This is the one bit that decides soundness (§6.1) and the direction of estimation bias (§6.4). It is measured by comparing the marginal divergence d against a high-order or model-free estimate of R on benign and attack traces; a deployment with $d > R$ has an unsound independence safety case and a dangerous estimation bias, and must be certified through $\hat{R}_{lo}$.
- (3) *The catch-rate frontier.* Theorem 4.14 predicts a threshold in $LR/\log \gamma$ at 1. Measuring the empirical catch rate against total evidence $L\hat{R}$, across attack lengths and audit budgets, tests the threshold and calibrates the $o(1)$ in practice.
- (4) *Order-sensitivity of $\hat{R}$.* The trajectory of $R^{[k]}$ as the fit order grows (Proposition 6.15) diagnoses which regime a deployment is in — decreasing toward R flags the dangerous $d > R$ case, increasing flags the safe one — and indicates when a low-order fit can be trusted.

These are the questions the present theory makes precise; answering them on real trusted-monitor traces is the empirical complement to this paper.

7 From the rate to a detector: what one actually accumulates

A reader who builds monitors will object that the escalation decision is never based on a single action’s histogram: one accumulates the evidence of a prefix and escalates when it is large. This is correct, and its answer is perhaps the most practical thing the theory has to say. “Accumulate the prefix evidence” names two different statistics, and the gap between them is exactly the gap between d and R .

Every detector below shares one skeleton — Page’s cumulative-sum recursion, with a threshold calibrated on benign traffic alone so that the benign mean time between alarms is at least γ — and they differ *only* in the per-step increment they accumulate.

```
# One CUSUM skeleton; only ‘increment’ differs (detectors A-D).
# ‘past’ abbreviates the history s[1:t-1]; f*_hist are score histograms.
calibrate(increment, benign, gamma):
    return smallest b: benign_ARL(increment, benign, b) >= gamma
run(increment, scores, b):
    W = 0
    for t = 1, 2, ...:
```

```

W = max(0, W + increment(t, scores))    # Page recursion
if W >= b: return ("escalate", t)

# A MARGINAL: score vs HISTOGRAMS; depends on s_t alone -> rate d
incrementA(t,s): return log f1_hist(s[t]) - log f0_hist(s[t])

# B CONDITIONAL, attack known: densities given past -> rate R
incrementB(t,s): return log f1(s[t]|past) - log f0(s[t]|past)

# C BENIGN-ANOMALY, no attack model: benign surprise -> rate R
# H0 = E_benign[-log f0(s[t]|past)] (so benign drift is 0)
incrementC(t,s): return ( -log f0(s[t]|past) ) - H0

# D CLASS/RED-TEAM, attack family {f_theta}: GLR -> rate R_theta
incrementD(t,s): return max over theta of
                    ( log f_theta(s[t]|past) - log f0(s[t]|past) )

```

Marginal versus conditional (A versus B). Detector A scores each action against the one-step *histograms*, as if the steps were independent draws; its statistic grows at rate d under attack. Detector B scores each action against the *predictive* densities given the recent history; it grows at R . The increments are identical but for the conditioning, and the two rates coincide exactly when the scores are independent. The consequences are the two halves of the soundness theorem (§6.1), now mechanical. When $d > R$ — benign and attack share temporal persistence, e.g. positively autocorrelated scores — a run of high benign scores feeds A three large increments where B, having predicted each high score from its predecessor, adds almost nothing; A’s benign statistic is noisier, so calibration is forced to a higher threshold and its realised detection slows, while the analyst who sized the budget from d has over-counted redundant evidence and written an optimistic safety case. When $d < R$ — in the extreme, a correlation-only attack with benign-identical marginals, $d = 0$ — A’s increment is *identically zero* and the detector cannot move, declaring the attack invisible; B’s increment is nonzero because the *dependence* changed, and it catches what A cannot see.

Attack known versus unknown (f_1). Detector B as written needs the attack’s conditional density $f_1(\cdot | \text{past})$, which a deployment does not possess: the adversary’s strategy is unknown, and an LLM monitor returns a scalar score, not a density. But f_0 and f_1 are not equally unknown, and the asymmetry is the whole practical lever. Benign traffic is abundant, so the benign law f_0 — including its temporal structure, the conditional $f_0(\cdot | \text{past})$ or its autocorrelation — is estimable; f_1 is not. Two deployable detectors exploit this, neither referencing the true f_1 . Detector C, the *benign-anomaly* CUSUM, accumulates the conditional surprise of each score under the benign predictive model, centred by the benign mean surprise H_0 so that a benign stream has zero drift (for an order-1 Gaussian benign model the increment is the centred squared residual $(r_t^2/v_0 - 1)/2$, $r_t = s_t - \mu_0(s_{t-1})$). It is an omnibus, one-sided test for departure from the benign *dynamics*: it realises R against such departures — including the correlation-only attack A is blind to — with no attack model whatsoever. Its price is that, not being matched to any attack, it is slower than a tuned detector, and a “too clean” attack pushes the statistic down (accumulate $|z_t|$ or a two-sided statistic if that matters). Detector D, when red-teaming yields

a parametrised family $\{f_\theta\}$, accumulates the generalized-likelihood increment $\sup_\theta \log f_\theta/f_0$ — the max-CUSUM of §6.3 — attaining the per-attack speed limit for whichever θ is live, at an additive $\log |\Theta|$ in the threshold. C and D are the two honest routes from B’s oracle benchmark to something one can run; in both, f_1 enters only when one wants to *forecast* the catch rate for a specific attack, through its rate R_θ , never to run or calibrate the detector.

Where this leaves the field. The distinction is not idle: it sorts existing monitors. The dominant *deployed* aggregators are order-independent — the maximum or mean of the per-step suspicion scores over a trajectory (Bhatt et al. 2025; Terekhov et al. 2025) — and these are Detector A in disguise: symmetric in the scores, blind to their dependence, hence governed by d and subject to its optimism when $d > R$. A recent and fast-growing line instead builds *stateful, sequential* monitors — carrying a running summary across ordered chunks (the Monitor Red-Teaming sequential monitor, Kale et al. 2025), maintaining evidence state across adaptively chosen windows (TRACE 2026), treating risk as an evolving trajectory state (TRACES 2026) — which are conditional in spirit, Detector B/C territory; their reported gains on tasks requiring long-range evidence linking are direct empirical confirmation that the signal lives in the temporal dependence the histograms discard. What neither line supplies, and what this paper adds, is the analytic layer above both: the divergence *rate* R as the governing scalar, the d -versus- R gap that quantifies how much an order-independent rule forfeits, and the characterisation (§6.1) of when order-independent reasoning is not merely suboptimal but *unsound* as a safety case. The theory does not prescribe a monitor; it says which quantity to measure to know whether a chosen aggregation can be trusted — and the first diagnostic of the experimental programme (§6.5), the autocorrelation of the benign score stream, is precisely the cheap test of whether the d - R gap is open in a given deployment.

8 Limitations and outlook

The theory is clean within a stated scope, and it is worth collecting, in one place, the boundaries it does not cross.

- (i) *Adaptive, closed-loop timing.* The attacker is assumed to commit to a stationary strategy and may choose its launch time, but not to throttle its malice in reaction to the monitor’s running output (§2.8). A fully adaptive attacker produces a non-stationary stream for which R need not exist; the stationary-class objective of §6.2 covers “the most evasive fixed pacing,” and a minimax-regret guarantee against arbitrary timing is a separate problem.
- (ii) *Deviation conditions for hidden state.* Conditions (U) and (L) are discharged with explicit constants for positive finite-state chains and the Gaussian AR(1) mean shift, but remain open for the renewal and hidden-Markov models, where they amount to a filter-stability property (Remark 3.13). The rate-level results of §5.4 hold regardless; only their conversion to delay guarantees awaits this.
- (iii) *Unbounded state with state-dependent increments.* For such models the worst-case-over-histories formalism can fail (Proposition 5.9); the rate and the achievability side survive, but a matching lower bound needs the typical-set reformulation of the delay (Remark 5.10), which we have not carried out.

- (iv) *Detection against an unknown attack, in general.* A monitor that does not know which attack is live runs a generalised-likelihood (max-CUSUM) or mixture procedure over the class $\mathcal{A}$. For finite and for smooth finite-dimensional $\mathcal{A}$ this attains the per-attack speed limit at a threshold cost of order $\log |\mathcal{A}|$ or $p \log(1/\delta)$, lower-order in $\log \gamma$ (§6.3). The general nonparametric case — $\mathcal{A}$ of infinite metric entropy — is open: the minimax lower bound still binds every detector, but no single detector need be uniformly first-order optimal, and characterising when one exists is left for future work.

The thesis the paper defends is narrow and, we think, robust: the difficulty of cheap monitoring is governed not by how different a malicious action looks in isolation, but by the *rate* at which that difference accumulates against the budget for human attention. The marginal divergence d that a histogram reports and the divergence rate R that governs detection are distinct quantities, and an adversary controls their ratio through correlation, pacing, and hiding. When the two coincide — memoryless scores — the familiar independence accounting is exact. When they part, and especially when $d > R$, that accounting is not merely imprecise but unsound, certifying as safe attacks that no budget-compliant detector can catch, by a factor that the closed forms make explicit and unbounded. A safety case for a deployed monitor must therefore measure the rate, not the histogram; the speed limit is the rate, and only the rate sees it.

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A The information rate: how Assumption 2.4 follows from ergodicity

Assumption 2.4 postulates that

$$R := \lim_{n \rightarrow \infty} \frac{1}{n} \Lambda_1^n, \quad \Lambda_1^n = \sum_{t=1}^n Z_t = \log \frac{d\mathbb{P}_{\text{att}}^{(n)}}{d\mathbb{P}_{\infty}^{(n)}}(S_{1:n}),$$

exists as a finite *constant*, $\mathbb{P}_{\text{att}}$ -almost surely and in $L^1(\mathbb{P}_{\text{att}})$, the second equality being the likelihood-ratio factorisation of Lemma 2.7. The point of this appendix is that, granted Assumptions 2.1–2.3 and one integrability condition, this is a *theorem*, not a further hypothesis. We define the notions involved, state the classical results we use with proof sketches, assemble the conclusion, and end with the exact circumstances under which it holds and the ways it breaks. Nothing in the body depends on the details here; this is for the reader who wants to see the engine with its cover off.

A.1 Measure-preserving systems, the shift, and ergodicity

The probability space, recalled. We restate the setting of §2.2 so that this appendix stands on its own. Scores take values in a standard Borel space $\mathcal{X}$ (a finite alphabet, the real line $\mathbb{R}$, or $\mathbb{R}^k$). The sample space is the sequence space $\Omega = \mathcal{X}^{\mathbb{N}}$: a single point $\omega = (\omega_1, \omega_2, \dots) \in \Omega$ is one entire infinite score trajectory. The *coordinate maps* $S_t(\omega) := \omega_t$ are the individual scores read

off that trajectory, and $\mathcal{F} = \sigma(\bigcup_t \mathcal{F}_t)$, with $\mathcal{F}_t = \sigma(S_1, \dots, S_t)$, is the σ -algebra of events they generate. A *probability law* (synonymously, a *probability measure*) $\mathbb{P}$ on the measurable space $(\Omega, \mathcal{F})$ assigns a probability to each event $A \in \mathcal{F}$; it is exactly what fixes the joint distribution of the whole score sequence $(S_t)_{t \geq 1}$. In the application $\mathbb{P}$ is one of the two laws of Definition 2.2 — the benign $\mathbb{P}_\infty$ or the attack $\mathbb{P}_{\text{att}}$ — and every statement in this appendix is proved for a generic such $\mathbb{P}$ and then applied to each in turn; we write the plain symbol $\mathbb{P}$ exactly when only stationarity and ergodicity, and not the specific benign/attack dynamics, are in play.

The shift. The one dynamical object on this space is the *shift* $\theta : \Omega \rightarrow \Omega$,

$$(\theta\omega)_t := \omega_{t+1} \quad (t \geq 1),$$

which erases the first coordinate and slides the rest down by one: $\theta(\omega_1, \omega_2, \omega_3, \dots) = (\omega_2, \omega_3, \dots)$. Read through the coordinate maps it advances time, $S_t \circ \theta^k = S_{t+k}$, so iterating θ is precisely “letting the clock run”: evaluating a function of the scores after applying θ^k returns the value of that same function k steps later. It is the shift, not the measure, that moves; the measure is what may or may not be left unchanged by that motion.

Measure-preserving — who preserves what. The shift θ *preserves* the law $\mathbb{P}$ — equivalently, $\mathbb{P}$ is θ -invariant, or the score process is *stationary* — when

$$\mathbb{P}(\theta^{-1}A) = \mathbb{P}(A) \quad \text{for every } A \in \mathcal{F}.$$

The map θ is the thing that is measure-preserving (it is the grammatical subject); the measure $\mathbb{P}$ is what gets preserved (the object); and the content of the equation is that advancing time by one step changes no probability, so the shifted sequence $(S_2, S_3, \dots)$ has the same law as $(S_1, S_2, \dots)$, and hence every finite block has a law that does not depend on where in time it sits. When this holds we call the quadruple $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ a *measure-preserving system*. Assumption 2.3 asserts exactly that both $\mathbb{P}_\infty$ and $\mathbb{P}_{\text{att}}$ form such systems; wherever an abstract theorem below opens with “let $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ be measure-preserving,” that is the hypothesis it is invoking, with $\mathbb{P}$ equal to one of those two laws.

Invariant sets, and ergodicity. A set $A \in \mathcal{F}$ is *invariant* if $\theta^{-1}A = A$ up to a $\mathbb{P}$ -null set — membership in A is unaffected by a one-step time-shift — and the invariant sets form a sub- σ -algebra $\mathcal{I} \subseteq \mathcal{F}$, the *invariant σ -algebra*. A function $g : \Omega \rightarrow \mathbb{R}$ is *invariant* if $g \circ \theta = g$ almost surely; the invariant functions are exactly the $\mathcal{I}$ -measurable ones. The system is *ergodic* if $\mathcal{I}$ is trivial: every invariant event has probability 0 or 1, equivalently every invariant $g \in L^1$ is almost surely a constant. Ergodicity is the formal content of “the process has no persistent hidden mode that the dynamics never average away.” It is exactly what forces a time-average — which, being unchanged by the shift, is always an invariant random variable — to collapse to a single deterministic number instead of depending on which mode happened to be drawn.

Examples and a non-example. A product (i.i.d.) law is ergodic: an invariant set lies in the tail σ -algebra, which is trivial by the Kolmogorov 0–1 law. An irreducible aperiodic finite-state Markov chain, started from its stationary law, is ergodic (indeed mixing). The instructive failure is a stationary *mixture*: flip one fair coin before time 1 and thereafter run attack dynamics A (with increment-mean R_A) if heads, dynamics B (mean $R_B \neq R_A$) if tails. The result is stationary

but not ergodic — the event “dynamics A ” is invariant with probability $\frac{1}{2}$, so $\mathcal{I}$ contains the coin — and, as we will see, its time-average rate is a genuine two-valued random variable rather than a constant. This is the case ergodicity rules out.

A.2 Birkhoff’s pointwise ergodic theorem

Birkhoff’s theorem turns a *time*-average along the shift into a *space*-average against the measure. We state it for an abstract measure-preserving system and specialise it immediately afterwards; in the application the system is $(\Omega, \mathcal{F}, \mathbb{P}_\infty, \theta)$ or $(\Omega, \mathcal{F}, \mathbb{P}_{\text{att}}, \theta)$ from §A.1, and the integrable function g is the per-step log-likelihood functional whose time-average is the rate R . For a function g the time-average $A_n g := \frac{1}{n} \sum_{k=0}^{n-1} g \circ \theta^k$ is the mean of g along the first n points of the orbit $\omega, \theta\omega, \dots, \theta^{n-1}\omega$ of ω under the shift.

Theorem A.1 (Birkhoff, 1931). *Let $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ be a measure-preserving system (§A.1) and $g \in L^1(\mathbb{P})$. Then the time-averages $A_n g := \frac{1}{n} \sum_{k=0}^{n-1} g \circ \theta^k$ converge $\mathbb{P}$ -a.s. and in $L^1(\mathbb{P})$ to $\mathbb{E}[g \mid \mathcal{I}]$. If the system is ergodic, the limit is the constant $\mathbb{E}[g]$.*

Proof sketch. The engine is the *maximal ergodic lemma* (Garsia’s form): for $g \in L^1$, writing $S_k = \sum_{j=0}^{k-1} g \circ \theta^j$ and $M_N = \max_{1 \leq k \leq N} S_k$, one has $\int_{\{M_N > 0\}} g d\mathbb{P} \geq 0$. (Sketch of this in turn: on $\{M_N > 0\}$, $M_N = g + \max(0, M_{N-1} \circ \theta) \leq g + M_N \circ \theta$, and integrating, using $\int M_N = \int M_N \circ \theta$ by invariance, leaves $\int_{\{M_N > 0\}} g \geq 0$.) Applying it to $g - \lambda$ yields the maximal inequality $\mathbb{P}(\sup_n A_n g > \lambda) \leq \mathbb{E}|g|/\lambda$. For pointwise convergence, set $\bar{g} = \limsup_n A_n g$ and $\underline{g} = \liminf_n A_n g$; both are invariant. For rationals $\beta < \alpha$ the set $E = \{\underline{g} < \beta < \alpha < \bar{g}\}$ is invariant, and applying the maximal lemma to $g - \alpha$ and to $\alpha - g$ *restricted to E* sandwiches $\alpha\mathbb{P}(E) \leq \int_E g \leq \beta\mathbb{P}(E)$; since $\alpha > \beta$ this forces $\mathbb{P}(E) = 0$. Taking the countable union over $\beta < \alpha$ gives $\bar{g} = \underline{g}$ a.s., so $A_n g$ converges a.s. to an invariant limit g_∞ . That $g_\infty = \mathbb{E}[g \mid \mathcal{I}]$ is its defining property: for $A \in \mathcal{I}$, $\int_A A_n g = \int_A g$ by invariance, and passing to the limit gives $\int_A g_\infty = \int_A g$. Finally the maximal inequality makes $\{A_n g\}$ uniformly integrable, so a.s. convergence upgrades to L^1 . Ergodicity makes $\mathcal{I}$ trivial, whence $\mathbb{E}[g \mid \mathcal{I}] = \mathbb{E}[g]$ a.s. $\square$

A.3 The finite-memory case: Assumption 2.4 in one line

For an order- m model — a finite-state chain (§2.6) or the AR(1) law, where the predictive density depends only on a bounded window — one has, for $t > m$,

$$Z_t = \log \frac{f_1(S_t \mid S_{t-m:t-1})}{f_0(S_t \mid S_{t-m:t-1})} = \psi(S_{t-m}, \dots, S_t) = \psi \circ \theta^{t-m-1}$$

for one fixed measurable window-functional ψ . Thus $\{Z_t\}_{t>m}$ is a stationary functional of the shift, and Theorem A.1 applied to $g = \psi$ gives $\frac{1}{n} \sum_{t=m+1}^n Z_t \rightarrow \mathbb{E}_{\mathbb{P}_{\text{att}}}[\psi]$ a.s. and in L^1 ; the m initial terms contribute $O(1/n) \rightarrow 0$. Ergodicity (Assumption 2.3) makes the limit the constant

$$R = \mathbb{E}_{\mathbb{P}_{\text{att}}}[\psi] = \mathbb{E}_{\mathbb{P}_{\text{att}}}[D(f_1(\cdot \mid S_{1:m}) \parallel f_0(\cdot \mid S_{1:m}))],$$

the stationary mean of the per-step conditional divergence. The only side condition is $\mathbb{E}_{\mathbb{P}_{\text{att}}}|\psi| < \infty$, which is automatic when the per-step log-likelihood ratio is bounded — a finite alphabet with strictly positive kernels, or the Gaussian model where ψ is quadratic with finite stationary moments. This is the “short Birkhoff computation” promised in Remark 2.5, and it already covers every model for which the paper exhibits a closed form.

A.4 The general case: relative-entropy Shannon–McMillan–Breiman

When the memory is unbounded the increments Z_t condition on a *growing* past $S_{1:t-1}$ and are no longer a stationary sequence, so Theorem A.1 does not apply to them directly. The bridge is martingale convergence.

Theorem A.2 (Lévy’s upward theorem). *If $\mathcal{F}_n \uparrow \mathcal{F}_\infty$ and $X \in L^1$, then $\mathbb{E}[X \mid \mathcal{F}_n] \rightarrow \mathbb{E}[X \mid \mathcal{F}_\infty]$ a.s. and in L^1 .*

Pass to the two-sided stationary extension of $\mathbb{P}_{\text{att}}$ on $\mathcal{X}^{\mathbb{Z}}$ (the natural extension of the shift; it exists for any stationary source and leaves all finite-dimensional laws unchanged). There the *infinite-history* increment

$$\tilde{Z}_t := \log \frac{f_1(S_t \mid S_{-\infty:t-1})}{f_0(S_t \mid S_{-\infty:t-1})} = \tilde{\psi} \circ \theta^t$$

is a genuine stationary, ergodic sequence, so by Theorem A.1, $\frac{1}{n} \sum_{t=1}^n \tilde{Z}_t \rightarrow \mathbb{E}_{\mathbb{P}_{\text{att}}}[\tilde{Z}_1] := R$ a.s. and in L^1 , provided $\tilde{Z}_1 \in L^1(\mathbb{P}_{\text{att}})$. The finite-history increment differs from $\tilde{Z}_t$ only in conditioning on the truncated past $S_{1:t-1}$ rather than $S_{-\infty:t-1}$; by Theorem A.2 this discrepancy tends to 0 a.s. and in L^1 as the available past grows, and a Cesàro argument (the technical core of Barron’s theorem) shows the averaged discrepancy $\frac{1}{n} \sum_{t \leq n} (Z_t - \tilde{Z}_t) \rightarrow 0$. Combining,

$$\frac{1}{n} \Lambda_1^n = \frac{1}{n} \sum_{t=1}^n Z_t \longrightarrow R \quad \mathbb{P}_{\text{att}}\text{-a.s. and in } L^1.$$

Theorem A.3 (Relative-entropy SMB; Barron 1985, Algoet–Cover 1988). *Let $\mathbb{P}_{\text{att}}$ be stationary and ergodic and $\mathbb{P}_\infty$ stationary, with $\mathbb{P}_{\text{att}}^{(n)} \ll \mathbb{P}_\infty^{(n)}$ for every n and finite relative-entropy rate $R = \lim_n \frac{1}{n} D(\mathbb{P}_{\text{att}}^{(n)} \parallel \mathbb{P}_\infty^{(n)})$. If $\tilde{Z}_1 \in L^1(\mathbb{P}_{\text{att}})$ then $\frac{1}{n} \log \frac{d\mathbb{P}_{\text{att}}^{(n)}}{d\mathbb{P}_\infty^{(n)}} \rightarrow R$ $\mathbb{P}_{\text{att}}$ -a.s. and in $L^1(\mathbb{P}_{\text{att}})$.*

The order- k truncations $R^{[k]} = \mathbb{E}_{\mathbb{P}_{\text{att}}}[D(f_1(\cdot \mid S_{1:k}) \parallel f_0(\cdot \mid S_{1:k}))]$ that approximate R are exactly the estimands of Proposition 6.15, with $R^{[0]} = d$ and $R^{[k]} \rightarrow R$; note their approach to R need *not* be monotone — the sign of $R^{[k]} - R$ tracks $d - R$, the same one-bit asymmetry that the soundness analysis turns on — which is precisely why the relative-entropy statement needs the martingale bridge above rather than the monotone-conditioning argument that proves the entropy AEP.

A.5 Assembly, and the precise hypotheses

Assumption 2.4 holds — R exists as a finite constant, $\mathbb{P}_{\text{att}}$ -a.s. and in L^1 — under exactly:

- (H1) *Stationarity* of $\mathbb{P}_{\text{att}}$ (and $\mathbb{P}_\infty$): Assumption 2.3. Gives existence of the time-average limit (Theorem A.1).
- (H2) *Ergodicity* of $\mathbb{P}_{\text{att}}$: Assumption 2.3. Collapses the limit from the invariant random variable $\mathbb{E}[\tilde{Z}_1 \mid \mathcal{I}]$ to the constant $\mathbb{E}[\tilde{Z}_1]$.
- (H3) *Finite-horizon absolute continuity* $\mathbb{P}_{\text{att}}^{(n)} \ll \mathbb{P}_\infty^{(n)}$: Assumption 2.1, which makes each Λ_1^n a.s. finite and equal to the log density ratio (Lemma 2.7).

(H4) *Integrability* $\mathbb{E}_{\mathbb{P}_{\text{att}}}|\tilde{Z}_1| < \infty$, equivalently a finite relative-entropy rate. Automatic when the per-step log-likelihood ratio is bounded.

Under (H1)–(H4): finite memory invokes Theorem A.1 directly (§A.3); unbounded memory invokes Theorem A.3 (§A.4). The L^1 half of the conclusion is what licenses the identification with the *divergence* rate that names the paper: $\frac{1}{n}D(\mathbb{P}_{\text{att}}^{(n)}\|\mathbb{P}_{\infty}^{(n)}) = \frac{1}{n}\mathbb{E}_{\mathbb{P}_{\text{att}}}\Lambda_1^n \rightarrow R$, since L^1 convergence permits exchanging the limit and the expectation.

A.6 When it fails

Each hypothesis is doing visible work, and dropping it breaks the conclusion in a specific way.

- *Non-ergodic* (drop (H2)). The limit still exists by Theorem A.1, but as the invariant random variable $\mathbb{E}[\tilde{Z}_1 \mid \mathcal{I}]$, not a constant. In the mixture of §A.1 it is R_A or R_B according to the initial coin: a rate that is itself random. The remedy is to condition on the ergodic component; every result then holds component-wise, with R the rate of whichever component is realised.
- *Infinite rate* (drop (H4)). If $\mathbb{E}_{\mathbb{P}_{\text{att}}}|\tilde{Z}_1| = \infty$ the average can fail to converge to a finite limit. This is the analytic face of the heavy-tailed cliff of Remark 2.20: an infinite-mean gap sends the per-step contribution's normalisation to zero, $R \rightarrow 0$ in the fresh-start reading, and the logarithmic speed limit becomes vacuous.
- *Non-stationary* (drop (H1)). A closed-loop attacker who times malice against the monitor's own running output breaks shift-invariance; there is then no single R , and the rate framework does not apply (§8).
- *One-sided subtlety*. The infinite-history increment $\tilde{Z}_1$ lives on the two-sided extension; for the one-sided stream this is the standard stationary coding of the infinite past, implicit in (H1), and the boundary effect of conditioning on $S_{1:t-1}$ rather than $S_{-\infty:t-1}$ is exactly what the Cesàro step of §A.4 shows to be asymptotically negligible.